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PureLine PQ EO



OPERATION AND MAINTENANCE MANUAL

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8
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12

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Contact Details

If, after reading all the sections of the manuals provided, you still experience problems with your UV system, contact your local Technical Support representative:

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Please have the system's serial number available (printed on the rating label on the front of the cabinet).

Visit us on the web: www.nuvonicuv.com

To order a further copy of this manual, use the contact details above quoting the code on the front cover.

Warranty

Nuvonic warrants their UV systems against defective materials and manufacturing defects for a period of 18 months from date of receipt.

Refer to our website for full details of the terms and conditions of warranty for systems, UV lamps, and other consumables.

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Passwords

The default user names and passwords required to gain access to the different system menus are listed below.

Menu	User name	Password
Operator	—	—
Supervisor	supv	9132
Engineer	engr	78014

The Operator menu allows basic operation of the system (switching on and off, displaying of readings, warnings and trips).

The Supervisor and Engineer menus allow changes to system settings which will affect how the system operates and performs. We therefore recommend that these passwords are only divulged to personnel who have been fully trained in operating and maintaining the UV system.

The Factory menu is for use by the manufacturer only.



System parameters are factory-set to suit the requirements of the process. Changing these parameters can lead to insufficient treatment levels or malfunctioning of the entire system, and may prevent the minimum performance requirements being met.

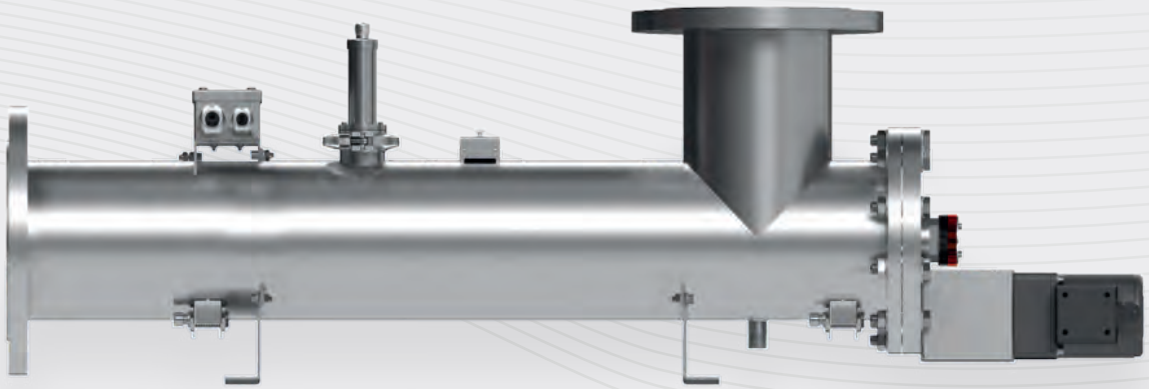
The configuration of the system should only be changed by personnel who are fully aware of the implications of the changes. For technical assistance and details of training courses, please consult technical support using the contact details given on the inside front cover of this manual.

Remove this page from the manual when appropriate to prevent unauthorised access to the software menus.

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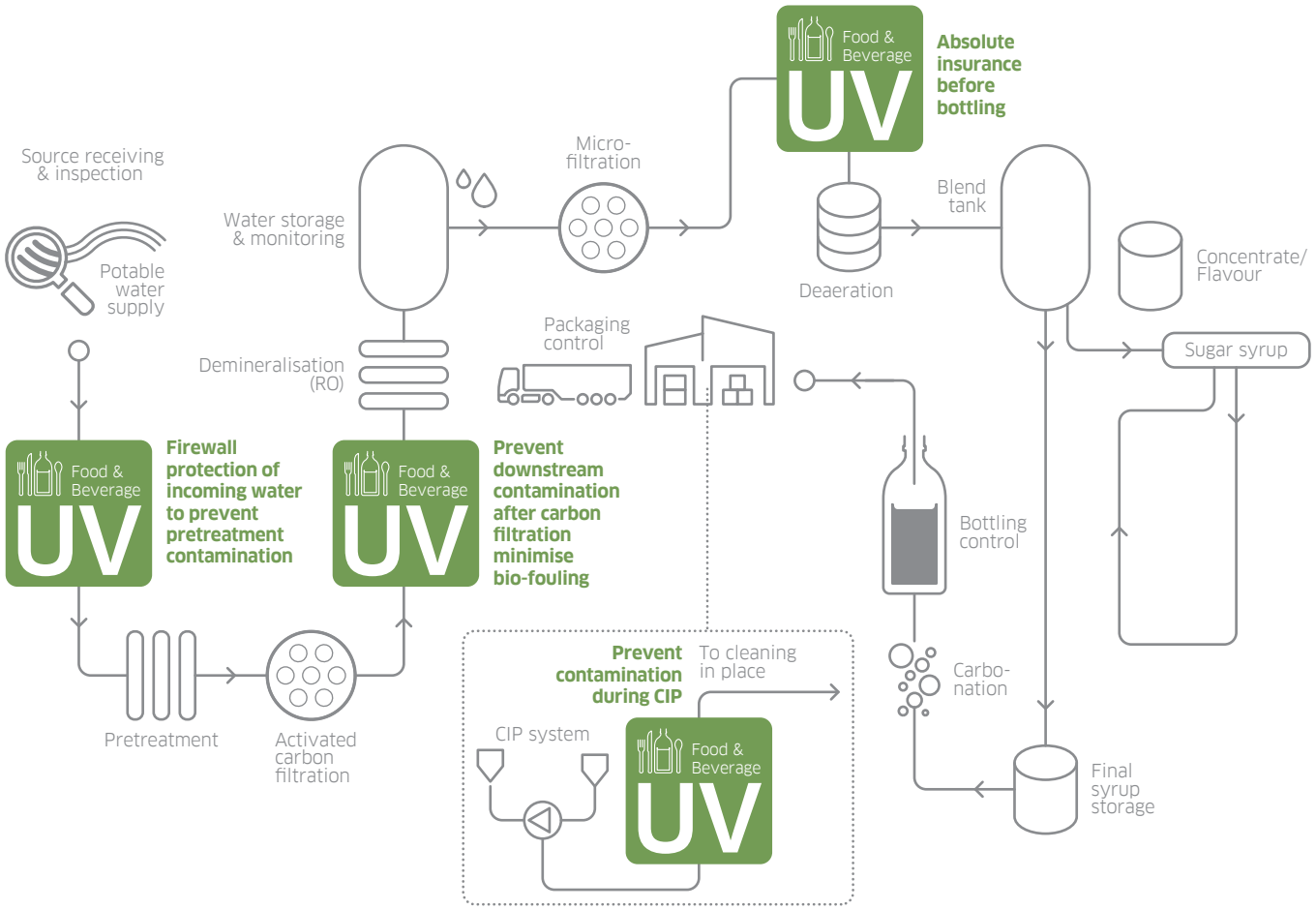
PureLine PQ EO

BIOASSAYED UV TREATMENT FOR FOOD & BEVERAGE

Our **PureLine PQ EO** UV systems are aimed specifically at providing third party bioassayed UV treatment for product and process waters used in the food and beverage industry. Integrating innovative single medium pressure lamp chamber design with sensors and intelligent control technology to automatically deliver optimum treatment performance with high operational efficiency. Eliminating harmful micro-organisms, reduce the bioburden, protect against bio-fouling, and lead to fewer CIP / SIP cycles. With a certified dry UV sensor that measures the germicidal output of the UV system and a UV dose read out makes it easy to monitor and log performance. The control system can take flow and transmittance meter inputs and calculate the UV dose based on real time operating conditions.

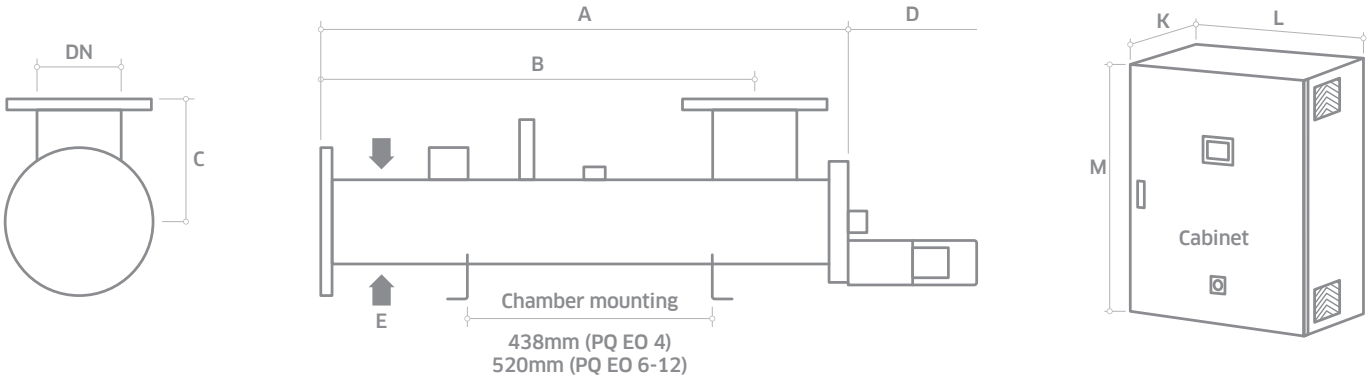
Application Optimised UV for Food & Beverage

POTENTIAL LOCATIONS OF THE PURELINE PQ EO™



KEY FEATURES	WHAT IT GIVES YOU	BENEFITS FOR YOU
INTELLIGENCE		
UV sensor	Continuous verification of performance with in-built low UV dose alarm	Easy to monitor and log system performance
UVGuard™ on UV sensor window	Protects against UV exposure when checking a UV duty sensor with a reference sensor while the system is operating	Ability to safely audit the UV performance without interrupting production
Flow and UV transmittance (UVT) meter inputs	Stepless adjustment of lamp power based on real time operating conditions	Optimised use of energy, saving operating costs
OPTIMISATION		
Single medium pressure lamp	Provides germicidal wavelengths to treat your product or process water	Does not affect taste and colour of final product
		No chemicals
		Protects pre-treatment equipment and RO filters from bio-fouling reducing CIP frequency and downtime
	High treatment capacity with a single lamp	Compact footprint and reduced operating cost
Innovative chamber design	Maximises the water's exposure to UV light	Reduces energy costs
Designed specifically for the food and beverage industry	Flanged connections, standard internal finish	Reduced system cost where sanitary design is not critical
	FDA and EC approved seals	Industry compliant materials
	*Automatic wiper	Self cleaning to maintain performance
INTEGRATION		
Designed for your process	*Skid mountable	Easy to install
	*UVShield™ power cut-out for lamp access	Enhanced operator safety when changing a lamp
	*Water leak detection	Increased product safety
	RS 485 Industrial Ethernet	Easy integration to SCADA or plant control systems

*Option



MODEL NUMBER	MAX POWER (KW)	MIN T10(%)	DIMENSIONS (MM)														APPROX WEIGHT (KG)		
	Starting		Chamber							Control Cabinet (fan cooled)			Control Cabinet (with A/C)			Chamber	Control Cabinet	
			Unwiped A	Wiped A	B	C	D	E	DN	K*	L	M**	K*	L	M**	(Empty)	Fan cooled	With A/C
PureLine PQ EO 4	4.5	80	1009	1232	823	165	900	114	100	400	800	1200	400	1250	1200	30	96	120
PureLine PQ EO 6	4.5	80	1035	1286	850	245	950	168	150	400	800	1200	400	1250	1200	44	96	120
PureLine PQ EO 8	4.5	80	1110	1361	875	320	1000	210	200	400	800	1200	400	1250	1200	65	96	120
PureLine PQ EO 10	4.5	80	1190	1441	903	430	1100	273	250	400	800	1200	400	1250	1200	96	96	120
PureLine PQ EO 12	6.8	80	1430	1685	1093	475	1325	324	300	400	800	1200	400	1250	1200	145	96	120

All dimensions are approximate for clearance purposes only. We have a policy of continuous product development, exact drawings are available on request.
All specifications are subject to change without notification. Your distributor or our account manager can advise on correct sizing and specification requirements.
* Allow dimension L in front of cabinet for door opening and panel access.
** M dimension includes the space for the cabinet mounting brackets but you need to allow space below the cabinet for cable entry and access (minimum of 250 mm).

UV CHAMBER	
Material:	StSt 316L / 1.4404
Internal finish:	As made pipe and tube, welds as laid, electropolished and passivated
External finish:	BS EN 10088-2 or 10088-3, 1J or 2J and ASTM No. 4
Process (mating) connections:	Flange EN 1092-1 PN16
Drain connection:	Tri-clamp blanked off
End plate:	Removable end plate
Degree of protection:	IP65 equivalent to NEMA 4 but not for outside use
Arc tube (lamp):	Medium pressure
Arc tube enclosure:	Doped quartz (F240)
Number of arc tubes (lamps):	1
Expected lamp life:	9000 hours
Temperature sensor:	Yes
UV sensor:	Calibrated DVGW compliant dry sensor with UVGuard™ sensor window
Working fluid temperature:	with UVGuard™ sensor window
Maximum CIP temperature:	95°C lamp off and CIP request acknowledged
Hydrostatically pressure tested:	Yes to PED requirements EN 13445
Chamber mounting:	Horizontal only
Operating pressure:	6 bar (positive pressure only)
Seals:	EPDM, ADI free, EC 1935/2004, FDA 21 CFR 177.2600 approved

OPTIONS
Document Support Pack
Cabinet: Stainless steel 304
Cabinet: Stainless steel 304 with air conditioning (5°-50°C), IP66 (NEMA 4X), relative humidity <95% non condensing
Cabinet: Stainless steel 316 with air conditioning with slooping roof (5°-50°C), IP66 (NEMA 4X), relative humidity <95% non condensing
Operation and Maintenance manual and printed Installation and Commissioning manual in Chinese, English, French, German and Spanish
Wiper: Automatic (electrically driven)
Flange options: ANSI 150, JIS, Table 'E' and tri-clamp
Chamber internal finish: <0.6 µm Ra or <0.38 µm Ra, welds polished out, electropolished and passivated
Lead length: 20 and 29 m
Max CIP temp: 130°C lamp turned off and CIP request acknowledged
Operating pressure: 10 bar or 16 bar
Vent valve: Manual valve hygienic design

OPTIONS (CONTINUED)
Aggressive water package: For 400 ppm to 20000 ppm chloride water
UVShield™: Power cut-out for lamp access
Bleed valve: Hygienic valve with tri-clamp connection
Skid mounting (not ship board or earthquake zone)
Welder Document Pack for chamber construction
Water leak detection: Detects water leaking from the UV lamp enclosure
UL 508A
In field UV reference sensor kit

CABINET (CONTROLLER UVTOUCH™)	
Material:	Polyester coated carbon steel
Degree of protection:	IP55 / NEMA 12
Supply voltages:	380 V to 480 V (-5% to +10%), 50/60 Hz
Operating temp range:	5°C to 40°C
Relative humidity:	<85% non-condensing
Cooling fans:	Yes

CABINET (GENERAL)	
Ballast power adjustment:	Stepless variable power (30 to 100% of maximum ballast rating
Interconnecting cable lengths:	10 m cabinet to chamber

CUSTOMER OUTPUTS	
4-20 mA passive outputs:	UV RED dose, UV intensity and chamber temperature
VFC outputs:	Lamp ready (enable flow), system running, common warning, common trip, low dose warning, water leak detected, system in remote, OK to CIP

CUSTOMER INPUTS	
4-20 mA active or passive inputs:	Flow meter and transmittance meter
VFC inputs:	Remote stop/start, remote reset, remote CIP request, reduce power
24 V dc pulsed inputs:	Start and stop

CUSTOMER COMMUNICATIONS PORT	
RS 485:	Industrial Ethernet

APPROVALS
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Contents

Passwords 3
1. Introduction 13
1.1. Glossary of Abbreviations/Terms 13
1.2. System Description 15
2. Safety 16
2.1. Risks Associated with the System 16
2.2. System Warning Labels 16
2.2.1. Cabinet 16
2.2.2. Chamber 17
2.3. Ultra Violet Light 17
2.4. Pressurised Liquid 18
2.5. Earthing 18
2.6. High Voltages 19
2.7. High Temperatures 19
2.8. Mercury 19
2.8.1. Mercury spillage procedure 20
2.9. Breakage of the Thimble 21
2.10. Risk Assessment 21
3. Installation and Commissioning 23
3.1. Parts Supplied 23
3.2. Handling and Storage Instructions 23
3.3. Installation Requirements 24
3.4. Installing the UV Lamp 29
3.5. Interface Wiring 31
3.5.1. Remote On/Off control 34
3.5.2. Communication using Ethernet 35
3.6. Connecting and Configuring Analogue Inputs 36
3.7. Connecting and Configuring Analogue Outputs 39
3.8. Configuring Digital Outputs 41
3.9. Configuring the Dose Operating Mode 43
3.9.1. Dose control in validated systems 44
3.10. Setting the Initial Intensity Value 46
3.11. Installing a Bleed Valve (option) 47
3.12. Adjusting the Water Leak Detector (option) 48
3.13. Setting System Date and Time 49
3.14. Commissioning Procedure 49
4. System Operation 50

4.1. Software Overview....51
4.2. Switching On and Off....52
4.3. Start/Stop Modes....53
4.3.1. Local mode (Push buttons)....53
4.3.2. Local mode (Timed)....54
4.3.3. Remote VFC mode....54
4.3.4. Remote Pulsed mode....54
4.3.5. Remote Comms mode....55
4.4. Wiper Operation (option)....55
4.5. Warnings and Trips....56
4.6. Reduced Dose....57
4.6.1. Local mode (Push buttons)....58
4.6.2. Local mode (Timed)....58
4.6.3. Remote mode (VFC or Pulsed)....58
4.6.4. Remote Comms mode....58
4.7. Operator Menu....59
4.8. Supervisor Menu....61
4.9. Engineer Menu....63
4.10. Data Logging....67
4.11. Using Recipes....68
5. Maintenance....69
5.1. Safe Practices for Maintenance Procedures....71
5.2. Service Schedule....72
5.3. UV Lamp Replacement....72
5.4. Quartz Thimble Replacement....76
5.4.1. Removing the quartz thimble....76
5.4.2. Inspecting and cleaning the thimble....79
5.4.3. Fitting the thimble....80
5.5. UV Sensor....81
5.5.1. Removing the UV sensor and housing....81
5.5.2. Cleaning the UV sensor and housing....83
5.5.3. Reinstalling the UV sensor and housing....83
5.5.4. Replacing the UV sensor housing window....85
5.5.5. UV sensor verification....90
5.6. Replacing the Chamber Temperature Sensor....92
5.7. Auto-Wiper (option)....93
5.7.1. Removing the auto-wiper assembly....93
5.7.2. Servicing the leadscrew bearing....94
5.7.3. Servicing the wiper yoke....97

5.7.4. Refitting the auto-wiper assembly 100
5.7.5. Replacing wiper limit switches 102
5.8. Replacing the Ballast 102
5.9. Replacing the Touchscreen Battery 105
5.10. Replacing the Cabinet Temperature Sensors 106
5.11. Maintenance of the Cooling Fans / Air Conditioner 106
5.12. Replacing the Surge Protection Module 107
5.13. Spares List 109
5.14. Clean-in-Place 111
6. Alarms and Fault-Finding 114
6.1. Fault Flowcharts 114
6.2. System Alarms 118
6.3. Ballast Troubleshooting 122
6.4. System Event Sequence 123
6.5. Testing Volt-Free Contacts 125
7. Ethernet Communications 126
7.1. Establishing Communication 126
7.2. Sending and Receiving Data 126
8. Effect of UV Light on Process Liquid 129
8.1. Ultraviolet Light 129
8.2. Intensity 129
8.3. Dose 129
9. Lamp / Quartz Breakage During Operation 130
10. Decommissioning 131
10.1. Temporarily Suspending Operation 131
Appendix A: System Record Table 132
Appendix B: Declarations of Conformity 133
Appendix C: Drawings 137

Tables

Table 1: Available Inputs and Outputs 32
Table 2: Configurable Output Alarm Options 42
Table 3: Fault-Finding Table 118
Table 4: Ballast Display Messages 122
Table 5: Ethernet Tags and Modbus Addresses 126

Figures

Figure 1: System Overview....15
Figure 2: System Connection Diagram....26
Figure 3: Incoming Supply Cable Connections....27
Figure 4: Inserting UPS Fuses (option)....28
Figure 5: User's Terminal Connections in Control Cabinet....32
Figure 6: Ethernet Connection Port in Control Cabinet....35
Figure 7: Procedure for Selecting Start/Stop Mode....36
Figure 8: Procedure for Configuring Analogue Sensor Inputs....38
Figure 9: Procedure for Configuring Default Analogue Outputs....40
Figure 10: Procedure for Configuring Optional Analogue Sensor Output....40
Figure 11: Procedure for Configuring Digital Outputs....41
Figure 12: Control of the Dose Using Set Points....43
Figure 13: Procedure for Configuring Average Dose Mode....45
Figure 14: Procedure for Configuring RED Dose Mode....45
Figure 15: Alternative Settings and Display for RED Dose Mode....46
Figure 16: Procedure for Setting Bleed Temperature....47
Figure 17: Procedure for Setting Date and Time....49
Figure 18: Cabinet Controls....50
Figure 19: UV Touch software structure....51
Figure 20: Start/Stop Buttons on Home Screen....54
Figure 21: Procedure for Setting the On/Off Timer....54
Figure 22: Procedure for Changing Auto-wiper Time Interval....56
Figure 23: Warning Indicators (example of Flow warning shown)....57
Figure 24: Procedure for Setting Reduced Dose by Timer....58
Figure 25: Micro SD Card Port for Data Logging....67
Figure 26: Procedure for Saving or Loading Recipes....68
Figure 27: Chamber Parts....70
Figure 28: Power Isolator Switch Locked Off....71
Figure 29: Procedure for Resetting Usage Counters for Consumables....75
Figure 30: UV Sensor Housing Parts....85
Figure 31: UV Sensor Test Procedure....91
Figure 32: Fitting the Chamber Temperature Sensor....92
Figure 33: Wiper Parts....93
Figure 34: Wiper Yoke Assembly Components....98
Figure 35: Position of Limit Switch after Replacement....102
Figure 36: CIP Procedure (Local Mode)....112
Figure 37: Clean-in-Place Timing Diagram....113

Figure 38: System Fault Flowchart 115
Figure 39: System Warnings and Trips Flowchart 116
Figure 40: Low/Failed UV Intensity Fault Flowchart 117
Figure 41: Timing Diagram for Normal Operation and Low UV Events (not to scale) 123
Figure 42: Operational State Sequence 124
Figure 43: Procedure for Testing VFCs 125

1. Introduction

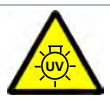
This manual is intended to provide detailed guidance on the installation, operation and maintenance of your UV system.

Please make this manual available for reference by any persons working with the system.

A section has been provided (Appendix A) to note updates, modifications and operating data. Ensuring this section is used and kept up to date is the responsibility of the equipment owner and will assist in ensuring the system is maintained in a good operating condition.

A printed Installation and Commissioning Manual is also provided with this system.

The following symbols are used in this manual (les symboles suivants sont utilisés dans ce manuel):

	This symbol indicates a requirement or warning. Indique une nécessité ou un avertissement.
	This symbol indicates a recommendation. Indique un conseil.
	This symbol indicates an electrical hazard. Indique un danger électrique.
	This symbol indicates a hazard due to ultraviolet (UV) light. Indique un danger en raison de la lumière ultraviolette (UV).
	This symbol indicates a hazard due to pressurised liquid within the chamber. Indique un danger dû à un liquide sous pression dans la chambre.
	This symbol indicates a hazard due to high temperature. Indique un danger en raison de températures élevées.

1.1. Glossary of Abbreviations/Terms

316L	A grade of 316 stainless steel with reduced carbon content
CFD	Computational Fluid Dynamics
COSHH	Control of Substances Hazardous to Health regulations
Dose	UV energy incident on unit surface area, measured in mJ/cm^2; also known as fluence
Drg	Drawing
H/F	High Frequency
ID	Internal Diameter
Intensity	UV power incident on unit surface area, measured in W/m^2
IPA	Iso-propyl alcohol (2-propanol)
LCD	Liquid Crystal Display
LED	Light-Emitting Diode

Log inactivation/ reduction	Reduction in the number of organisms that can reproduce, expressed as a logarithm (1 = 90% reduction, 2 = 99%, 3 = 99.9%, etc.)
OD	Outer Diameter
Qualification	Certification package used to demonstrate system performance according to industry standards
P/N	Part number
RED	Reduction Equivalent Dose – measured UV dose in mJ/cm^2 delivered by the system; known as VRED if divided by validation factor to correct for uncertainties in measurement
Thimble	A quartz sleeve which surrounds the UV lamp and is sealed at one end
UV	Ultraviolet
UVT	Ultraviolet Transmittance; the percentage of UV light transmitted through a material, usually over a distance of 10 mm (0.4 inches); also abbreviated as TX
VFC	Volt-Free Contact
VP	Variable Power

1.2. System Description

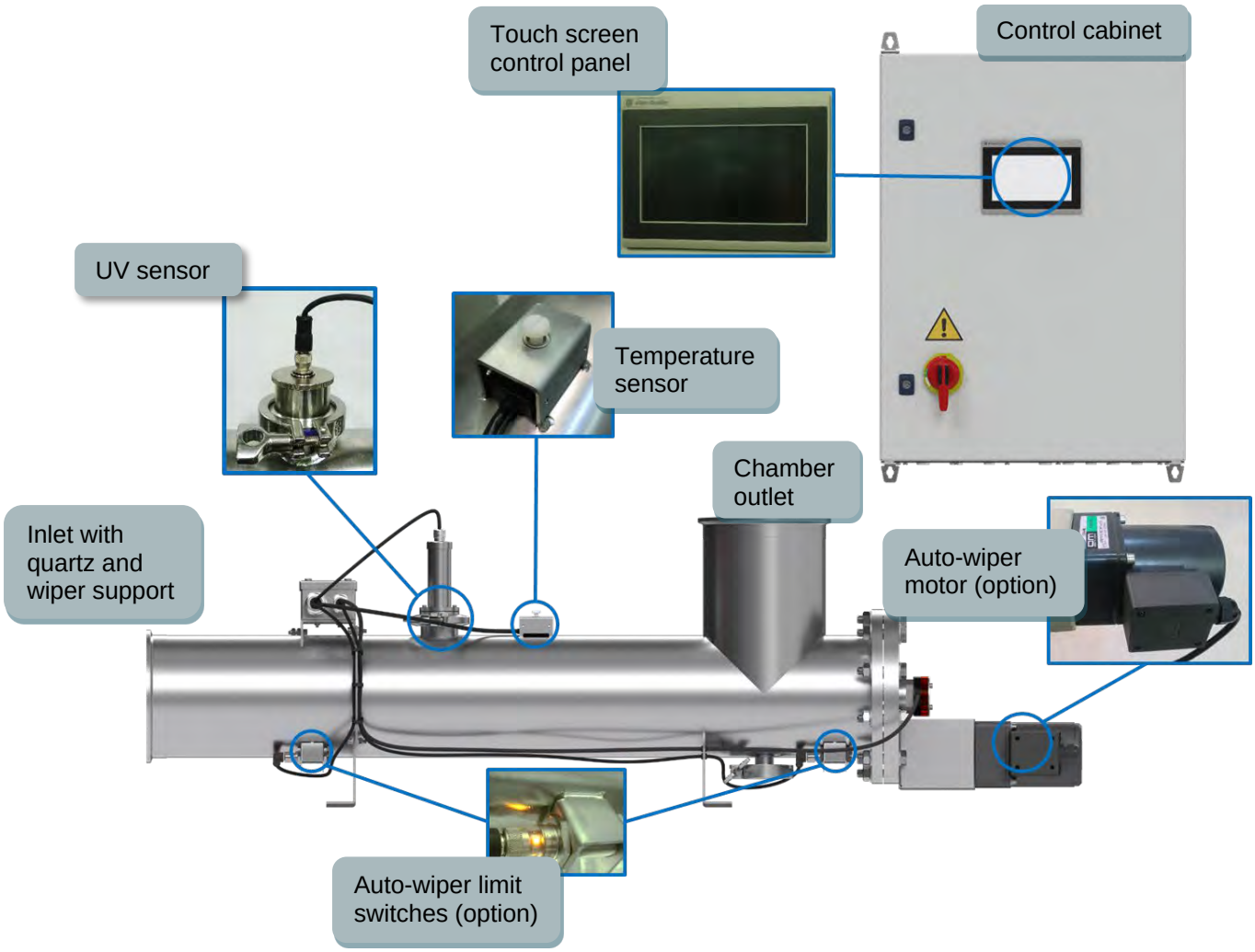
The UV system provides treatment of process liquid by exposing it to high intensity UV light, which inactivates viruses and bacteria, rendering them harmless. The UV light is generated by a medium-pressure UV lamp inside a stainless steel chamber, which is optimised to ensure that the process stream receives a specified dose of UV to achieve the required level of treatment. Sensors attached to the chamber send data to the UV Touch controller to allow continuous monitoring and optimisation of the system's performance.

An automatic wiper is available as an option to remove deposits from the quartz thimble surrounding the UV lamp. A wiper ring travels up and down the thimble and in the process cleans any deposits from the surface. Limit switches detect the end of each wipe stroke and stop the motor.

Where systems are used in critical applications, additional parts and documentation are available, including certified lamps, higher internal chamber finishes and tri-clamp fittings.

The primary components of the system are shown in Figure 1.

Figure 1: System Overview



2. Safety

2.1. Risks Associated with the System

The protection that the equipment provides may be impaired if used in a manner not specified in this manual.

Incorrect operation or use for other than the intended purpose may:

- endanger the health of the operator;
- damage the unit and other equipment;
- prevent efficient operation of the unit.

All persons concerned with the installation, commissioning, operation, maintenance and repair of the unit:

- **must** be suitably qualified;
- **must** observe the operating instructions;
- **should** attend a specific training course on the system's operation and maintenance.

Take the time to read and understand the Safety AND the System Operation sections in this manual before attempting to use the system in order to ensure correct and safe operation.

	**IF IN DOUBT - ASK.**

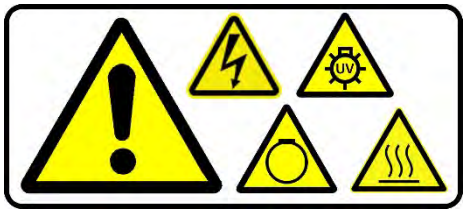
	**System parameters are factory-set to suit the requirements of the process. Changing these parameters can lead to insufficient treatment levels or malfunctioning of the entire system, and may prevent the minimum performance requirements being met.** **The configuration of the system should only be changed by personnel who are fully aware of the implications of the changes. For technical assistance and details of training courses, please consult technical support using the contact details given on the inside front cover of this manual.**

2.2. System Warning Labels

2.2.1. Cabinet

	Label affixed on front of cabinet. General warning, caution to be taken. Access allowed only by qualified personnel. Risk of electric shock. Refer to the Operation and Maintenance Manual before commencing any servicing procedures. Étiquette apposée sur le devant de l'armoire et sur le côté du ballast. Avertissement général, précautions à prendre. Accès autorisé au personnel qualifié uniquement. Risque de choc électrique. Reportez-vous au manuel d'utilisation et d'entretien avant toute intervention.

2.2.2. Chamber

<table>
<tr><td rowspan="6">  </td><td colspan="2">Label provides warnings for:
Les étiquettes préviennent contre les dangers suivants :</td></tr>
<tr><td>  </td><td>Chamber to be accessed only by authorised personnel.
Accès au réacteur restreint aux personnes autorisées uniquement.</td></tr>
<tr><td>  </td><td>Electrical hazard
High voltages; risk of electric shock. DO NOT remove any covers or fittings whilst the UV lamp is ON.
Danger électrique
Hautes tensions ; risque de choc electrique. Ne pas retirer de caches ou de raccords avec la lampe UV allumée.</td></tr>
<tr><td>  </td><td>Optical radiation hazard (UV)
UV light is harmful to eyes and exposed skin. DO NOT remove any covers or fittings whilst the UV lamp is ON.
Danger rayonnement optique (UV)
Les rayons UV sont nocifs pour les yeux et la peau exposée. Ne pas retirer de caches ou de raccords avec la lampe UV allumée.</td></tr>
<tr><td>  </td><td>Pressurised system warning
Chamber may be under pressure. Release liquid pressure and drain before servicing.
Avertissement : système sous pression
Le réacteur peut être sous pression. Relâcher la pression et vidanger avant entretien.</td></tr>
<tr><td>  </td><td>High temperature (Hot surface)
The outside surface of the chamber may be hot. Care should be taken when handling.
Température élevée (surface chaude)
La surface extérieure du réacteur peut être chaude. Prendre ses précautions lors de la manipulation.</td></tr>
<tr><td>  </td><td colspan="2">Label affixed to chamber adjacent to lamp access point. Provides warning of UV light hazard (see above) and risk of electric shock. Access allowed only by qualified personnel.
Etiquette apposée sur le réacteur de manière adjacente au point d’accès à la lampe. Fournit un avertissement du danger lié au rayonnement UV et au risque de choc électrique. Accès au réacteur restreint aux personnes autorisées uniquement.</td></tr>
<tr><td>  </td><td colspan="2">Label affixed adjacent to wiper motor power supply connections. Provides warning that motor may start automatically on a timed cycle, and of risk of electric shock if cover removed. Isolate at the control panel.
Etiquette apposée de manière adjacente aux connexions d’alimentation moteur du nettoyeur automatique. Le moteur peut démarrer automatiquement sur un programme minuté. Fournit un avertissement du risque de choc électrique si le couvercle est retiré. Isoler au niveau du panneau de commande.</td></tr>
</table>

2.3. Ultra Violet Light

The UV lamp emits wavelengths principally in the 200–315 nm region. The emissions in this region are invisible and highly damaging to eyes and exposed skin.



DO NOT ATTEMPT TO OPERATE THE UV LAMP OUTSIDE THE CHAMBER.

DO NOT ATTEMPT TO LOOK AT AN OPERATING UV LAMP.

N'ESSAYEZ PAS DE FAIRE FONCTIONNER LA LAMPE UV EN DEHORS DE LA CHAMBRE.

N'ESSAYEZ PAS DE REGARDER UNE LAMPE UV EN FONCTIONNEMENT.

2.4. Pressurised Liquid

During operation, the pressure of the liquid in the chamber can reach up to 6 bar (600 kPa or 87 psi) (optionally 10 or 16 bar (145 or 232 psi)). If the quartz thimble nipple is removed before the system is depressurised, the thimble can be ejected violently from the chamber and may cause serious injury to personnel. This could also occur with other parts connected to the chamber including the UV sensor housing.



ALWAYS ISOLATE AND DRAIN THE CHAMBER BEFORE CHANGING THE QUARTZ THIMBLE OR ANY OTHER WETTED PARTS, AND ENSURE NO-ONE IS STANDING IN LINE WITH THE THIMBLE WHILE THE QUARTZ NIPPLE (THIMBLE FITTING) IS REMOVED.

Refer to section 5.4 for further details.

TOUJOURS ISOLEZ ET VIDANGEZ LA CHAMBRE AVANT DE CHANGER LA GAINE DE QUARTZ OU TOUT AUTRE PIÈCES MOUILLÉES, ET VÉRIFIEZ QUE PERSONNE NE SE TIENT EN LIGNE AVEC LA GAINE DE QUARTZ PENDANT QUE LA BAGUE D'APPUI EST SUPPRIMÉE.

Reportez-vous au section 5.4 pour plus de détails.



2.5. Earthing



THE EQUIPMENT MUST BE EARTHED CORRECTLY USING A CONDUCTOR OF ADEQUATE CROSS-SECTION, AND THE SUPPLIED LEADS.

Refer to section 3 for installation details.

L'ÉQUIPEMENT DOIT ÊTRE CORRECTEMENT MIS À LA TERRE À L'AIDE D'UN CONDUCTEUR DE SECTION TRANSVERSALE ADÉQUATE AINSI QUE LES CÂBLES FOURNIS.

Pour plus de détails concernant l'installation, reportez-vous à la section 3.

2.6. High Voltages



ENSURE INSTALLATION HAS BEEN CARRIED OUT IN ACCORDANCE WITH THIS MANUAL BEFORE OPERATING EQUIPMENT.

There may be voltages of up to 480 V in the cabinet and UV lamp connections.

Wait a minimum of five (5) minutes after the isolator has been turned off before gaining access to the cabinet, ballast or UV chamber connections.

Before attempting to service a treatment chamber, ensure that the power supply is completely disconnected by means of the door isolator. The isolator should be padlocked in the OFF position, and local lock-off procedures followed.

AVANT DE FAIRE FONCTIONNER L'ÉQUIPEMENT, ASSUREZ-VOUS QUE L'INSTALLATION A BIEN ÉTÉ RÉALISÉE CONFORMÉMENT AUX INSTRUCTIONS DE CE MANUEL.

L'armoire et les connexions de la lampe UV peuvent présenter des tensions jusqu'à 480 V.

Après extinction de l'isolateur, attendre un minimum de cinq (5) minutes avant d'avoir accèss à l'armoire ou aux connexions de la chambre de traitement.

Avant d'intervenir sur une chambre de traitement, s'assurer que l'alimentation électrique est complètement déconnectée par le sectionneur de la porte. Le sectionneur doit être cadenassé en position OFF et les procédures locales de verrouillage doivent être respectées.



ALL INSTALLATION AND MAINTENANCE WORK SHOULD BE CARRIED OUT BY QUALIFIED PERSONNEL.
TOUTES LES TÂCHES D'INSTALLATION ET DE MAINTENANCE DOIVENT ÊTRE RÉALISÉES PAR DES EMPLOYÉS QUALIFIÉS.

2.7. High Temperatures

The UV lamp wall reaches high temperatures during normal operation and the lamp retains heat for a considerable time after it has been switched off.



TAKE EXTREME CARE WHEN HANDLING HOT UV LAMPS.
FAITES EXTRÊMEMENT ATTENTION LORSQUE VOUS MANIPULEZ DES LAMPES UV CHAUDS.

2.8. Mercury

UV lamps contain a small quantity of mercury (<250 mg / 0.009 ounces).

Cotton gloves should be worn at all times when handling lamps to prevent the glass being marked with fingerprints, and care should be taken to avoid breakage. New lamps should be stored in their original packaging until required for use.

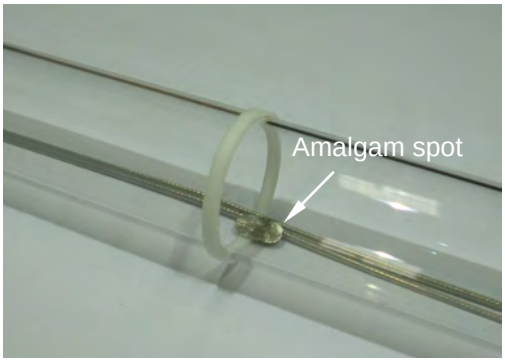
Old UV lamps must be disposed of safely. On request, your supplier representative can supply details of suitable disposal companies.

2.8.1. Mercury spillage procedure

In the event of a UV lamp breakage, carry out the following procedure.

Check List
Items needed to clean up a small mercury spill:
Zip-Lock type plastic bags
Rubber or latex gloves
Paper towels
Cardboard or squeegee
Eyedropper, syringe or pipette
Duct tape, or shaving cream and small paint brush
Mercury Spillage Cleaning kit (optional)

1. Put on rubber or latex gloves.
2. If there are any small broken pieces of glass or sharp objects, pick them up with care. Place all broken objects on a paper towel. Fold the paper towel and place in a zip lock bag, secure and label as broken glass.
3. Locate visible mercury beads or amalgam spots. Use a squeegee or cardboard to gather mercury beads. Use slow sweeping motions to keep mercury from becoming uncontrollable.



   Note: Mercury can move surprising distances on hard-flat surfaces, so be sure to inspect the entire room when "searching."
4. Use the eyedropper to collect or draw up the mercury beads. Slowly and carefully squeeze mercury onto a damp paper towel. Place the paper towel in a zip lock bag, secure and label as spilt mercury.
5. After you have removed the larger beads, put shaving cream on top of a small paint brush and gently "dot" the affected area to pick up smaller hard-to-see beads. Alternatively, use duct tape to collect smaller hard-to-see beads. Place the paint brush or duct tape in a zip lock bag and secure.
6. OPTIONAL STEP: It is optional to use commercially available mercury cleaning kits that contain powdered sulphur to absorb the beads that are too small to see. The sulphur performs two functions: (1) it makes the mercury easier to see since there may be a colour change from yellow to brown and (2) it binds to the mercury so that it can be easily removed and suppresses the vapour of any residual mercury. Mercury spill kits can normally be purchased from laboratory, chemical supply and hazardous materials supply manufacturers.
7. Mercury waste should be properly disposed of in accordance with local laws for hazardous waste.

2.9. Breakage of the Thimble

The thimble that encapsulates the UV lamp is made from quartz, a highly purified form of silica. Cotton gloves should be worn at all times when handling thimbles and care should be taken to avoid breakage. The thimble may crack or shatter if subjected to sudden impact forces from sharp edges. The edges of the pieces of broken quartz will be extremely sharp and should be handled very carefully. Ensure that the chamber is thoroughly cleaned out after a breakage to remove quartz fragments.

Provided that it is changed as recommended in the service schedule, the quartz thimble is able to withstand considerable compressive forces when loaded evenly, so the thimble is very unlikely to break due to the force exerted by the pressurised liquid in the chamber. It can also withstand high thermal gradients, and will not be affected by the temperature difference between the operating UV lamp and the liquid being treated.

2.10. Risk Assessment

OPERATION OR PROCEDURE: Commissioning, Servicing, Maintaining and Operating Medium Pressure UV Systems

Hazards	Persons at Risk	Controls and Precautions
 High Voltages (Cabinet and UV lamp)	Service Engineer Operator, Installer	1. Door switch safety interlock. 2. ‘High voltage’ warning signs. 3. Circuit breakers in OFF position. 4. Statement in this guide referring to high voltages (section 2.6). 5. Secured keyhole plate to prevent access to UV lamp or UVShield™ (optional) to cut power when attempting to access lamp.
 Ultraviolet light (UV lamp)	Service Engineer Operator, Installer	1. Hazard warning label on UV chamber. 2. Instruction to switch off, isolate system and lock off isolator prior to replacing UV lamp. 3. Statement in this guide referring to exposure to eyes and skin (section 2.3). 4. UVGuard™ shutters to prevent exposure to UV light when checking UV sensor. 5. Automatic cut-out of lamp if UVShield™ unit (optional) is removed. 6. Visual indicator on UVShield™ unit (optional) to show if lamp is lit.
 Pressurised liquid (UV chamber)	Service Engineer Operator, Installer	1. Hazard warning label on UV chamber. 2. Instruction to switch lamp off, isolate and drain chamber prior to replacing thimble. 3. Instruction to stand to one side of chamber when removing thimble.
 High temperatures (UV chamber)	Service Engineer Operator, Installer	1. Hazard warning label on UV chamber. 2. Statement in this guide referring to high chamber temperatures (section 2.2.1), with recommendation for risk assessment.

	High temperatures (UV lamp)	Service Engineer Operator, Installer	1. Instruction to allow lamp to cool before removal. 2. Statement in this guide referring to high temperatures (section 2.7).
	Mercury (UV lamp)	Service Engineer Operator, Installer	1. Follow COSHH or local safety guidelines. 2. Mercury spillage procedure (section 2.8.1).
	Heavy weights (Cabinet and UV chamber)	Installer	1. Weights given in data sheet at front of manual. 2. Recommendation to use suitable lifting equipment when installing.
	Noise	Operator	1. The noise level of the machinery is below 70 dB(A) at any time during all foreseeable uses of this machine.

3. Installation and Commissioning



ALL INSTALLATION WORK MUST BE CARRIED OUT BY QUALIFIED PERSONNEL USING APPROPRIATE SAFE WORKING PRACTICES.*

READ THE SAFETY SECTION BEFORE COMMENCING INSTALLATION.

TOUT TRAVAIL D'INSTALLATION DOIT ÊTRE EFFECTUÉ PAR DU PERSONNEL QUALIFIÉ EN UTILISANT DES PRATIQUES DE TRAVAIL SURES ET APPROPRIÉES.*

VEUILLEZ LIRE LA SECTION TRAITANT DE LA SÉCURITÉ AVANT DE COMMENCER L'INSTALLATION.

* Refer to local safety standards and guidelines as appropriate, such as NFPA 70E in the USA and the current edition of the IEE Wiring Regulations in the UK.

3.1. Parts Supplied

The system components are:

- 1 × Control cabinet (either fan-cooled or air-conditioned)
- 1 × UV chamber with optional auto-wiper mechanism
- 1 × UV sensor
- 1 × Quartz thimble (preinstalled in the chamber)
- 1 × UV lamp
- 1 × set of UV leads (prewired to the cabinet)
- 1 × Installation and Commissioning Guide (printed copy)
- 1 × Operating and Maintenance Manual (electronic copy)
- 1 × Panel locking key
- 1 × 3 mm hex head driver (required to change UV lamp)

Also required (but not supplied) are:

- An adjustable wrench with jaw opening at least 46 mm (required for the quartz nipple removal).
- A section of plastic pipe, 1.5–2 m (4.9–6.6 ft) long and 25 mm (1 inch) in diameter (to assist with quartz thimble removal).

3.2. Handling and Storage Instructions

1.	Prior to installation, store the equipment in a clean dry area at -20ºC to 55ºC (-4ºF to 131ºF), with a relative humidity of less than 95% (non-condensing).
2.	The UV system equipment is delivered in protective packaging but it should be protected from shocks and heavy weights. When unpacking the equipment, check that all components are present and undamaged. Contact your supplier representative if there is any damage.
3.	The cabinet and chamber are large and heavy (refer to the data sheet at the front of this manual for specific weights and dimensions). Before unpacking the equipment, perform a risk assessment to identify a safe method of lifting and moving the items. Use suitable and properly maintained lifting equipment operated by trained personnel.

4.	**Do NOT touch the quartz thimble or UV lamp with bare hands** – use cotton or non-marking disposable gloves to prevent the glass being marked with fingerprints.

3.3. Installation Requirements



The performance of this UV system has been validated for a defined range of operating conditions. It must be installed with specified inlet and outlet pipework to achieve this performance. Contact your supplier for further information.

La performance de ce système UV ont été validées pour une gamme définie de conditions d'utilisation. Il doit être installé avec une canalisation d'entrée et de sortie spécifiée pour atteindre cette performance. Contactez votre fournisseur pour plus d'informations.

The UV chamber should be installed in a readily accessible location to enable maintenance of the system.



Access space equal to or greater than the chamber length MUST be allowed at the opposite end to the chamber's water inlet for UV lamp and quartz thimble replacement.

À préciser toutefois que l'on DOIT prévoir un dégagement égal ou supérieur à la longueur de la chambre à l'extrémité opposée de l'orifice d'entrée de l'eau de la chambre pour le remplacement de la lampe UV et de la gaine de quartz.

Other requirements when installing the system are as follows:

1.	**Install the equipment in an indoor environment**, out of direct sunlight.
2.	**Do NOT install a UV chamber upstream from the process pump**, as negative pressure can cause damage to the internal components, in particular the quartz thimble.
3.	The **ambient temperature** in the installation area should be between **5ºC and 45ºC (41ºF and 113ºF)** with a **relative humidity** of less than **85% (non-condensing)** for fan-cooled cabinets and **95% (non-condensing)** for air-conditioned cabinets.
4.	The cabinet and the chamber are large and heavy; refer to the data sheet at the front of this manual for specific weights and dimensions. Before installing the system: • **Identify a suitable fixture or structure** against which the items can be mounted which will safely support their weight. • **Identify suitable fittings** for attaching the items to the supporting fixtures. • **Perform an assessment of the safest method of lifting and supporting the items** while they are attached to the supporting fixtures. Use suitable and properly maintained lifting equipment operated by trained personnel. If the system is mounted on a skid, **ensure that the skid is fixed to the floor securely and that the floor is suitable for supporting the system**.
5.	**Ensure that the chamber can be safely isolated and drained for servicing**. We recommend that lockable valves are fitted at the inlet and outlet of the chamber, and that a bypass is installed to allow the process liquid to flow during servicing.

6. The installed system must match the installation used during validation testing, to ensure that the delivered UV dose is equal to or greater than the validated dose. Therefore, **a minimum length of straight pipe must be connected to the chamber inlet**. This pipe should contain no reducers/expanders or valves which might affect the liquid flow. The dimensions of the inlet pipe are shown in the table below. We also recommend that **at least two pipe diameters of straight pipe are fitted to the chamber outlet** before any reducers or other components are installed.

Model	Internal Diameter (mm)	Minimum Straight Length at Inlet (mm / inches)
PQ EO 4	100	500 / 20
PQ EO 6	150	750 / 30
PQ EO 8	200	1000 / 40
PQ EO 10	250	1250 / 50
PQ EO 12	300	1500 / 60

7. **If the installation uses plastic pipework**, we recommend using stainless steel pipework for the first metre at the inlet and outlet to minimise the exposure of the plastic pipework to UV light. Any bends installed as light traps should be at least 2 pipe diameters from the connection point.

8. **Ensure that the displays on the control panel are at eye level** or otherwise easily viewable.

9. **Mount the control cabinet clear of the floor** as a precaution against water ingress.

10. **Ensure that the top of the chamber is easily accessible** in case access to the sensors and junction box is required during maintenance of the system.

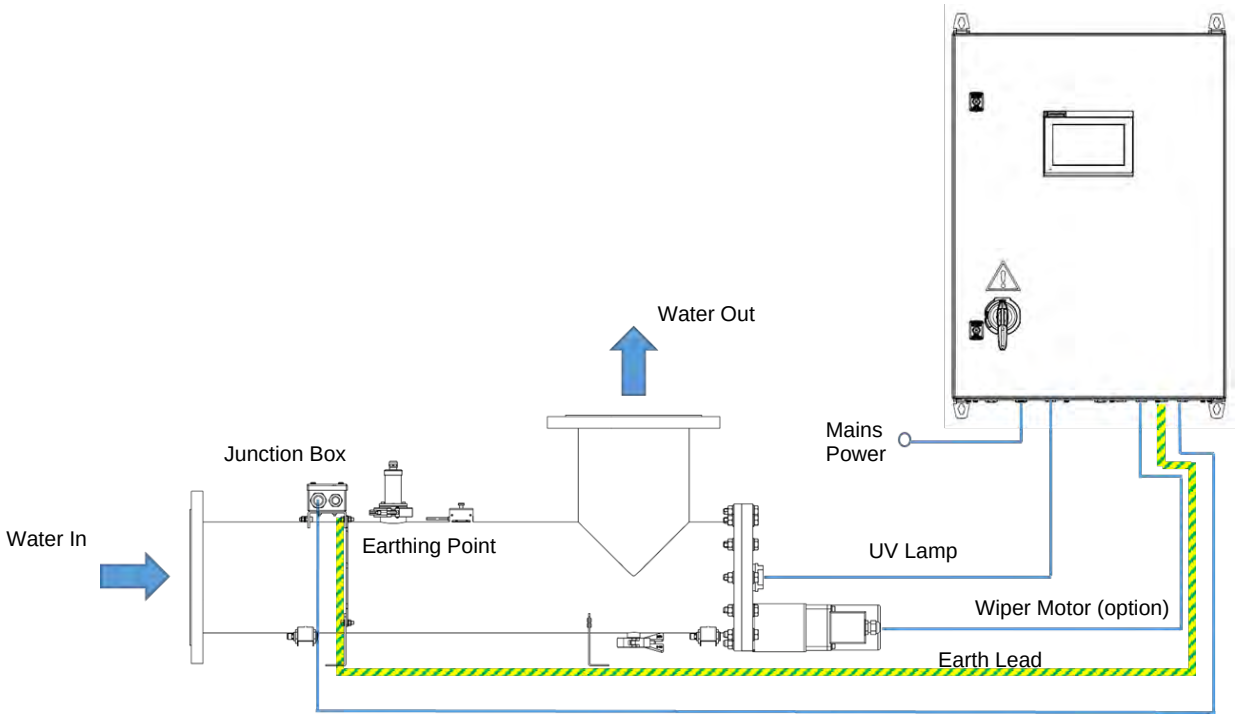
11. **Perform a risk assessment and take suitable precautions to prevent accidental contact with the surface of the chamber** in case it becomes hot during operation. This is particularly important when the alarm temperature is set above 50ºC (122ºF) or if the system ever operates without liquid flow.

12. The **standard cable length** between system components is 10 m (32.8 ft). If the cables provided are longer than required, we recommend that they are shortened by a competent electrician. **Do not coil the cables.** If longer cables are required, please contact your supplier representative.

    Install cables to minimize mechanical stress and exposure to water and other fluids, and in accordance with local regulations.

13. Connect the leads between the system components as shown in Figure 2 and the system drawings in Appendix C. Use the bracket provided to keep the junction box lead away from the chamber surface. If the optional wiper is fitted, **separate the wiper wiring** from the high tension and power cables.

Figure 2: System Connection Diagram

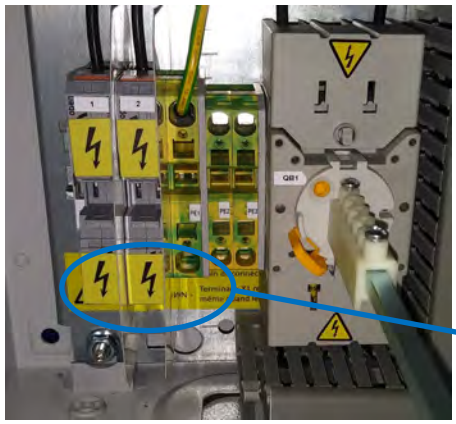


14. Ensure that the **earth lead** (green/yellow) from the cabinet is fitted to the UV chamber earthing point (Figure 2). Use the bracket provided to keep the lead away from the chamber surface.

15. **If the ETA Plus ballasts in the cabinet have been unused for more than 2 years**, elevated leakage currents may arise on the internal electrolytic capacitors which can damage the ballast. As a preventative measure, apply 115–230 VAC to the ballast main supply for one hour to reform the capacitors BEFORE reconnecting the normal power supply and operating the ballast.

16. The mains supply cable to the control cabinet must be wired on site. This is connected to the cabinet through the available inlet ports on the bottom of the panel and is wired to the -X1 terminal block as shown in the circuit diagrams. Install the system in accordance with local electrical safety guidance and requirements, such as the National Electric Code (NEC) in the USA and the current edition of the IEE Wiring Regulations in the UK, and refer to these for methods to incorporate derating factors such as cable type, temperature and bunching, etc.

Figure 3: Incoming Supply Cable Connections



Notes:

1. The **mains power supply must be stable to within half a cycle** and have a voltage tolerance within ±10% of the allowed range. Systems installed in locations with power supplies that do not meet these criteria must be equipped with an Uninterruptible Power Supply (UPS) operating continuously. Failure to observe this requirement could result in the lamp stopping frequently, causing the system to trip.
2. **Do not connect the mains power supply to the control cabinet via an Earth Leakage Circuit Breaker (ELCB)**. This is because the cabinet is fitted with an EMC filter that is connected to earth and the leakage current will cause an ELCB to trip. The ballast that drives the UV lamp has a built-in Earth Leakage Breaker.

17. Refer to the circuit diagrams at the back of this manual for information about the power consumption of each system and the recommended circuit breakers to be fitted to the mains electrical power supply.

The system must only be connected to the correct mains power supply as indicated on the rating label on the front of the control cabinet. If the correct voltage is not available, DO NOT connect the supply to the cabinet. Contact your supplier representative.

Le système doit être raccordé à l'alimentation secteur adéquate comme indiqué sur la plaque signalétique placée sur le devant de l'armoire. Si la tension adéquate n'est pas disponible, NE raccordez PAS l'alimentation à l'armoire. Contactez le représentant de votre fournisseur.

18. If your system is equipped with a **UPS** inside the cabinet, you will find two 25 A fuses provided in a sealed bag attached to the door of the UPS. Just before you switch on the mains power supply, open the UPS, insert these fuses as shown in Figure 4, and close the door.

19. **Check the system for leaks BEFORE fitting the UV lamp.**

20. Ensure that the **PLC inside the cabinet is switched to RUN** (the switch shown below must be pushed to the left).



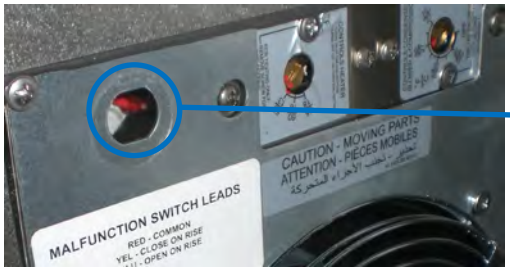
21. Ensure that the **control cabinet locking key** is kept in a convenient, safe location for future access to the cabinet components.

Figure 4: Inserting UPS Fuses (option)





On control cabinets cooled by air-conditioning units, check that the phase monitor window inside the cabinet is lit when the system is switched on. If not, disconnect the main power supply, swap any two of the power supply cables and then switch power on again – the window should now be lit. If the phases are not correct, then the compressor will not operate and the cabinet will not be cooled (even though all other components will operate normally).



Phase monitor window (must be lit when power is switched on)

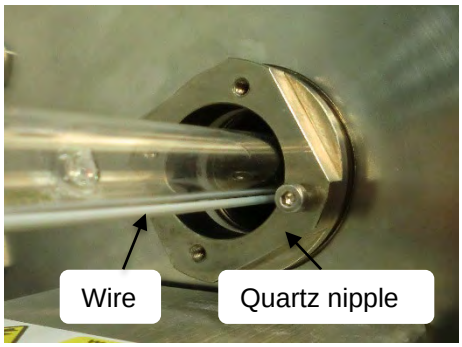
3.4. Installing the UV Lamp

WEAR COTTON OR NON-MARKING DISPOSABLE GLOVES WHEN HANDLING THE UV LAMP AND SPACER TO PREVENT THE GLASS BEING MARKED WITH FINGERPRINTS.

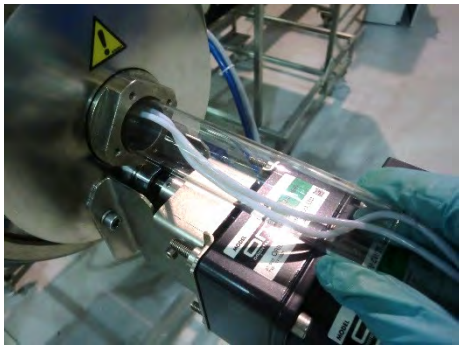
PORTEZ DES GANTS DE COTON OU DES GANTS JETABLES NE MARQUANT PAS POUR MANIPULER LA LAMPE UV ET LA VIROLE EN QUARTZ POUR EMPÊCHER QUE LE VERRE SOIT MARQUÉ PAR DES EMPREINTES DIGITALES.

1. Carefully insert the UV lamp (supplied separately) into the thimble through the quartz thimble fitting (nipple).

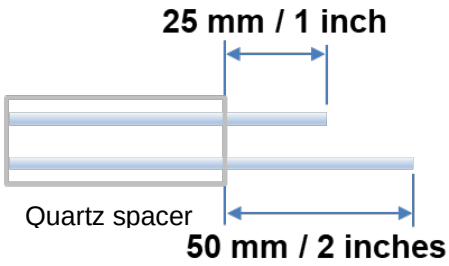
   Note: Rotate the lamp so that the wire from the far end is underneath the quartz bulb.



2. Slide the lamp spacer over the lamp wires and push it against the end of the lamp.



3. Check the lamp wire lengths and, if necessary, cut the lamp wires to the lengths shown, measured from the end of the lamp spacer.

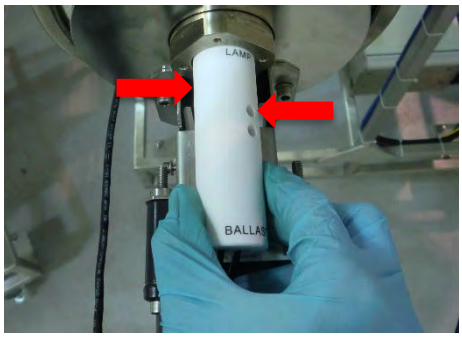


4. If necessary, strip 10 mm (0.4 inches) of outer insulation from the end of each wire. If preferred, securely crimp a ferrule to each wire for ease of connection to the terminals in the ceramic block. Trim any excess wire from the ends of the ferrules.



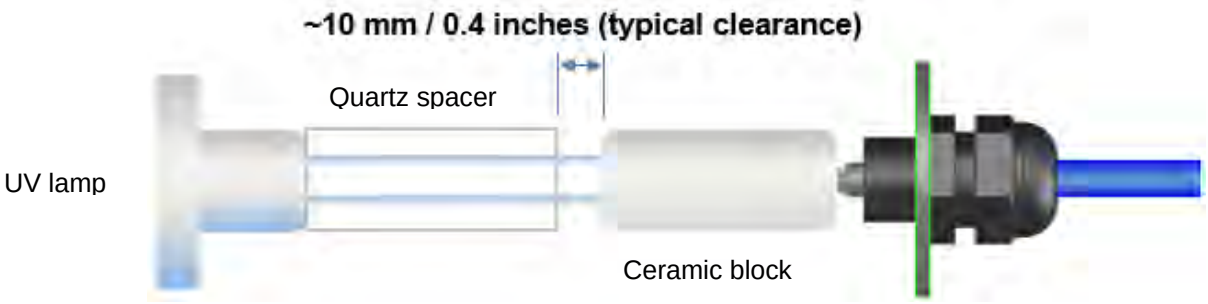
5. Connect the lamp wires to the terminals in the ceramic block at the end marked LAMP (indicated with red arrows).

   It does not matter which way around the wires are connected.



6. Push the ceramic block and spacer into the end of the thimble.

   When installed, the lamp position should be set by the combined length of the spacer and the ceramic block as shown. **Note that the quartz spacer must be pressed against the end of the lamp.**



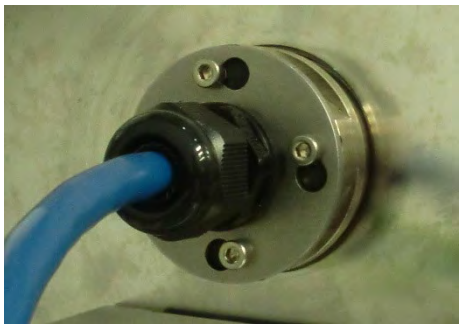
DO NOT COIL LAMP CABLES.

If the cables provided are longer than required, we recommend that they are shortened by a competent electrician.

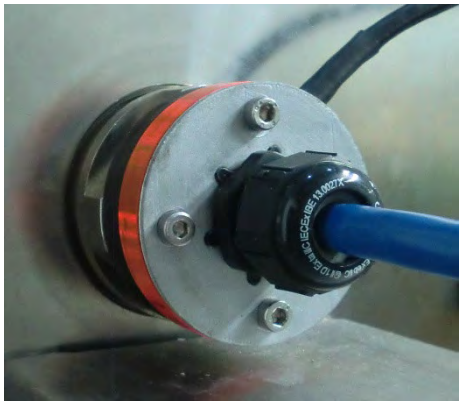
7. Screw the lamp connector onto the quartz nipple.

   If a keyhole plate is fitted then rotate the plate before tightening the screws.

   A UVShield™ unit (which also contains an optional leak detector) may be installed on some systems. This glows to indicate when the UV lamp is on, and disconnects power to the lamp if it is removed from the quartz nipple or if a water leak is detected through the quartz thimble.



Keyhole plate



UVShield™ unit

3.5. Interface Wiring

Remote control and monitoring of the UV system is achieved using the user's interface terminals located in the control cabinet (Figure 5). These connections are described in Table 1, and shown in the system site wiring diagrams in Appendix C.

Figure 5: User's Terminal Connections in Control Cabinet

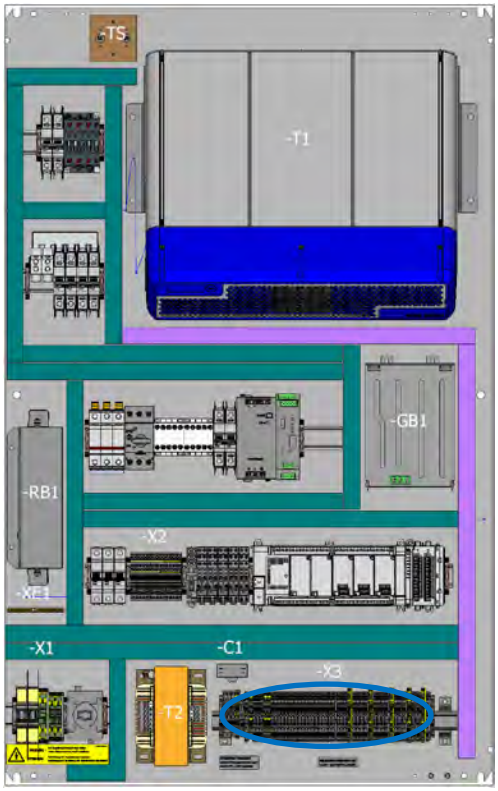


Table 1: Available Inputs and Outputs

Label on Drawing	Terminals	I/O Type	Function
Customer's SCADA (Ethernet RS-485)	KEB1.1	RJ-45 input to control panel.	Allows the user to control and interrogate the system remotely over a Local Area Network using Ethernet.
Remote Stop/Start	X3:14, 15	Digital input (volt-free contact closure) from user's control system.	Closing a volt-free (dry) contact across these terminals will start the lamp when in remote. Opening the contact will stop the lamp.
Remote Reset	X3:16, 17	Pulsed digital input (volt-free contact closure) from user's control system.	Closing a volt-free (dry) contact across these terminals will return the system to the Available state after a Trip. Only operates when system in Remote modes (section 4.5). Open contact once system reset.
Remote CIP Request	X3:18, 19	Digital input (volt-free contact closure) from user's control system.	Closing a volt-free (dry) contact across these terminals will prepare the system for a Clean-In-Place operation if the CIP function is enabled (section 5.14). An OK to CIP signal (see below) is sent when system is ready.

Label on Drawing	Terminals	I/O Type	Function
Reduce Power	X3:20, 21	Digital input (volt-free contact closure) from user's control system.	Reduces risk of overheating during periods when the flow through the system is recirculated or stopped. HIGH (closed) input triggers Reduced Dose mode; lamp runs at reduced dose and Low Dose alarms are set to the lower dose limit. Only operates when system in Remote modes (section 4.6).
Remote Pulse Start/Stop	X3:22 (START) X3:23 (STOP)	Digital input (24 V dc) from user's control system.	Pulsed signals (minimum 1 second duration) used to switch the lamp on and off.
System Set for Remote Control	X3:35	Digital output (volt-free contact) to user's control system.	HIGH (closed) if system is in Remote (VFC, Pulsed or Comms) mode, and goes LOW (open) if mode is changed to Local or Timer mode.
Available	X3:36	Digital output (volt-free contact) to user's control system.	HIGH (closed) if system is in Available state, LOW (open) if not.
Warming	X3:37	Digital output (volt-free contact) to user's control system.	HIGH (closed) if system is in Warming state, LOW (open) if not.
Flow Enable	X3:38	Digital output (volt-free contact) to user's control system.	HIGH (closed) if system is in Flow Enable state, LOW (open) if not. Used to initiate valve or pump start sequences from the user's control system.
Running	X3:39	Digital output (volt-free contact) to user's control system.	HIGH (closed) if system in Running state, LOW (open) if not.
Cooling	X3:40	Digital output (volt-free contact) to user's control system.	HIGH (closed) if system in Cooling state, LOW (open) if not.
Common Alarm Warning	X3:41	Digital output (volt-free contact) to user's control system.	Initially LOW (open) and goes HIGH (closed) after the system has a non-critical fault (e.g., Low UV). Alarm must be rectified before the output goes LOW (open).
Common Alarm Trip	X3:42	Fail-safe digital output (volt-free contact) to user's control system.	Initially HIGH (closed) and goes LOW (open) after the system has a critical fault (e.g., chamber over-temperature). Alarm must be rectified and system reset before the output goes HIGH (closed).
Low UV Dose	X3:43	Digital output (volt-free contact) to user's control system.	Initially LOW (open) and goes HIGH (closed) when the system has a Low UV fault.
Water Leakage Trip	X3:44	Fail-safe digital output (volt-free contact) to user's control system.	Initially HIGH (closed) and goes LOW (open) when the water leak detector detects a leak. Alarm must be rectified and system reset before the output goes HIGH (closed). This can be used for special shutdown sequences or alarms as it can indicate a broken quartz thimble.

Label on Drawing	Terminals	I/O Type	Function
OK to CIP	X3:45	Digital output (volt-free contact) to user's control system.	Initially LOW (open) and goes HIGH (closed) when the system is in the CIP Cycle state. A Clean-In-Place process can now be performed on the system (section 5.14).
Configurable Outputs in Software	X3:46 X3:47 X3:48	Digital output (volt-free contact) to user's control system.	Use the Engineer menu to select the specific details of each output (section 3.8).
Flow	X3:60-63 and adjacent PE terminal	Analogue 4–20 mA input to cabinet.	From a flow meter or from the user's control system. Input can be active or passive (see section 3.6 and the system site wiring diagrams).
UV Transmission	X3:64-67 and adjacent PE terminal	Analogue 4–20 mA input to cabinet.	From an online transmittance meter or from the user's control system. Requires T_{10} value. Input can be active or passive (see section 3.6 and the system site wiring diagrams).
Spare Analogue I/Ps	X3:68-71 or X3:72-75 and adjacent PE terminals	Analogue 4–20 mA input to cabinet.	
Dose (RED) O/P (see note below)	X3:76-77 and adjacent PE terminal	Analogue 4–20 mA output to user's control system.	Active analogue output of RED calculated from dose model (section 3.7).
Dose (Avg) O/P (see note below)	X3:78-79 and adjacent PE terminal	Analogue 4–20 mA output to user's control system.	Active analogue output of average dose calculated from dose model (section 3.7).
UV Intensity	X3:80-81 and adjacent PE terminal	Analogue 4–20 mA output to user's control system.	Active analogue output of measured UV intensity (section 3.7).
Configurable	X3:82-83 and adjacent PE terminal	Analogue 4–20 mA output to user's control system.	Active analogue output selected from following options: UV transmittance, chamber temperature, flow rate, % lamp hours run, % quartz hours run and % wiper service hours remaining. Refer to section 3.7.

A reduced UV dose is measured briefly when the sensor housing window is wiped. If you are monitoring the UV dose remotely then ignore a dip of 15 seconds duration if your system is wiped.

Une dose d'UV réduite est mesurée brièvement lorsque la fenêtre d'accès de capteur est essuyée. Si vous surveillez la dose d'UV à distance, vous pouvez ignorer un pendage de 15 secondes si de votre système possède un nettoyeur.

3.5.1. Remote On/Off control

To switch the system on and off remotely, make a connection either to X3:14 and 15 for a continuous 24 V dc signal or to X3:22 and 23 for a 24 V dc pulse (minimum 1 second duration). The system then needs to be set to the appropriate Remote mode (**VFC** or **Pulsed**) (Figure 7).

3.5.2. Communication using Ethernet

The system can be controlled and interrogated remotely using an RJ-45 Ethernet port available inside the cabinet (Figure 6). See section 7 for further information on communicating with the system using Ethernet.

Figure 6: Ethernet Connection Port in Control Cabinet

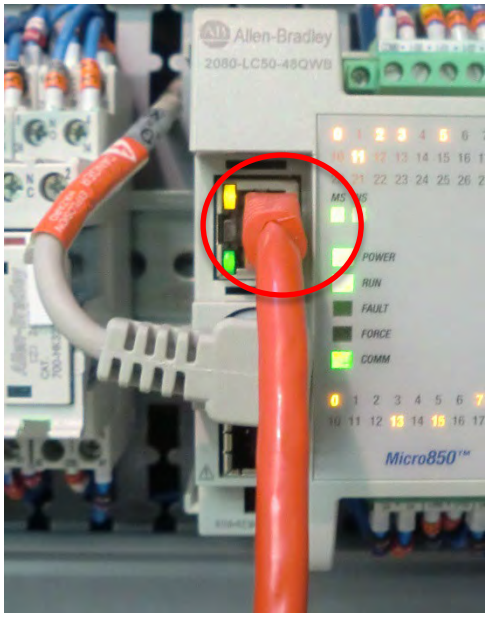
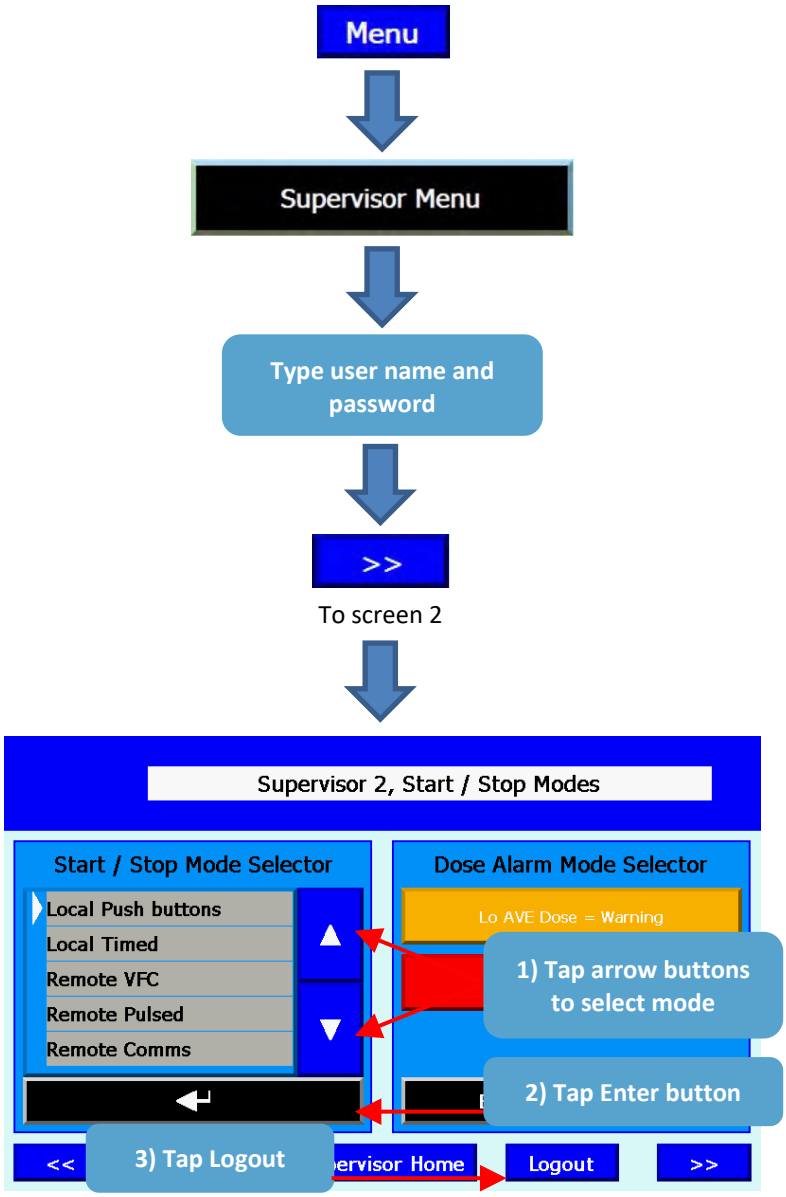


Figure 7: Procedure for Selecting Start/Stop Mode



Refer to section 4.3 for further details of the start/stop modes.

3.6. Connecting and Configuring Analogue Inputs

A number of analogue (4–20 mA) inputs are available to allow a user to send process stream parameter values to the UV system, which can then be used to adjust the lamp output and control the delivered UV dose to the correct value. Inputs for the flow and UV transmittance are available (see Table 1 above). These can either be fixed values entered into the software, or live data supplied from an external sensor.

External sensors can either be passive or active, depending on whether or not they provide their own 24 V dc voltage supply and ground connections. This affects how the sensor should be connected to the system's control panel. Ideally, the external sensors will have accompanying literature to show whether they have passive or active outputs and over what

range they are able to measure. However, in cases where this information is not available, the following procedure may help with connecting the sensors to the UV control panel:

1. Connect a loop calibrator or process meter across the sensor outputs and determine if the voltage supply is internal (active) or external (passive).

2. If the maximum value measured by the meter is known, enter it as the 20 mA setting in the software, and enter zero as the 4 mA setting (Figure 8). Go to step 4. If the maximum value is not known, go to step 3.

   Note: It is good practice to ensure that the maximum value measured by the sensor is above the normal operating value of the system.

3. If the maximum value is not known, use an ammeter to measure the output current from the sensor in mA (*A*) for a known measured value (*B*), and calculate the maximum value (*C*) using the following equation:

$$C = \frac{B}{(A-4)} \times 16$$

   Enter the value of *C* as the 20 mA setting in the software and zero as the 4 mA setting (Figure 8).

4. Use the loop calibrator to simulate the sensor by connecting to the appropriate inputs of the control panel and varying the current output setting between 4 and 20 mA to check that the control panel displays the corresponding values correctly.

   The connections in the UV system cabinet should be configured as shown in the diagrams, depending on whether the sensor in question is passive or active.

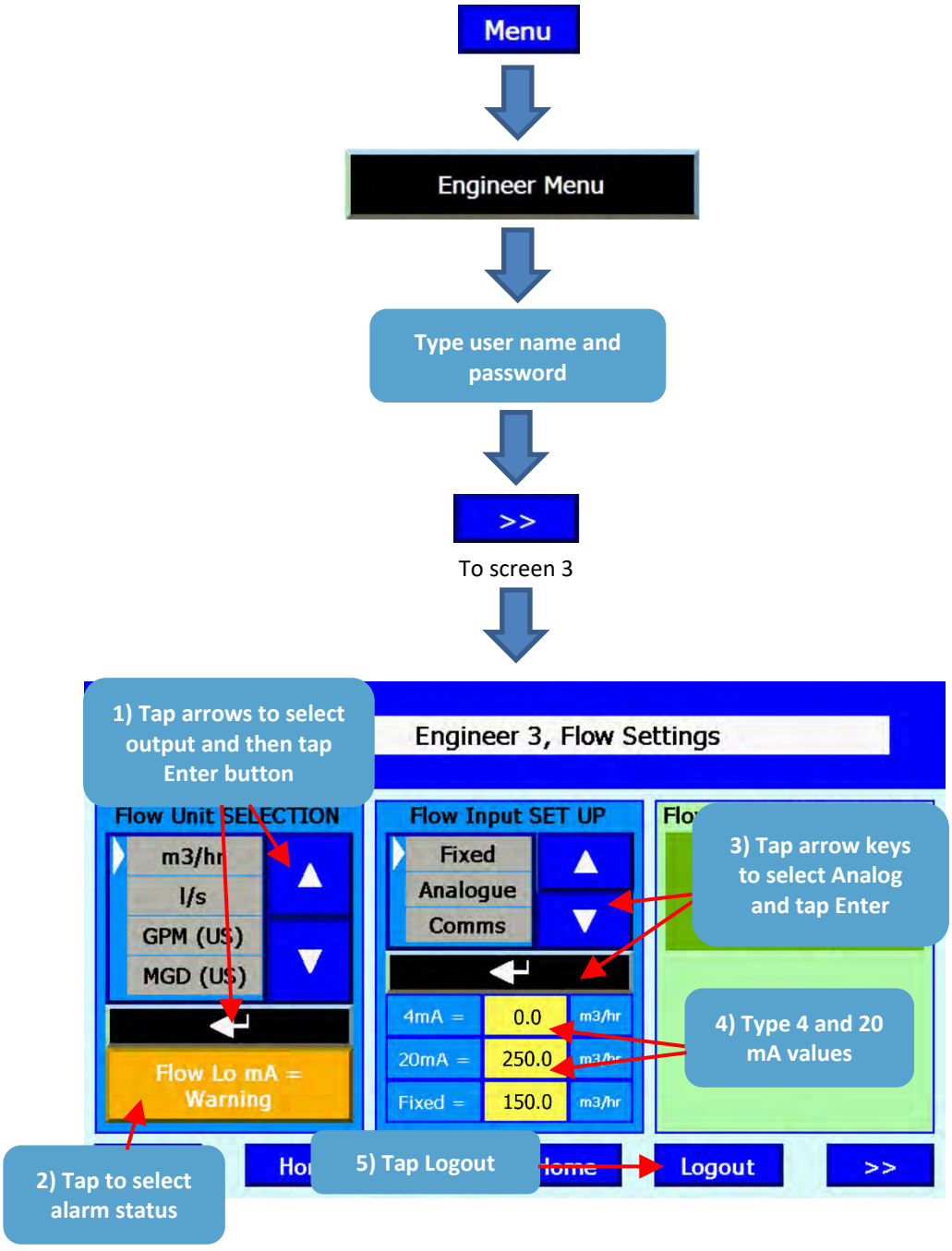
5. Connect the actual sensor and check that the control panel displays the expected values.

The outputs from the analogue sensors are measured by the control system using a 4-20 mA scale which needs to be configured in the software. To configure a flow meter input, follow the procedure in Figure 8 after referring to the warning in section 2.1.

For example, assume the flow meter can measure between 0 and 40 m^3/h. In this case, the **20 mA value** menu setting should be set to 40 m^3/h and the **4 mA value** should be set to 0.

Note: Set the Input to **Comms** to send the fixed input value using an Ethernet connection (section 7.2).

Figure 8: Procedure for Configuring Analogue Sensor Inputs



Follow a similar procedure when required to configure the transmittance meter using screen 4 of the Engineer menu. Similarly, the temperature and UV intensity sensor inputs, which are already configured on your system, can be adjusted using screens 1 and 2 respectively.

- **Ensure that the units for flow and temperature are selected <u>before</u> setting the 4 and 20 mA limits.**
- **If you change the temperature units, adjust the values for the warning and trip temperatures in screen 10 of the Engineer menu, and check the**

	bleed temperature if a bleed valve is installed (section 3.11). • **<u>Do not</u> change the temperature units when the lamp is running as the system may trip.**

3.7. Connecting and Configuring Analogue Outputs

Four analogue outputs are available for reporting data to an external data monitoring system if required (see Table 1). These outputs are all active (they contain their own 24 V dc voltage supply) and so they must be connected to passive terminals in the external monitoring system as shown in the system site wiring diagrams.

Values for the UV doses (RED and Average) and intensity can be configured in screen 5 of the Engineer menu by selecting the appropriate 4 and 20 mA setpoints (Figure 8). Use the right side of the screen to check that the output signal is converted to the correct value.

The fourth selectable output can be configured in screen 6; follow the procedure in Figure 10.

	**Configure the analogue outputs while the system is operating with the lamp switched on as the conversion can easily be checked.**

Figure 9: Procedure for Configuring Default Analogue Outputs

Follow the procedure described in Figure 8 to enter the Engineer menu and scroll to screen 5.

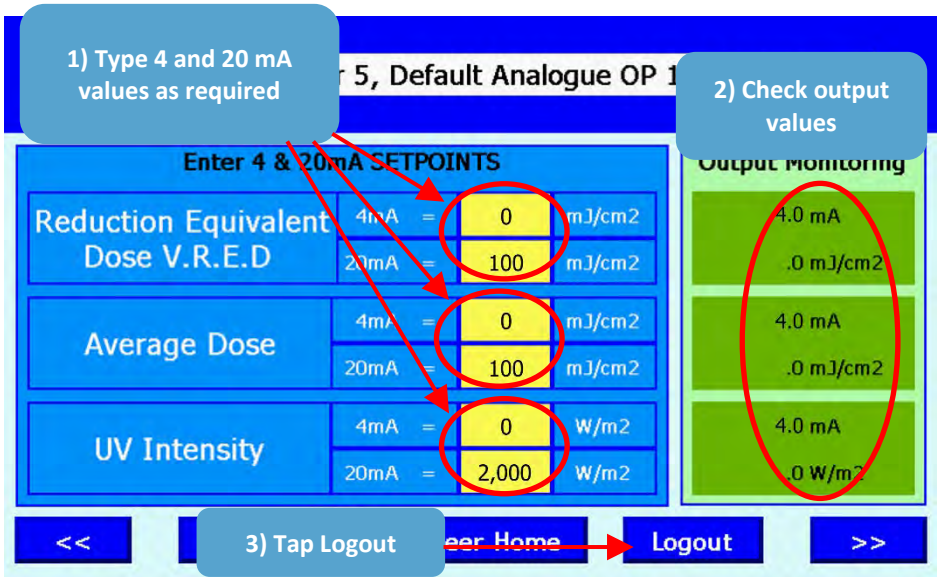
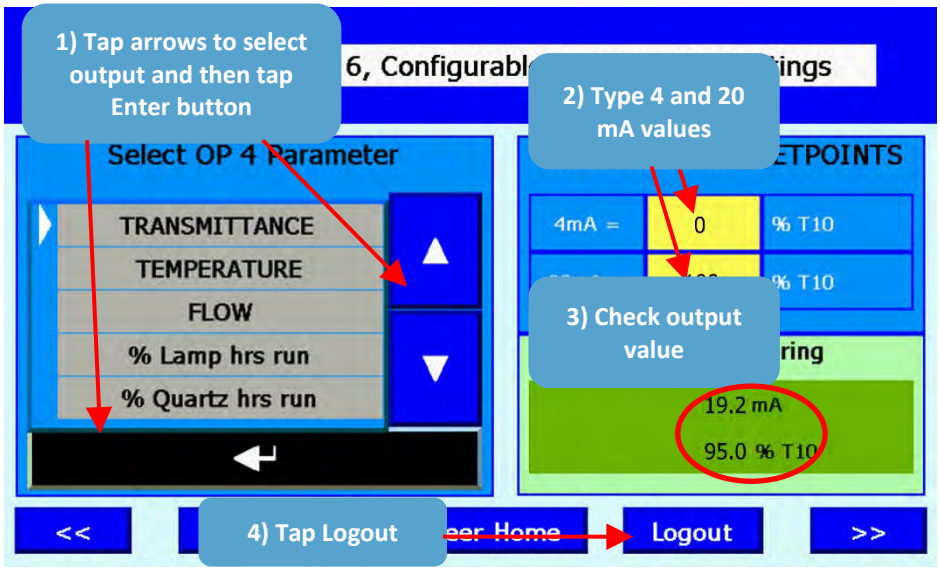


Figure 10: Procedure for Configuring Optional Analogue Sensor Output

Follow the procedure described in Figure 8 to enter the Engineer menu and scroll to screen 6.



3.8. Configuring Digital Outputs

Three digital output terminals are available in the control cabinet (X3:46-48). Each of these can be configured to monitor specific warnings or trips by following the procedure in Figure 11. Refer to Table 2 for the list of selectable options.

Figure 11: Procedure for Configuring Digital Outputs

Follow the procedure described in Figure 8 to enter the Engineer menu and scroll to screen 7.

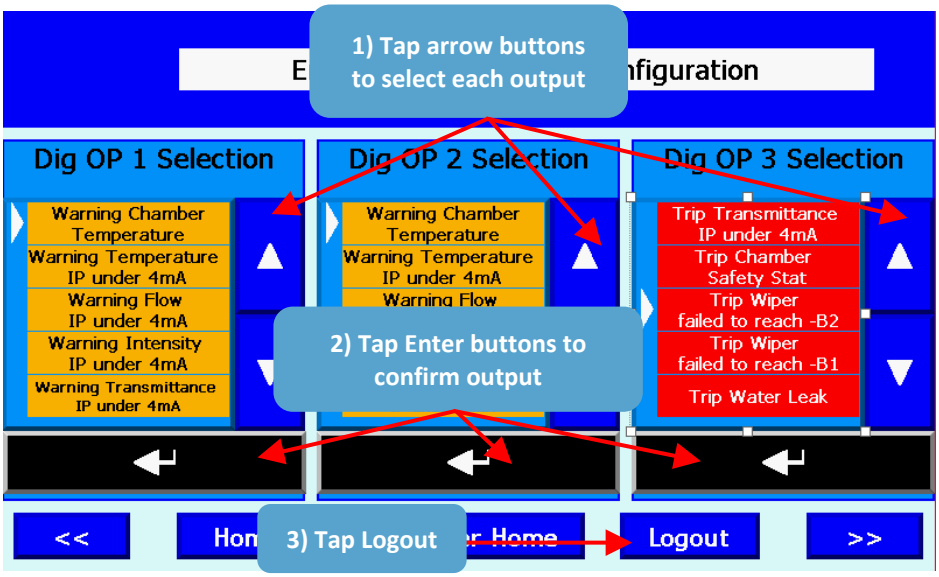


Table 2: Configurable Output Alarm Options

Configurable Output Alarm Setting	Notes
Warning: Chamber temperature	Chamber temperature exceeds warning limit
Warning: Temperature IP under 4 mA	Loss of signal from temperature sensor/flow meter/UV intensity sensor/transmittance meter (alarm set as Warning)
Warning: Flow IP under 4 mA	
Warning: Intensity IP under 4 mA	
Warning: Transmittance IP under 4 mA	
Warning: Wiper failed to leave -B1	Wiper yoke has not moved or arrived at limit switch as expected
Warning: Wiper failed to reach -B2	
Warning: Wiper failed to leave -B2	
Warning: Wiper failed to reach -B1	
Warning: Low DOSE ave	
Warning: Low DOSE red	
Warning: Flow above upper design limit	
Warning: Transmittance below design limit	
Trip: Chamber temperature	Chamber temperature exceeds trip limit
Trip: Temperature IP under 4 mA	Loss of signal from temperature sensor/flow meter/UV intensity sensor/transmittance meter (alarm set as Trip)
Trip: Flow IP under 4 mA	
Trip: Intensity IP under 4 mA	
Trip: Transmittance IP under 4 mA	
Trip: Chamber Safety Stat	Safety thermostat activated
Trip: Wiper failed to reach -B2	Wiper yoke has not arrived at limit switch as expected
Trip: Wiper failed to reach -B1	
Trip: Water leak	If optional leak detector fitted
Trip: Lamp cover removed –B5	If optional UVShield™ fitted
Trip: Low DOSE ave	
Trip: Low DOSE red	
Trip: Ballast earth leakage	
Warning: Cabinet getting warm	Cabinet internal temperature exceeds 50ºC / 122ºF
Trip: Cabinet too hot	Cabinet internal temperature exceeds 60ºC / 140ºF

Refer to Table 3 in section 6.2 for further details on these alarms.

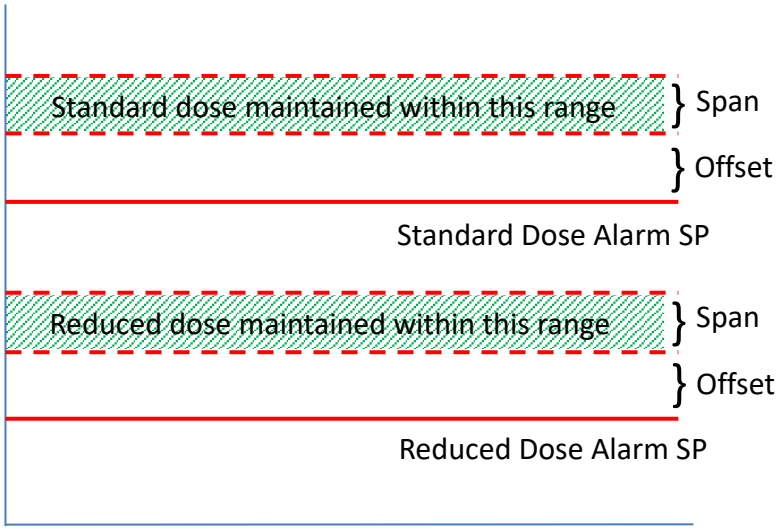
3.9. Configuring the Dose Operating Mode

The dose operating mode allows the system to vary the lamp power to control the measured UV dose. The Home screen displays the dose being used to control the system.

Two dose modes are available, one using the average dose and the other using a calculated Reduction Equivalent Dose (RED). The system can also monitor both dose parameters simultaneously and adjust the lamp output to ensure that both remain above their specified minimum levels. In this case, both doses are displayed in the Home screen.

Each dose mode is controlled by a combination of selected values. The *Standard Dose Alarm SP* (set point) is the minimum permitted value of the dose before an alarm is triggered. The alarm level plus the *Offset* gives the lower control limit for the dose, and the addition of the *Span* gives the upper control limit, as shown in Figure 12. Similar set point controls are used if the system is delivering a reduced dose (section 4.6).

Figure 12: Control of the Dose Using Set Points



For example, to control the system to deliver an average dose of between 32 and 35 mJ/cm^2 and trigger an alarm if the dose falls below 25 mJ/cm^2, enter the following values:

Standard Ave Dose Alarm set point = 25
Ave Offset = 7
Ave Span = 3

The average dose mode can be selected and configured in screen 11 of the Engineer menu. Follow the procedure in Figure 13. Before changing these settings, refer to the warning in section 2.1.

Note: If you are using average dose mode, you must set the initial intensity (S0) value as described in section 3.10.

The RED dose mode can be selected and configured in a similar way in screen 12, although you must make sure you select the target pathogen and the required log reduction as these allow the software to calculate the alarm set point. Follow the procedure in Figure 14. Before changing these settings, refer to the warning in section 2.1.

3.9.1. Dose control in validated systems

This independently-validated UV system can only deliver an RED dose within a defined range of liquid flow rate and UV transmittance (UVT) values. These values were defined during the validation testing. This affects the UV dose calculation as follows:

- The system is considered as being out of specification if the flow rate exceeds the maximum allowed value or the UVT is below the minimum allowed value, and an alarm is triggered.
- When operating in low flow or high UVT conditions, the dose calculation is constrained to use the minimum allowed flow rate and maximum allowed UVT from the validation testing.
- There is also a maximum value set for the log reduction that the system can achieve for a particular pathogen.
- For these reasons, there may be some instances when a change in the operating conditions causes no change in the RED dose or log reduction displayed in the software. In these cases, the system is still performing as intended and no action is needed.

➢ **If an alarm is triggered**, adjust the flow rate and/or UVT to return them to the normal range of values for your system to maintain the required level of treatment.

Figure 13: Procedure for Configuring Average Dose Mode

Follow the procedure in Figure 8 to enter the Engineer menu and scroll to screen 11.

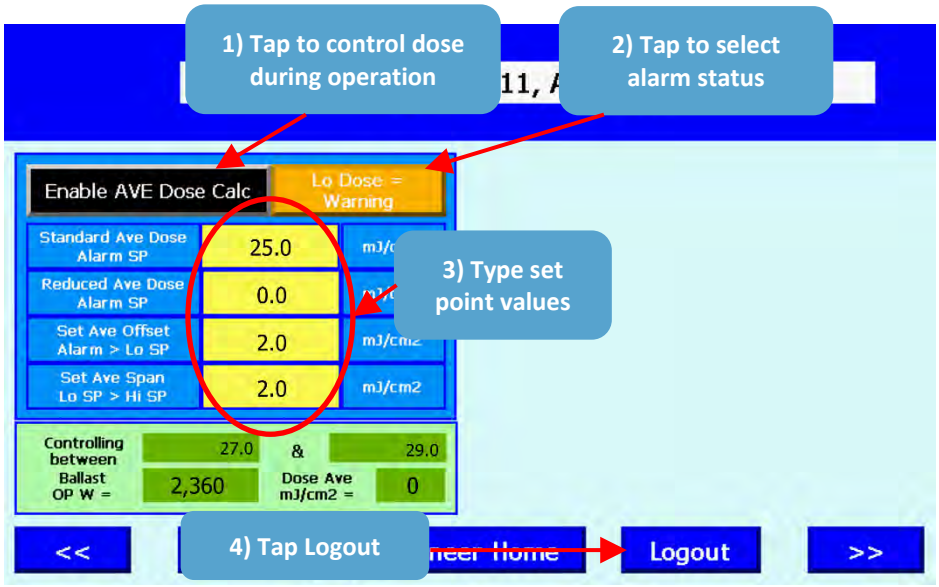
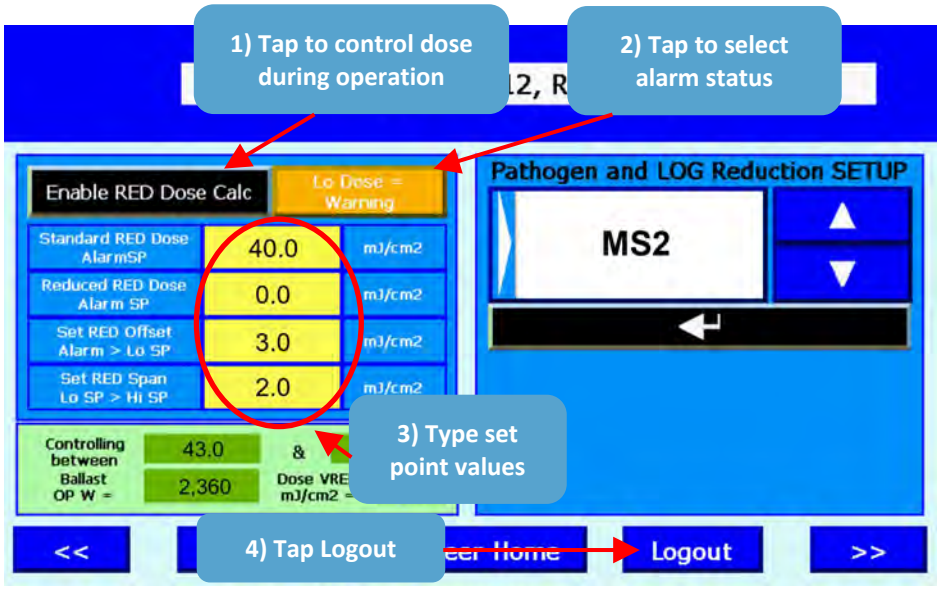


Figure 14: Procedure for Configuring RED Dose Mode

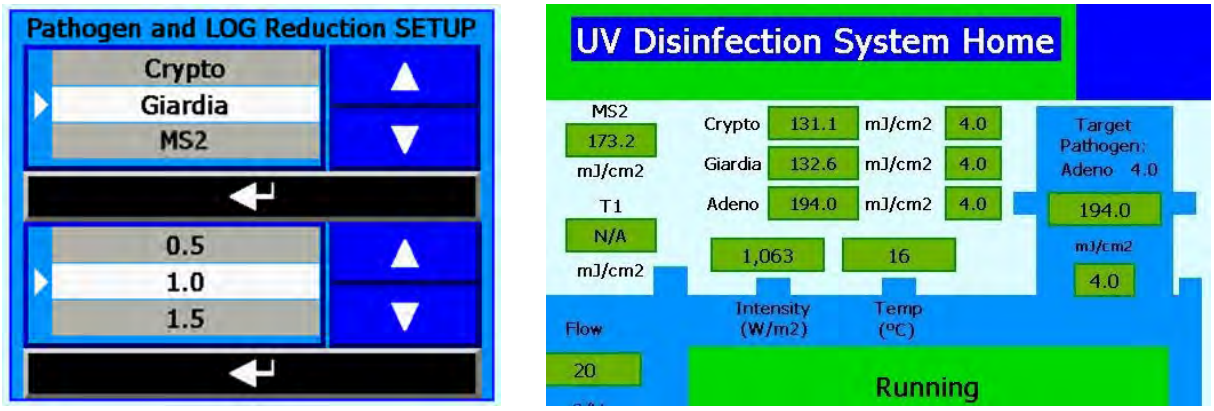
Follow the procedure in Figure 8 to enter the Engineer menu and scroll to screen 12.



In some systems, you may have the option of selecting the target pathogen and the required log reduction in Screen 12, which sets the alarm level for the RED (Figure 15, left). In these cases, the Home screen displays the calculated doses and (where applicable) the log reductions for a range of pathogens, although the UV dose is only controlled to target the pathogen shown on the right of the screen (Figure 15, right). Under certain operating conditions, it may not be possible to calculate the dose for every pathogen, in which case 'N/A' is shown in the Home screen.

Note: In systems where the target pathogen and log reduction can be selected, **you do not have to set the initial intensity (S0) value**.

Figure 15: Alternative Settings and Display for RED Dose Mode



3.10. Setting the Initial Intensity Value

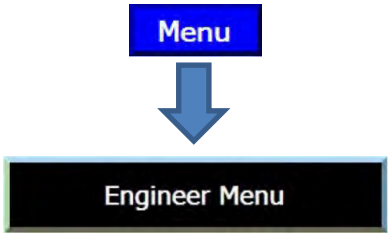
Note: This procedure is not required if your system is set up to deliver a known log reduction of a specific pathogen (see Figure 15).

Once the system is installed and running, the initial intensity value measured by the UV sensor (known as S0) must be entered into the software to allow the dose to be calculated correctly. The system may generate Low UV alarms until this is completed.

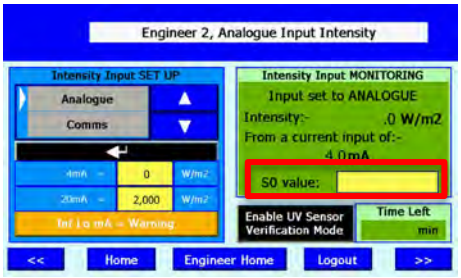
Changing the S0 setting will affect the average UV dose delivered by the system and so it is important that this procedure is completed during installation and commissioning.

The procedure involves entering the UV intensity that is displayed when the lamp is operating at its maximum running power (in the Flow Enable state).

Follow the steps below:

1.	Ensure that the chamber is full of water and switch on the lamp (section 4.2).	
2.	Wait until the system switches from the Warming state (yellow background) to the Flow Enable state (light green background). Note the value of the UV intensity (in W/m^2) displayed on the Home screen when the system has **0.5 minutes remaining** in the Flow Enable state.	
3.	Tap the keys shown here and enter the user name and password for the Engineer menu.	
4.	Press >> until screen 2 of the Engineer menu is displayed.	

5. Tap the yellow box marked **S0 value** and enter the intensity in W/m² that you noted in step 2. Tap the Enter key ↵ to save the setting.



6. Tap Logout to exit the Engineer menu.



3.11. Installing a Bleed Valve (option)

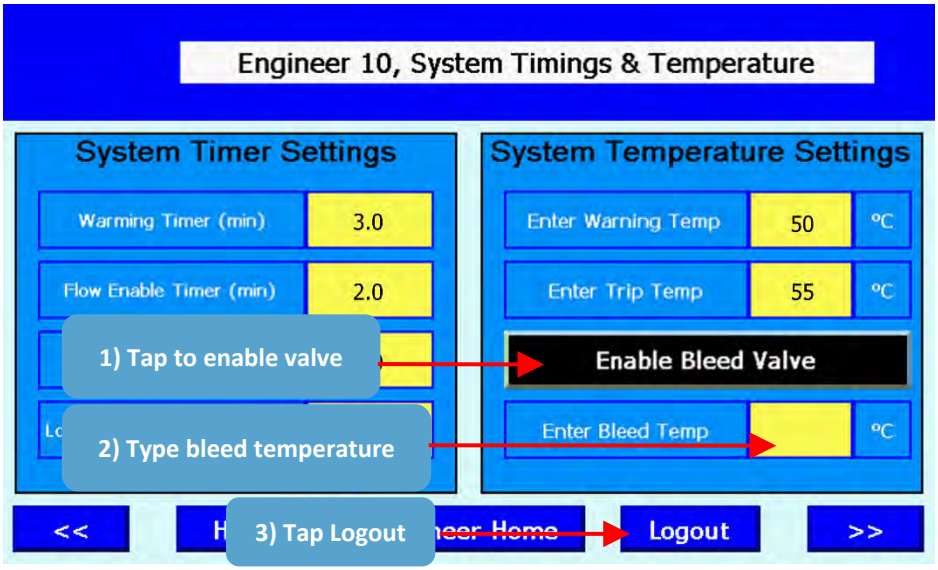
If, during operation, there are likely to be periods of reduced or no flow through the chamber, this may cause the chamber to overheat, at which point the lamp will be automatically switched off. This can be avoided by installing a bleed valve, which drains liquid from the chamber (usually from near the outlet) to lower the temperature. 24 VDC and 230 VAC bleed valve options are available.

Connect the cable supplied between the control cabinet and the bleed valve mounted on the chamber (refer to the system site wiring diagrams). This will open the valve when the bleed valve temperature set point is reached. Follow the procedure in Figure 16 to set the bleed valve temperature.

We recommend that the bleed temperature should be at least 10°C (18°F) below the alarm temperature of the system.

Figure 16: Procedure for Setting Bleed Temperature

Follow the steps described in Figure 8 to enter the Engineer menu and scroll to screen 10.



3.12. Adjusting the Water Leak Detector (option)

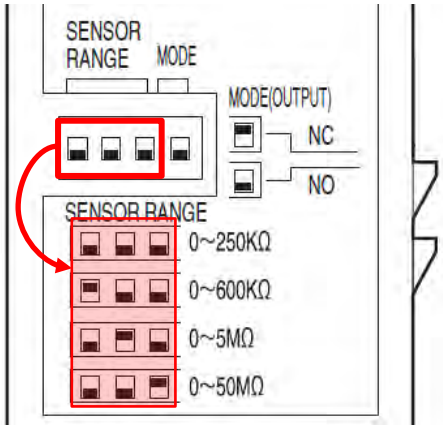
On systems where a UVShield™ unit is fitted and the integrated water leak detector is enabled, you can adjust the sensitivity of the relay in the control cabinet if necessary to suit your application.

1. Locate the DIP switches on the side of the water leak detector relay inside the cabinet (component numbering may vary from that shown).

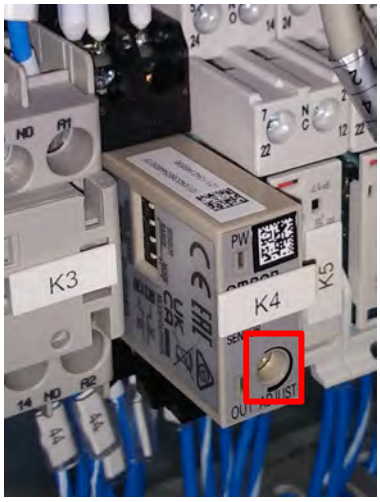


2. Use a small screwdriver to adjust the three switches from the left to change the range of resistance that can be detected.

   For applications using high purity water, we recommend using the 0–50 MΩ setting (Up Up Down) to ensure that a water leak will be detected. A lower setting may be sufficient for other applications and to avoid the system tripping unnecessarily.



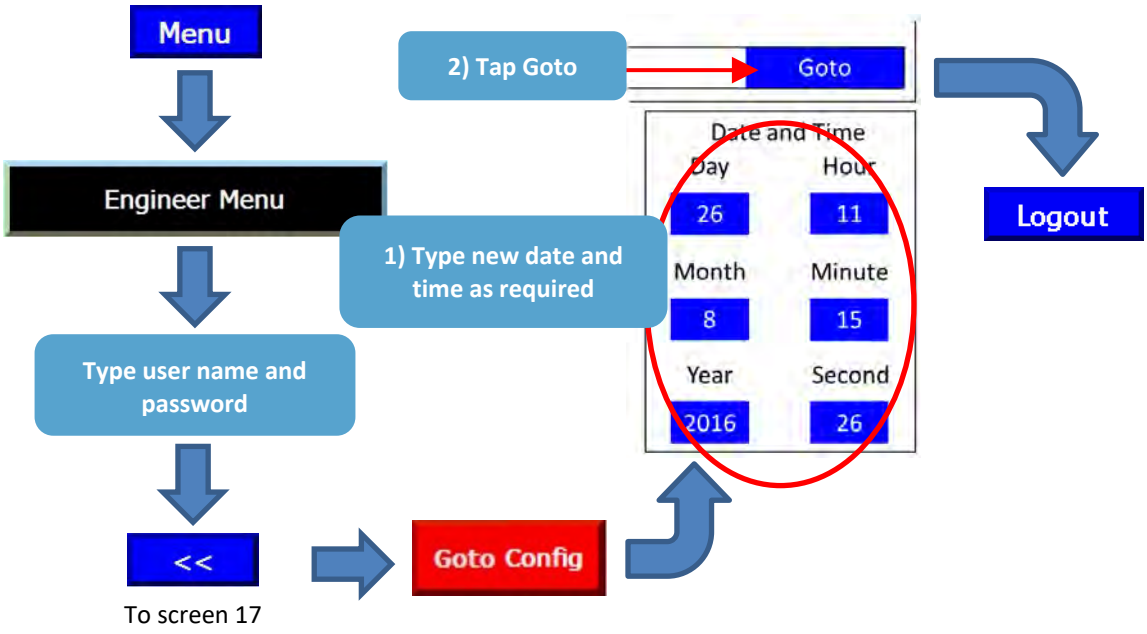
3. If necessary, further adjust the detected resistance value within the selected range using the ADJUST control on the top of the relay (turn clockwise to increase, anticlockwise to reduce).



3.13. Setting System Date and Time

If the date and time needs adjusting, follow the procedure in Figure 17.

Figure 17: Procedure for Setting Date and Time



3.14. Commissioning Procedure

Commissioning of the system should be carried out by a suitable qualified and trained engineer in accordance with the recommendations of this manual, with particular attention to safety.

See the Installation and Commissioning Guide for the commissioning checklist.

In order to check the system parameters and settings, refer to the Acceptance Test Record supplied with your system.

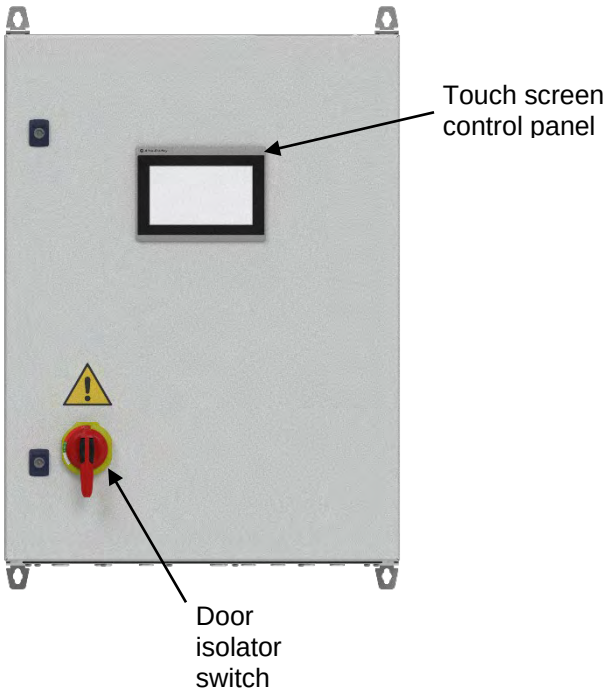
4. System Operation

This section provides detailed information on operation of the system.

	DO NOT USE THE EQUIPMENT IN A MANNER OTHER THAN THAT SPECIFIED IN THIS MANUAL, OTHERWISE THE PROTECTION PROVIDED BY THE EQUIPMENT MAY BE IMPAIRED. **N'UTILISEZ PAS L'ÉQUIPEMENT D'UNE MANIÈRE AUTRE QUE CELLE SPÉCIFIÉE DANS CE MANUEL, AUTREMENT LA PROTECTION FOURNIE PAR L'ÉQUIPEMENT PEUT ÊTRE ALTÉRÉE.**

	Before attempting to start the system, ensure that the chamber is full of water. **Avant de tenter de démarrer le système, assurez-vous que la chambre est pleine d'eau.**

Figure 18: Cabinet Controls



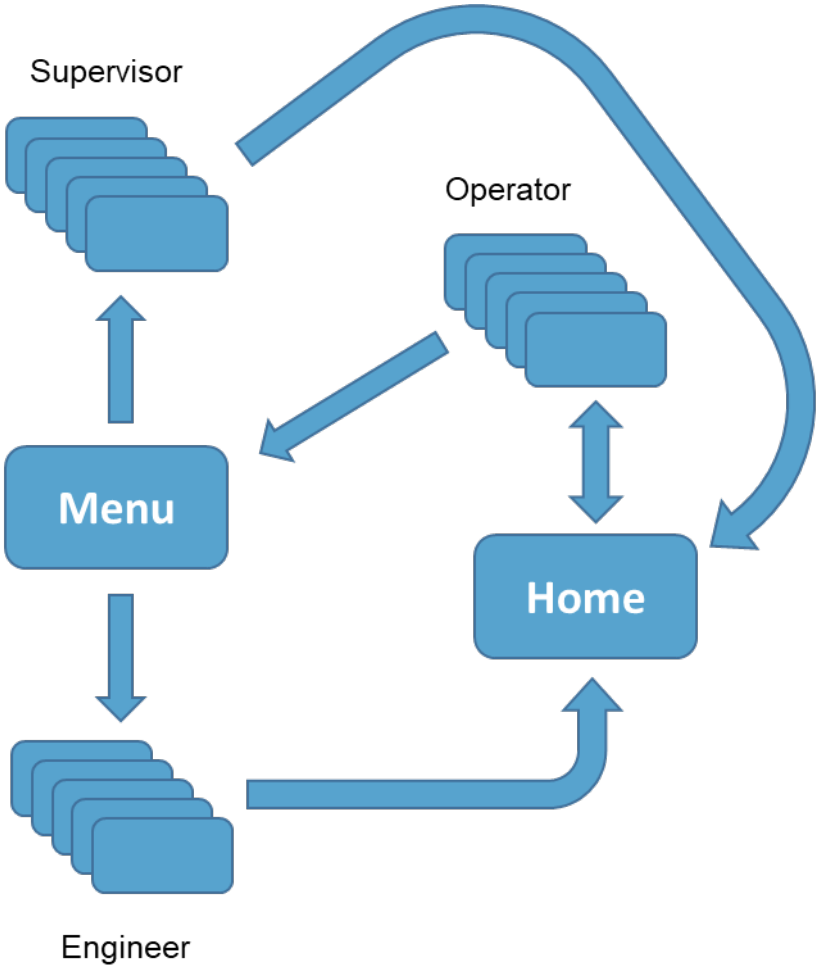
4.1. Software Overview

The control panel contains four menus:

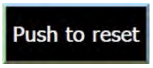
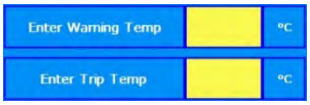
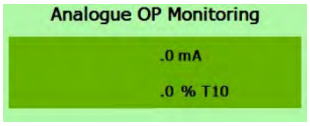
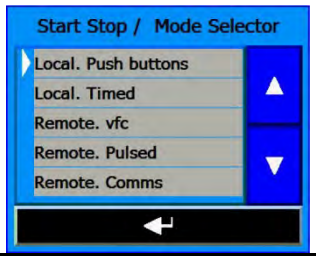
Operator menu	This menu is available to all users and provides operational data, both actual and historical (section 4.7). The menu allows the user to view additional information and to reset the system following a TRIP condition (provided the condition has been cleared).
Supervisor menu	This menu allows access to a number of system parameters and sensor settings (section 4.8), in addition to all Operator menu screens. It is password-protected. Refer to the warning in section 2.1.
Engineer menu	This menu allows access to settings for alarms, sensors and data outputs (section 4.9), in addition to all Supervisor menu screens. It is password-protected. Refer to the warning in section 2.1.
Factory menu	This menu is for use by the system supplier only.

You can navigate between the menus using the Menu screen, as shown in Figure 19. You can also return to the Home screen from any point in the software.

Figure 19: UV Touch software structure



The screens contain common features as follows:

	Tap these buttons to scroll between the screens of the menus.
	Tap black buttons to enter commands or choices.
	Tap yellow areas to type values using the on-screen keyboard, and then tap the Enter key  to save the setting.
	Green areas are used for displaying live data from the system.
	To select options: 1. Use the Up and Down arrow buttons to move the white marker to the required option. 2. Tap the black Enter button to select the option. The selected option is then highlighted in white.
	Tap to change alarm status from Warning (amber) to Trip (red).
	Tap to go to the menu screen with access to the Supervisor and Engineer menus.
	Tap to display useful information on the screen displayed.
	Tap to return to the home screen of the Operator menu. Similar buttons in the Supervisor and Engineer menus display the first screens of these menus.
	Tap to log out from the current menu.

4.2. Switching On and Off

Switch on the system using the door isolator switch. Wait for the software to start; this may take a few minutes.

When the Home screen is displayed, the banner colour will indicate the system state (see below).

- If the system is Available, start the lamp using the signal appropriate to the start/stop mode selected (section 4.3).

Refer to Figure 42 in section 6.4 for further details of the sequence of system states. Where applicable, the time remaining before the state changes (for example, Warming, Flow Enable or Cooling) is shown on the screen in decimal minutes.

State	Colour	Comments
Available		The lamp can be started when the system is in this state.
Warming		The lamp is warming up to its full power. UV intensity is displayed but UV dose is not calculated.
Flow Enable		Start the flow of liquid through the chamber. UV dose is calculated and displayed if a measurable flow is present.
Running		
Cooling		The lamp cannot be started while the system is in this state.
Tripped		If there are any active trip alarm conditions, the display will show TRIPPED and it will not be possible to start the lamp. Refer to section 4.5 for more information on solving Trip conditions.
CIP Cycle		A CIP process should only be performed when the system is in this state. Refer to section 5.14 for further details.

When the lamp is switched off, the system can safely be left in the Available state or can be switched off at the door isolator.

4.3. Start/Stop Modes

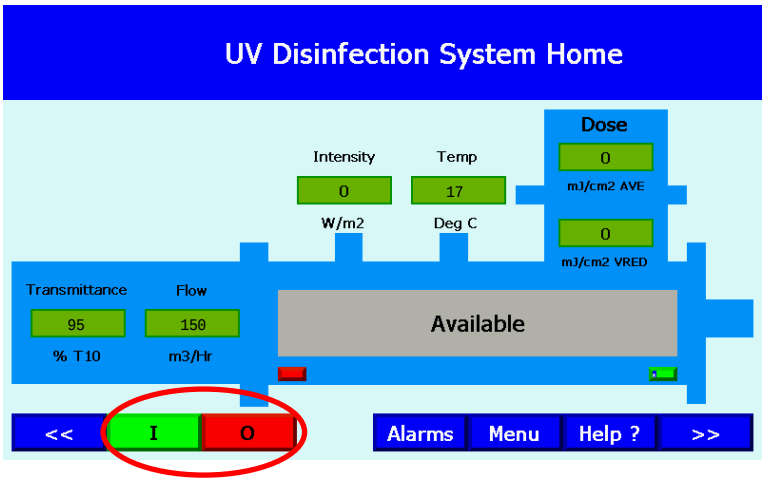
The system has five start/stop modes, which can be selected as shown in Figure 7. The mode can be changed without interrupting operation if the selected mode is set to switch the lamp on. For example, you can switch from Local mode (Push buttons) to Remote VFC while Running if the Remote VFC signal is Hi.

The start/stop modes are described in the following sections.

4.3.1. Local mode (Push buttons)

The lamp can be switched on and off and the system reset after a trip using the buttons on the Home screen (Figure 20). When in Local mode, the system will supply information but will not execute any command sent remotely (except for a Remote CIP Request; see section 5.14).

Figure 20: Start/Stop Buttons on Home Screen

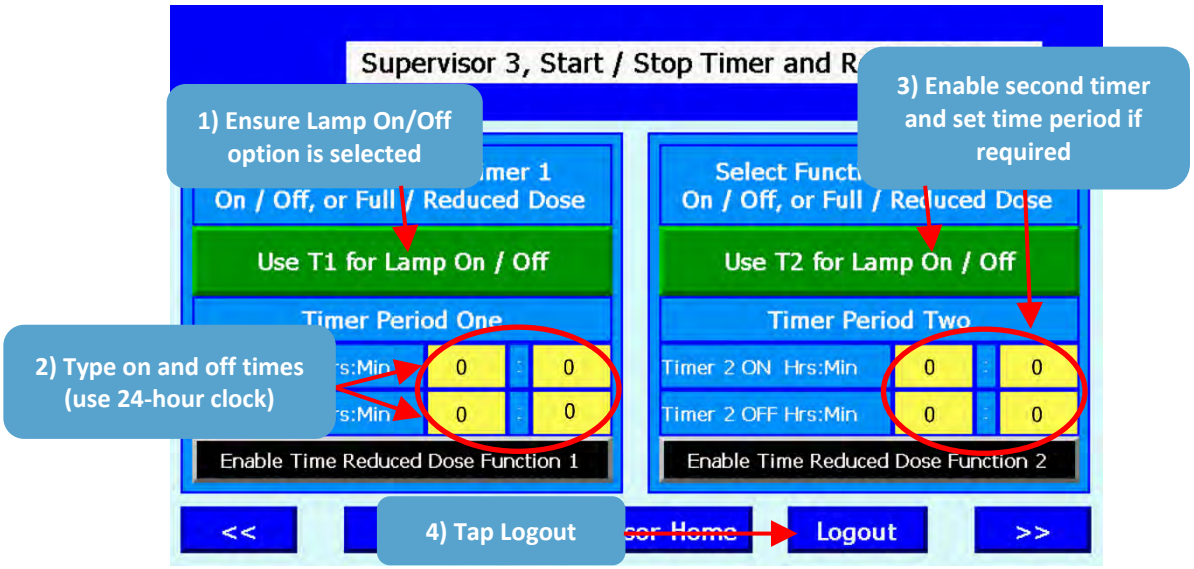


4.3.2. Local mode (Timed)

In this mode, you can set the local time for switching the lamp on and off.

Figure 21: Procedure for Setting the On/Off Timer

Follow the procedure in Figure 7 to enter the Supervisor menu and scroll to screen 3. You can use this screen to set two separate on/off periods.



4.3.3. Remote VFC mode

In Remote mode, the system is switched on by closing a contact across inputs X3:14 and 15. Opening the contact will switch the lamp off.

4.3.4. Remote Pulsed mode

In this mode, the lamp can be switched on by applying a 24 V dc pulse (minimum 1 second duration) to input X3:22. Applying a similar pulse to X3:23 switches the lamp off.

4.3.5. Remote Comms mode

In this mode, the lamp is switched on and off by sending commands to the control panel via the Ethernet communications interface (section 7).

4.4. Wiper Operation (option)

An automatic wiper is used to remove deposits from the quartz thimble. A wiper ring travels along the thimble and in the process removes any deposits from the thimble surface. The ring is driven along a leadscrew that runs alongside the thimble, operated by a motor. Limit switches detect the end of each wipe and stop the motor.

If the wiper yoke is not parked at its home position (limit switch 1) when power is switched on to the system, it will drive the motor to complete the wipe cycle and park the yoke in this location. The LED on this limit switch will go out, and it will be shown as green in the Home screen.

	**Fill the chamber with water before switching power on to the system. Before switching on the lamp, ensure that the wiper ring is positioned at one of the limit switches to prevent unnecessary heating of the ring (perform a wiping cycle).** **Remplissez la chambre avec de l'eau avant de mettre le système sous tension. Avant d'allumer la lampe, assurez-vous que l'anneau du nettoyeur est placé à l'un des commutateurs de limite pour empêcher le chauffage inutile de l'anneau (effectuer un cycle de nettoyage).**

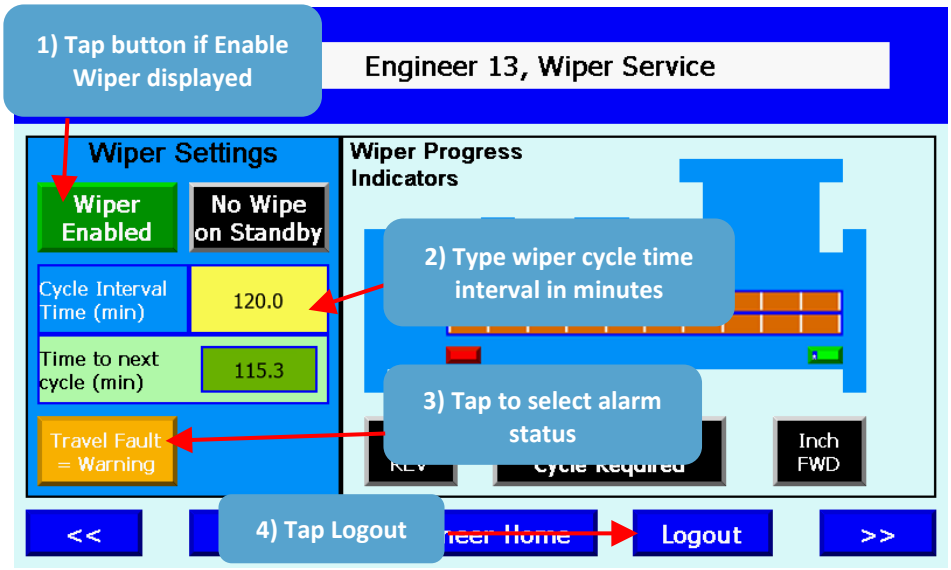
A wipe cycle is initiated according to the system's wiper interval settings or when a low UV alarm is first detected (when two cycles will automatically be performed). Refer to section 6 for possible causes if the low UV alarm remains active.

The time interval between wipe cycles is factory-set. If you want to change this setting, follow the procedure in Figure 22, after referring to the warning in section 2.1.

	**The wiper time interval is factory-set at 4 hours. If required you can set the interval according to the process conditions so that deposits on the thimble are removed regularly (refer to section 4.4). Deposits can bake on to the thimble quite quickly when a system is running and may then be difficult for the wiper to remove.** **L'intervalle de temps du nettoyeur est préréglée en usine à 4 heures. Si nécessaire vous pouvez régler l'intervalle selon les conditions du traitement afin que les dépôts sur la virole en quartz soient éliminés régulièrement (consultez la section 4.4). Les dépôts peuvent cuire sur la gaine assez rapidement lorsqu'un système est en cours d'exécution et sont alors difficiles pour supprimer avec le nettoyeur.**

Figure 22: Procedure for Changing Auto-wiper Time Interval

Follow the procedure in Figure 8 to enter the Engineer menu and scroll to screen 13.



Other features of the Wiper Service screen are:

- Tap **Enable Wiper** to enable the wiper function on the system if it is disabled.
- Tap **No Wipe on Standby** to maintain wiping when the UV lamps are off (this prevents deposits from building up on the quartz thimble if the chamber remains filled with water).
- Tap **Reset** (when available) to reset the system after a trip (once the trip condition has been rectified).
- Tap **Manual Wipe Cycle Required** to perform an immediate wipe cycle; this does not affect the timing of the automatic wipe cycles.
- Adjust the position of the wiper yoke forwards and backwards manually using the **Inch** buttons. **Note:** the Inch buttons will only function when the wiper is **not** enabled.

During a wipe cycle, the red grid on both this screen and screen 3 of the Operator menu gradually turns green as the wiper yoke moves along the quartz thimble. The limit switches are shown in green when the yoke is at that location, otherwise they are shown in red. A message also appears showing whether the yoke is moving forwards (**FWD ON**, away from the wiper motor) or reversing (**REV ON**).

Refer to section 5.7 for details of wiper servicing.

4.5. Warnings and Trips

When the system encounters unexpected conditions, an alarm is raised. This can either be a Warning or a Trip. In the case of a Warning, it is possible to continue the operation of the system. In the case of a Trip, the lamp is switched off and cannot be restarted until the alarm condition is cleared.

When a warning or a trip has occurred, a red banner is displayed at the top of the Home screen which describes the nature of the alarm. While this message is displayed, the other screen controls are disabled.



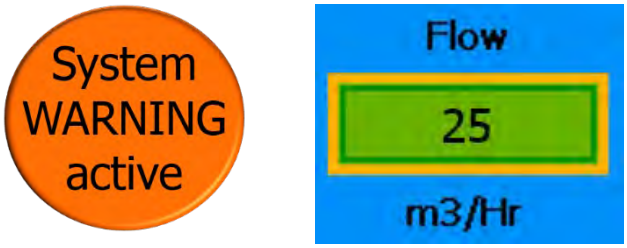
- Tap the **Ack** button to acknowledge the alarm (this action is recorded in the Alarms screen; see section 4.7).
- Tap the **Clear** button to remove the specific alarm notification without acknowledging the alarm.
- Tap the **Close** button to clear all alarm notifications from the screen without acknowledging any alarm.

Refer to section 6 for detailed information on the alarms and how to resolve them.

If a trip remains active after the message has been cleared, all screens will display a red banner and the Home screen will show that the system is in the Tripped state. Once the cause of the trip has been rectified, the system can be reset by either tapping the **Push to reset** button in Local mode or sending a signal to the Remote Reset input in Remote mode. The lamp can then be restarted; it may restart automatically if the system is in Remote VFC mode or Local mode (Timed). See section 4.3 for further details.

If a warning remains active after the message has been cleared, then the orange symbol in Figure 23 is displayed in the top left corner of the screen until the cause of the warning is removed. The parameter relating to the warning is also outlined in orange on the Home screen.

Figure 23: Warning Indicators (example of Flow warning shown)



Alarms are recorded in screen 5 of the operator menu (section 4.7).

4.6. Reduced Dose

If, during operation, there are likely to be periods of reduced or zero flow through the chamber, this may cause the chamber to overheat, at which point the lamp will be automatically switched off. To mitigate this problem, the system can be operated in Reduced Dose mode, which lowers the power to the lamp.

When running at reduced dose, the system will control the lamp power to keep the dose above the *Reduced Ave/RED Dose Alarm SP* value set in the Engineer menu (see section 3.9). The actual dose delivered may be higher than the set point if the lamp power cannot be reduced any further without causing the lamp to switch off.

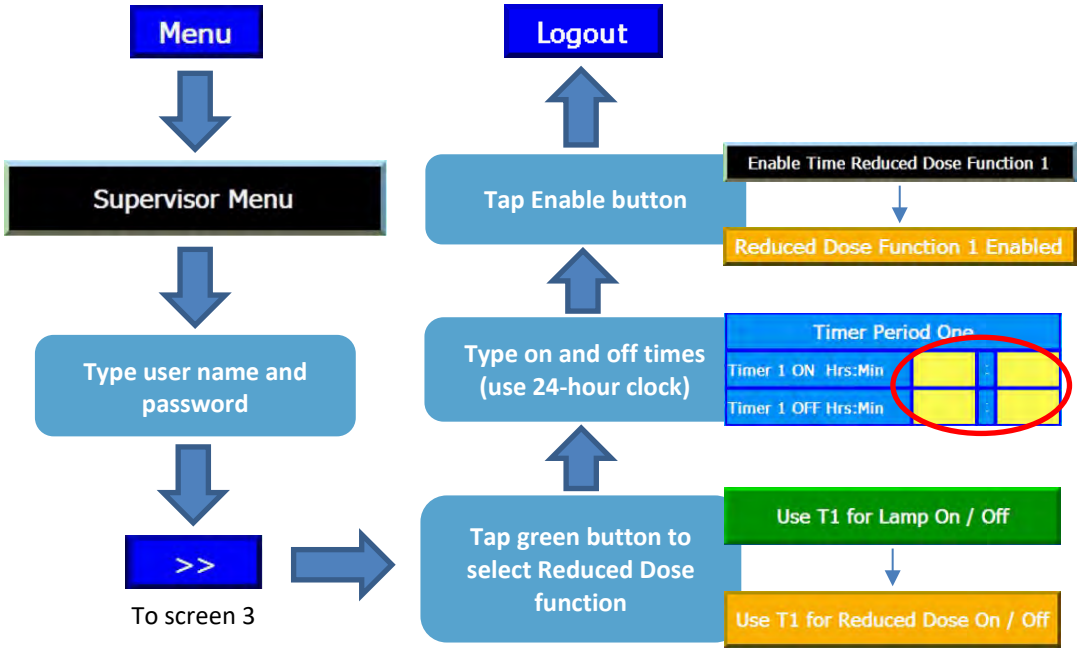
If operating in Reduced Dose mode fails to prevent the chamber from overheating, then a bleed valve should be installed, which drains liquid from the chamber (usually from near the outlet) to lower the temperature. Refer to section 3.9 for further details.

The Reduced Dose setting can be selected in several ways, depending on the system's start/stop mode, as shown in the following sections.

4.6.1. Local mode (Push buttons)

Follow the procedure in Figure 24 to set local on/off times for operating in Reduced Dose mode using Timer Period One. A second period within the same 24 hours can be defined using Timer Period Two.

Figure 24: Procedure for Setting Reduced Dose by Timer



4.6.2. Local mode (Timed)

Follow the procedure in Figure 24 but **use Timer Period Two for reduced power**, as Timer Period One is needed to turn the lamp on and off (section 4.3.2).

4.6.3. Remote mode (VFC or Pulsed)

In Remote mode (VFC or Pulsed), there are two ways of selecting Reduced Dose:

- Follow the procedure in Figure 24 (note that the Lamp On/Off option has no effect in Remote mode).

  OR

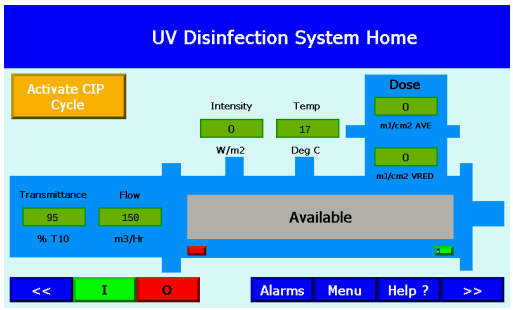
- Connect a signal to the Reduce Power input on the user's interface terminals (refer to Table 1 and the system drawings in Appendix C).

4.6.4. Remote Comms mode

Send a command via the site control system using the Ethernet connection (refer to section 7).

4.7. Operator Menu

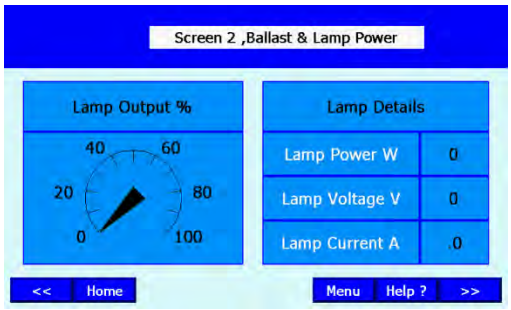
You can use the Operator menu to view the system's status, run the auto-wiper and clear trips.



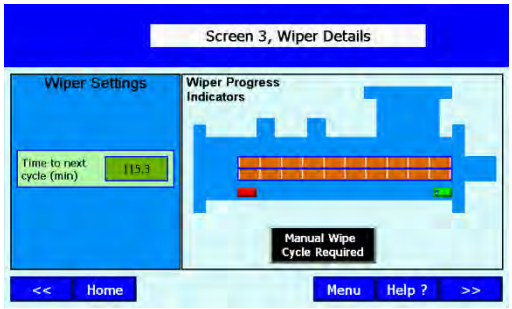
The Home screen displays the essential measurements for the system's operation, and the current state which is also indicated by the colour of the title banner (section 4.2).

If the CIP function is enabled on the system, tap **Activate CIP Cycle** to switch the system to the CIP Cycle state (section 5.14). (This will switch off the UV lamp.)

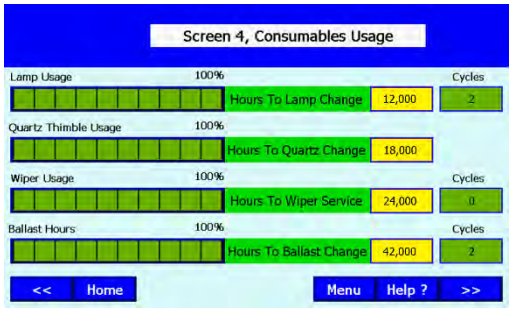
Tap **Push to reset** to reset the system after a trip has occurred and been rectified.



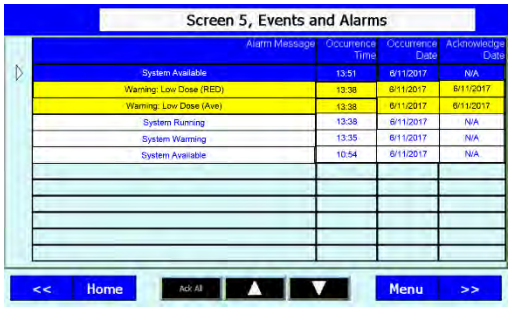
This screen displays the lamp's electrical measurements during operation, and the ballast output as a percentage of the maximum running power.



Use this screen, which is active only if the wiper is enabled, to run a manual wipe cycle, and to reset the system after a trip has occurred and been rectified. The screen also displays the status of the wiper and the time remaining until the next automatic wipe cycle.

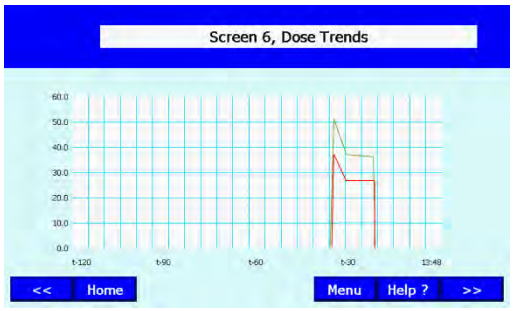
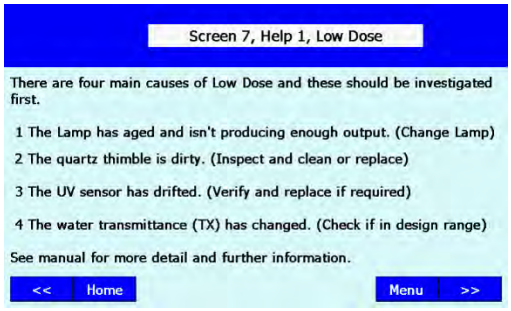
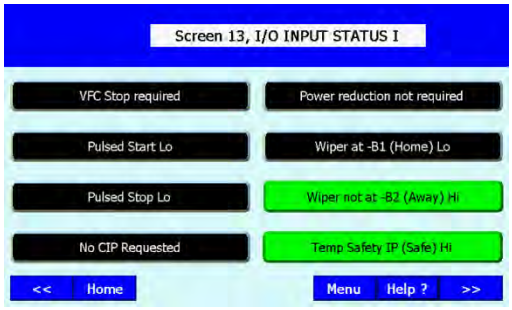
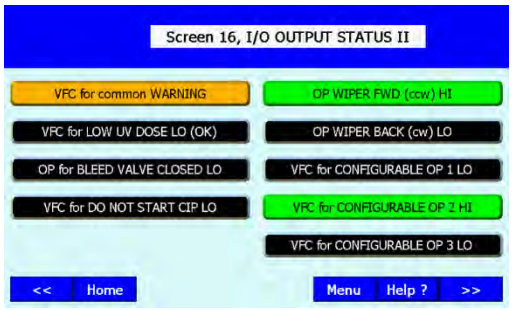


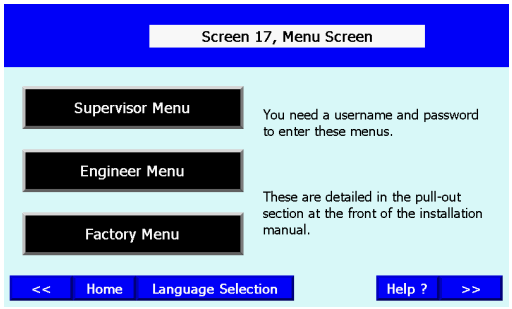
This screen displays the expired lifetime of the system's consumables and how much time remains until they require changing or servicing. The number of operating cycles completed for the lamp, ballast and wiper (if fitted) are also displayed.



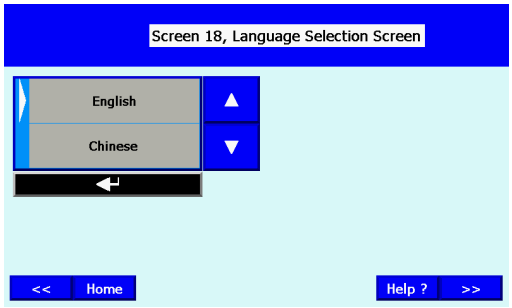
This screen lists the recent system events and alarms. Available, Warming and Running states are shown in white, with the current state of the system highlighted in blue. Active alarms are highlighted in red; once they are rectified they are shown in red letters. When they are acknowledged they are highlighted in yellow.

The screen can display up to 100 entries. Once it is full, the oldest entry will automatically be deleted

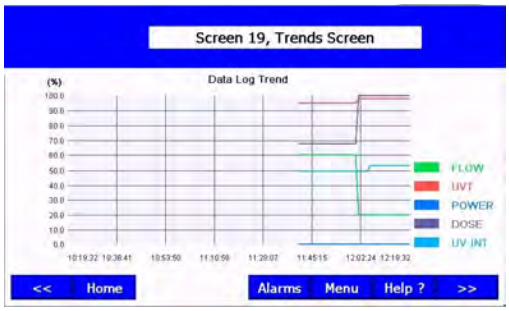
	when a new event is added.
	The trends in the doses for the previous three hours of continuous operation are plotted on this screen. RED is shown as a green line, and average dose is shown as a red line. Tap **All Trends** to view other important system parameters in screen 19 (see below).
	Screens 7–11 are a series of Help screens give guidance on low dose alarms, start/stop modes, flow and transmittance measurements, and CIP.
	This screen shows local support contact details. These can be entered and amended using screen 16 of the Engineer menu (section 4.9).
	Screens 13 and 14 display the states of the digital inputs, with a green background for HIGH (closed) and black for LOW (open). Inputs with an amber background indicate an active warning (HIGH), and inputs with a red background indicate an active trip (LOW).
	Screens 15 and 16 display the states of the digital outputs, with a green background for HIGH (closed) and black for LOW (open). Outputs with an amber background indicate an active warning (HIGH), and outputs with a red background indicate an active trip (LOW).



This screen, which can also be displayed by tapping the **Menu** button, allows access to the settings menus using the appropriate username and password. Tap the **Language Selection** button to go to screen 18 and change the display language.



Use this screen to change the language used on all screens. Tap the up and down arrows to select the language and then tap ↵.



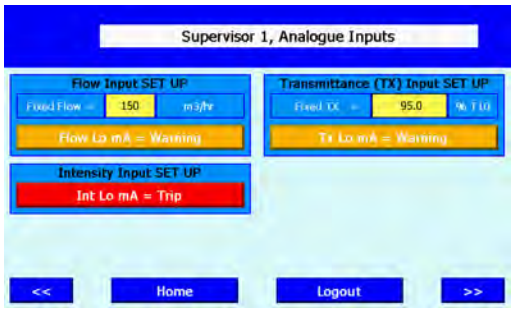
Screen 19 displays a plot of the analogue outputs from the control cabinet during the past two hours of operation. Each 4-20 mA output is displayed as a percentage (4 mA = 0%, 20 mA = 100%).

The scaling of the outputs can be adjusted in the Engineer menu.

If data logging alarms are enabled (section 4.10), then the status of the SD card is also displayed here.

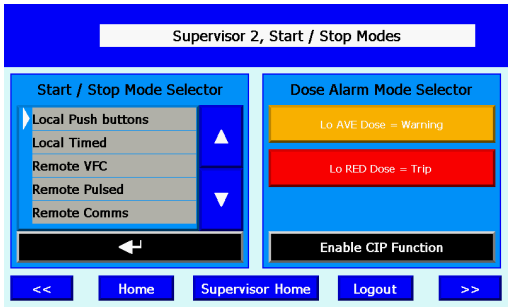
4.8. Supervisor Menu

This menu allows the user to select start/stop modes and times, fixed values for analogue inputs, and certain settings for the chamber temperature, UV intensity sensor and the auto-wiper (if fitted).



Enter fixed values for the flow and transmittance of the process liquid. Tap the buttons to select whether the alarm levels have warning or trip status.

Use the Engineer menu (section 4.9) to configure these inputs on a 4-20 mA scale.

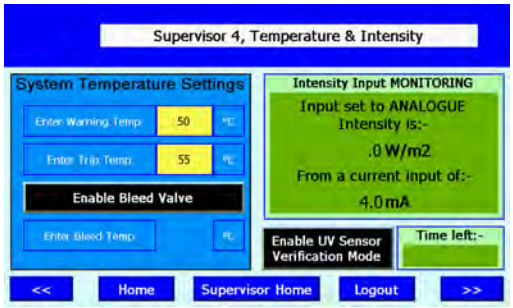


Select the start/stop mode (refer to section 4.3 for details). Also use this screen to select whether each low dose alarm has warning or trip status (section 4.5).

If CIP processes are to be performed on the system, tap **Enable CIP Function** to allow the system to be switched into the CIP Cycle state (section 5.14).



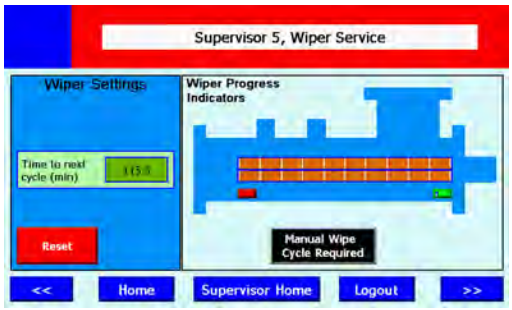
Set time intervals for the lamp to operate (in Local Timed mode, section 4.3.2) and for the system to operate in Reduced Dose mode (section 4.6).



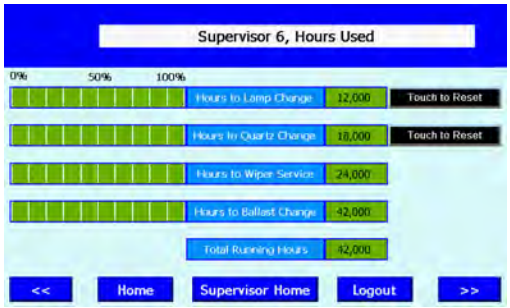
Set warning and trip temperatures and enable and configure the bleed valve if required. (If required, change the units in the Engineer menu first.)

The Trip temperature should be a maximum of 60ºC (140ºF) when a wiper mechanism is fitted, and 80ºC (176ºF) for unwiped systems.

Also use this screen to enable the UV sensor verification mode (section 5.5.5).



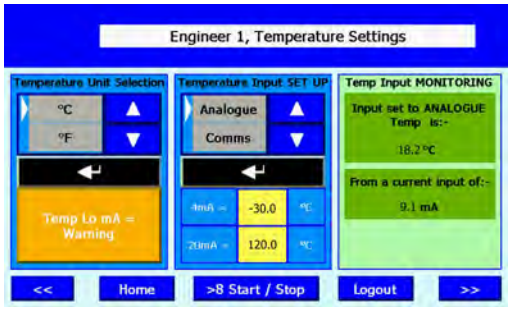
This screen is identical to screen 3 in the Operator menu. Use the **Reset** button to reset the system following a trip (once the trip condition has been rectified).



Reset the lamp and quartz usage counters; the other counters can be reset using screen 14 of the Engineer menu (Figure 29).

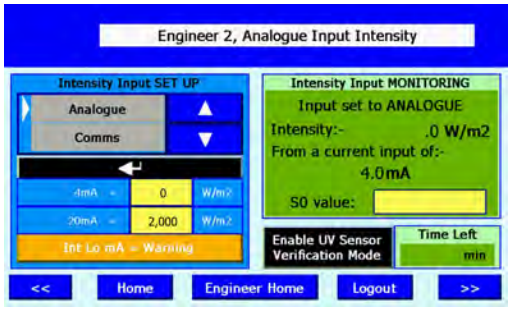
4.9. Engineer Menu

This menu allows control of analogue inputs and outputs, digital outputs and dose modes. Also use this menu for testing VFCs and loading or saving settings recipes.



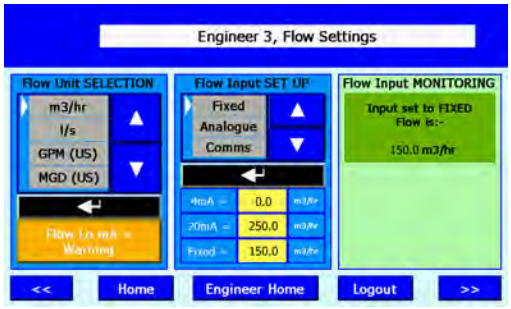
Adjust the chamber temperature sensor input. Select the units before setting the 4 and 20 mA values (and before setting the warning and trip temperatures in the Supervisor menu).

If units of Fahrenheit are selected, then a conversion to Celsius is also displayed.



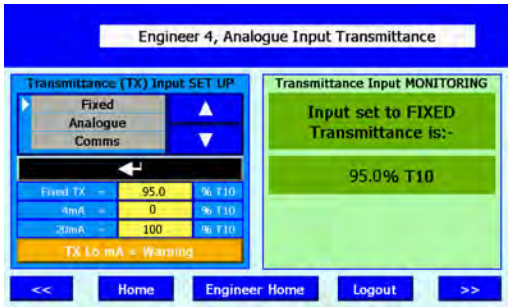
Configure the UV sensor input and select the status of the low intensity alarm (section 3.6).

Also use this screen to enable the UV sensor verification mode (section 5.5.5), and set the initial intensity value (section 3.10).

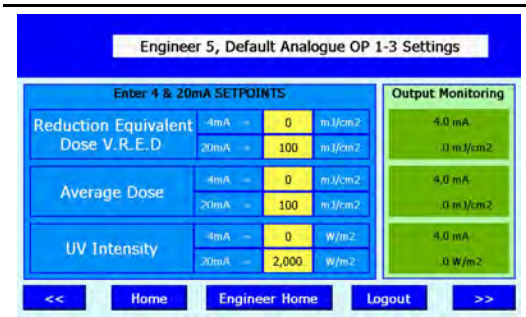


Configure the optional flow meter input and select the status of the associated alarm (section 3.6).

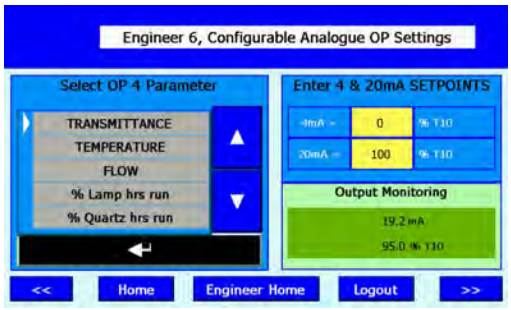
If units other than m^3/hr are selected, then a conversion is also displayed.



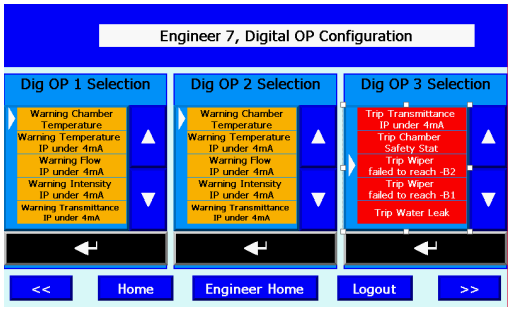
Configure the optional transmittance meter input and select the status of the associated alarm (section 3.6).



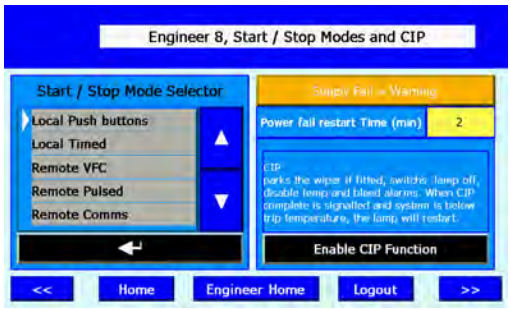
Configure the three default analogue outputs (section 3.7). The two dose outputs can be configured whether or not they are enabled on the system (see screens 11 and 12).



Select and configure the parameter of the fourth available analogue output (section 3.7).



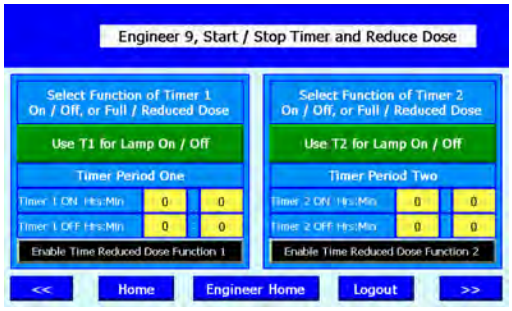
If required, select three parameters that can be monitored from the configurable digital output terminals in the control cabinet (section 3.8).



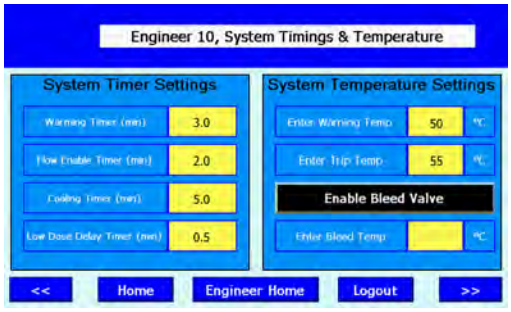
Select the start/stop mode (section 4.3).

Tap the top right button to select how the system responds to an interruption in the mains power supply. If the button shows **Supply fail = Warning**, the system will restart automatically after the software has reinitialised and the **power fail restart time** has elapsed, providing the system is not in the Cooling state. (Change the restart time in the yellow box if required.) If the button shows **Supply fail = Trip (NVR)**, the system must be reset before it can be restarted as normal (section 4.5).

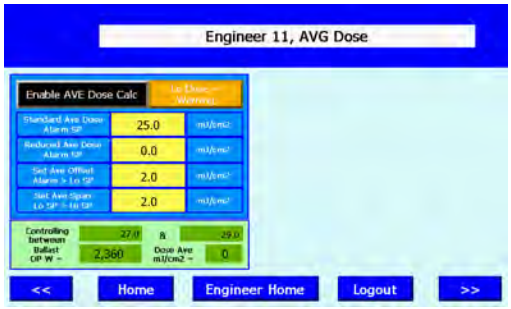
If CIP processes are to be performed on the system, tap **Enable CIP Function** to allow the system to be switched into the CIP Cycle state (section 5.14).



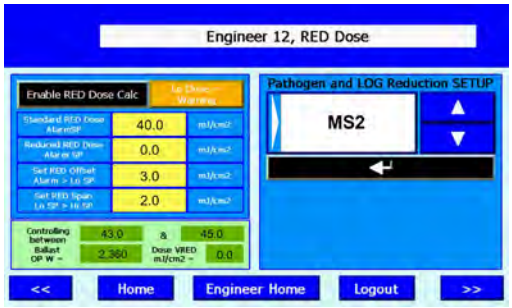
Use this screen to set time intervals for the lamp to operate (in Local Timed mode, section 4.3.2) and for the system to operate in Reduced Dose mode (section 4.6).



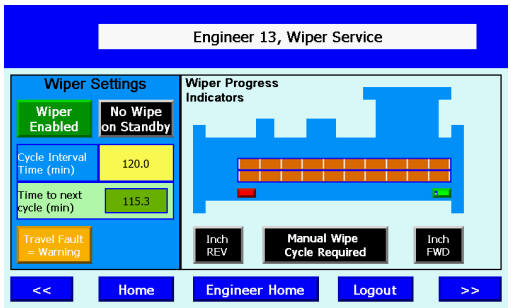
Enter durations for the Warming, Flow Enable and Cooling states on the system (see section 4.2 for details of these states). Set the time delay for signalling a low dose alarm, and the chamber warning and trip temperatures. Enable and configure the bleed valve if required (section 3.9).



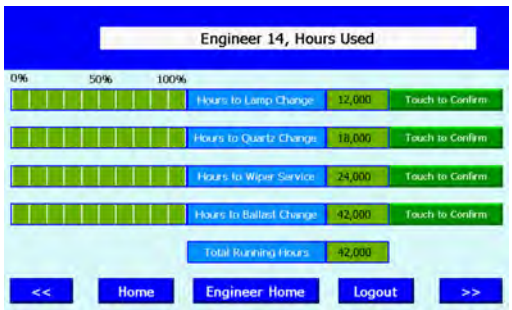
Select and configure the average dose alarm set points (section 3.9).



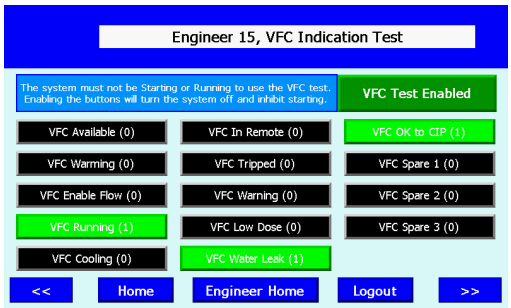
Select and configure the RED dose alarm set points (section 3.9).



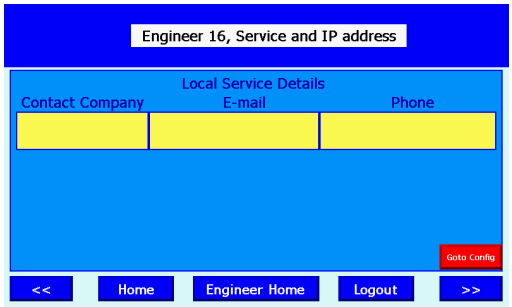
This screen gives full control of the wiper, including enabling its operation, maintaining wiping when on standby, setting the wipe cycle interval time and moving the position of the wiper yoke forwards and backwards manually (section 4.4). You can also select the status of the wiper fault alarm. Tap **Reset** to reset the system following a trip (once the trip condition has been rectified).



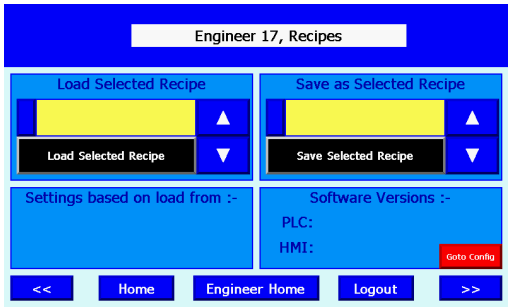
Reset the counters for usage of consumables to their initial values (see Figure 29).



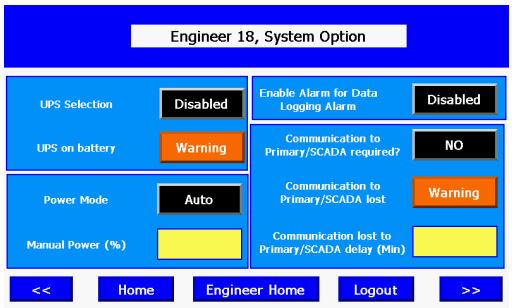
Test the volt-free contacts on the system during commissioning and troubleshooting (this will switch off the lamp and prevent it being restarted). A black background indicates the contact is open (LOW), and a green background indicates it is closed (HIGH). See section 6.5 for further details.



Enter contact details for the customer's local technical support which are then displayed in screen 12 of the Operator menu.



Recipes (settings for the system) can be loaded and saved using this screen. See section 4.11 for more information. The software applications installed on your system are also displayed in this screen. These applications relate to the system controller (PLC) and the touchscreen (HMI).



This screen contains the settings for monitoring of communications between the control cabinet and a primary controller or a site SCADA system. Tap **NO** to display an alarm if communications are lost (the button then changes to **YES**). You can then set the status of the alarm (warning/trip) and the delay time before the alarm is triggered.

You can also enable data logging alarms from this screen (section 4.10), as well as configuring the ballast power settings and the optional UPS (if fitted).

4.10. Data Logging

The data from the analogue outputs can be written to a micro SD card that is inserted into a slot on the back of the touchscreen controller inside the cabinet (Figure 25). Data is written to the card once per second. The data is saved as a *.csv file.

Note: Data is only written to the card when it is inserted. If you remove the card from the controller then the data for the period when the card is not inserted will be lost. When the card is re-inserted, the data logging continues using the same file as before.

Figure 25: Micro SD Card Port for Data Logging



You can optionally choose to enable data logging alarms in screen 18 of the Engineer menu (section 4.9). Tapping **Enable Alarms** will cause an error message to be displayed if the card is missing or cannot be accessed. The status of the SD card is also shown in the Trends screen by one of the following messages:

- No info *(card cannot be read)*
- Card not installed
- Card installed
- Card installed – access error

4.11. Using Recipes

Recipes are files containing all the software settings used to control the UV system. They can be saved at any time and then reloaded later to restore the system settings if required. Three types of recipe are available for you to use on your system:

Factory Acceptance Test (FAT): load this recipe to restore the system's original factory settings.

Site Acceptance Test (SAT): use this recipe to save the settings as they are configured when the system is first installed and commissioned.

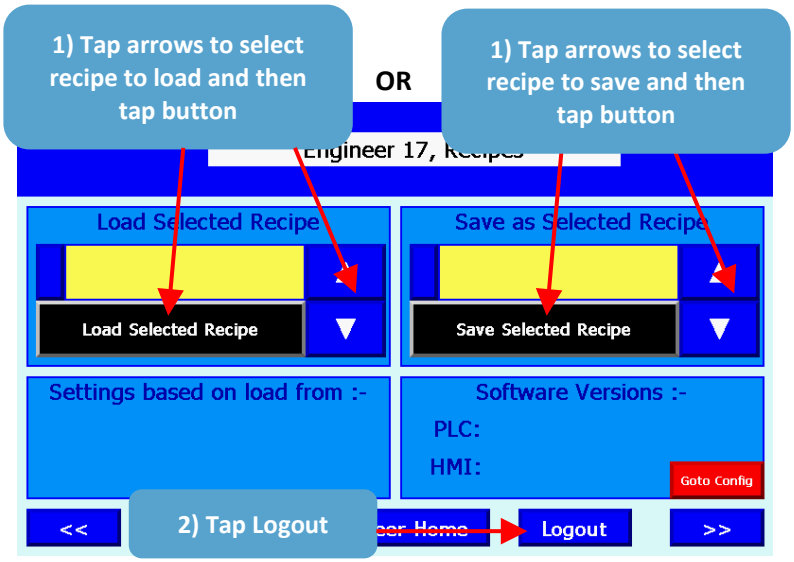
Customer Acceptance Test (CAT): use this recipe if you want to save a set of alternative settings for reuse at various times during the system's service lifetime.

We recommend that the FAT and SAT are retained unchanged after commissioning so that the system can be restored to these configurations at any time.

Follow the procedure in Figure 26 to load or save a recipe.

Figure 26: Procedure for Saving or Loading Recipes

Follow the procedure in Figure 8 to enter the Engineer menu and scroll to screen 17.



5. Maintenance

This section of the manual describes the maintenance procedures for the following components of the UV system:

COMPONENT		PROCEDURE	PAGE
UV lamp		Replacement	72
Quartz thimble		Removal	76
		Cleaning	79
		Refitting / Replacement	80
UV sensor and housing		Removal	81
		Cleaning	83
		Refitting	83
		Replacing window	85
		Verification	90
Chamber temperature sensor		Refitting / Replacement	92
Auto-wiper		Removal	93
		Servicing	94
		Refitting	100
		Adjustment	101
Ballast and other cabinet components		Replacement	102

Record the maintenance procedures carried out in the table in Appendix A.

ALL MAINTENANCE WORK MUST BE CARRIED OUT BY QUALIFIED PERSONNEL.

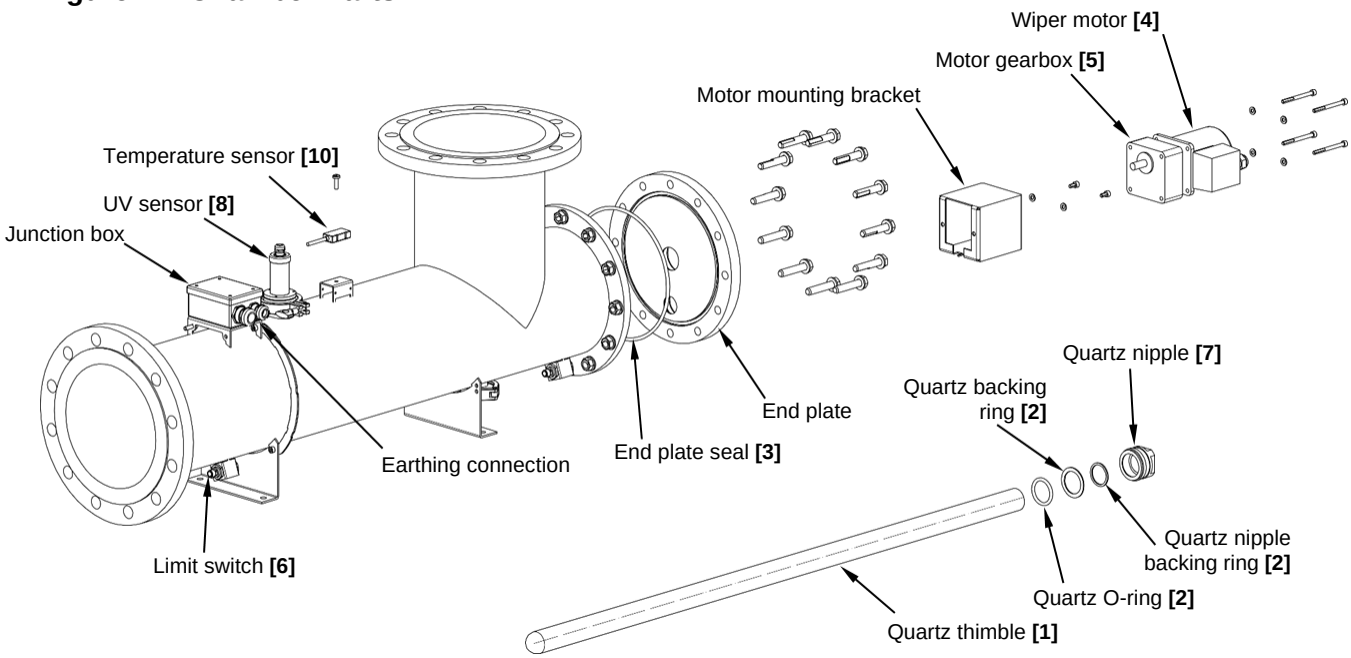
READ THE SAFETY SECTION.

TOUT TRAVAIL D'INSTALLATION DOIT ÊTRE EFFECTUÉ PAR DU PERSONNEL QUALIFIÉ.

VEUILLEZ LIRE LA SECTION TRAITANT DE LA SÉCURITÉ.

Figure 27 shows an exploded view of the chamber with the key parts identified. Wiper parts are optional.

Figure 27: Chamber Parts



Note: Figures in square brackets refer to the table of spare parts in section 5.13. Wiper parts are optional.

5.1. Safe Practices for Maintenance Procedures

	**PRIOR to carrying out ALL maintenance tasks on the UV chamber or control panel, ensure that the isolator on the front of the control panel is switched to OFF and LOCKED OFF (see Figure 28).** **AVANT de réaliser TOUTE tâche de maintenance sur la chambre UV ou sur le panneau de commande, assurez-vous que l'isolateur situé à l'avant du panneau de commande est ÉTEINT et qu'il est DÉVERROUILLÉ (voir Figure 28).**

Figure 28: Power Isolator Switch Locked Off



	**The UV chamber inlet and outlet valves should be fully closed and locked off. The chamber should then be drained to relieve any water pressure. Follow safe start-up procedures once the maintenance has been completed.** **Les vannes d'entrée et de sortie de la chambre UV doivent être entièrement fermés et verrouillées. La chambre doit alors être vidée pour soulager toute pression de l'eau. Suivez les procédures locales de démarrage sécurisé une fois que la maintenance est terminée.**

	**WEAR COTTON OR NON-MARKING DISPOSABLE GLOVES WHEN HANDLING THE UV LAMP AND QUARTZ THIMBLE TO PREVENT THE GLASS BEING MARKED WITH FINGERPRINTS.** **PORTEZ DES GANTS DE COTON OU DES GANTS JETABLES NE MARQUANT PAS POUR MANIPULER LA LAMPE UV ET LA GAINE DE QUARTZ POUR EMPÊCHER QUE LE VERRE SOIT MARQUÉ PAR DES EMPREINTES DIGITALES.**
	**Ensure that there is a clean, dry and safe working space available to set down components and carry out cleaning operations.** **Assurez-vous de disposer d'un espace de travail propre, sec et sûr pour y déposer les composants et réalisez les opérations de nettoyage.**

5.2. Service Schedule

We recommend that at least two spare UV lamps, a spare wiper seal kit and a spare thimble are always kept available for each chamber in order to service the system with the minimum downtime.

Il est recommandé qu'au moins deux lampes UV, un jeu de joints et une gaine de quartz rechange de rechange soient toujours disponibles pour chaque chambre pour l'entretien du système avec des temps d'arrêt minimal.

Period	Maintenance Procedure
9,000 hrs/Low UV warning*	Replace lamp
12 months	Clean thimble
	Clean UV sensor housing window
	Replace wiper parts including seals
	Replace UV sensor housing window O-ring and backing ring
	Perform UV sensor verification**
24 months	Replace quartz thimble and UV sensor housing window***

* Whichever is sooner
** Check sensor performance at least every 12 months
*** Replace every 12 months in high dose applications

The UV chamber is designed and tested in accordance with the Unfired Pressure Vessel standard EN 13445. Once in service, we recommend that the chamber is retested at 22 bar (2200 kPa / 319 psi) test pressure after it has passed through 500 pressurising/depressurising cycles.

La chambre UV est conçue et testée conformément à la norme EN 13445. Une fois en service, il est recommandé que la chambre soit retestée à 22 bar (2200 kPa) pression d'épreuve après son passage par le biais de 500 cycles de mettre sous pression/dépression.

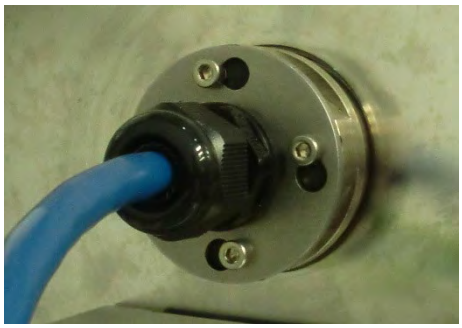
5.3. UV Lamp Replacement

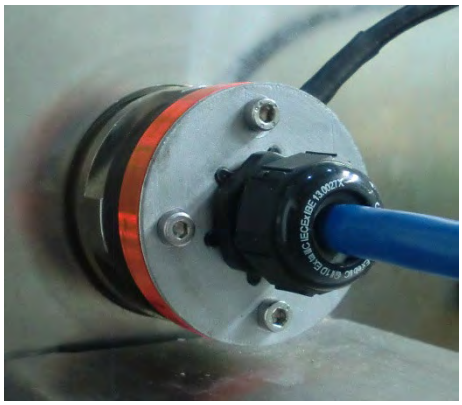
The UV lamp can be replaced without draining the system.

1. Ensure that the system is completely switched off, the cooling period has elapsed and the power isolator switch is locked off.

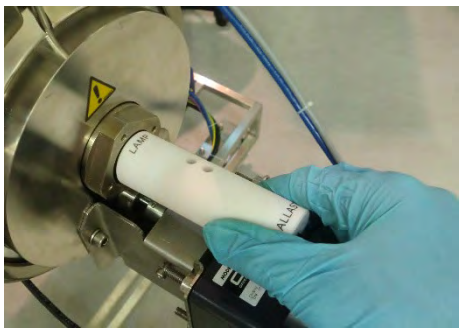
2. Loosen or remove the screws of the lamp connection on the chamber as needed, and then remove the cover.

   The lamp connection can either be a keyhole plate (top) or a UVShield™ unit (bottom).





3. Gently pull the length of cable out of the unit until the ceramic block can be accessed (the end of the lamp will be pulled out of the chamber by this action). Unscrew the lamp wires from the ceramic block.

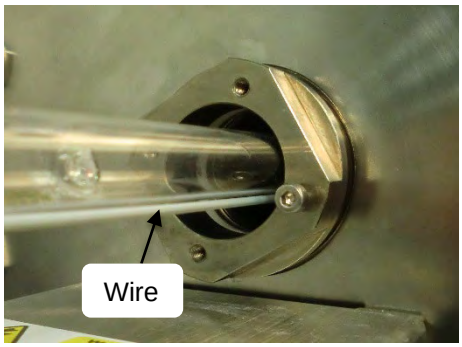


4. Carefully withdraw the quartz spacer and the UV lamp.

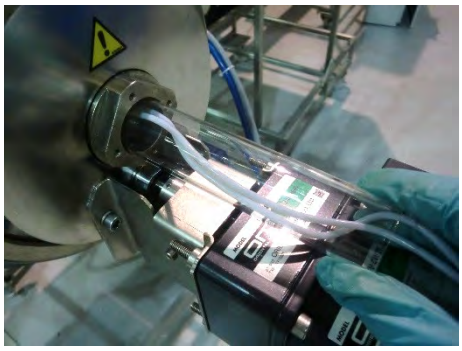
   Wear gloves when handling the UV lamp and take care in case its surface is still hot. The ceramic connections or pinch points can be used to handle the lamp if necessary.

5. Carefully insert the new UV lamp into the thimble through the quartz thimble fitting (nipple).

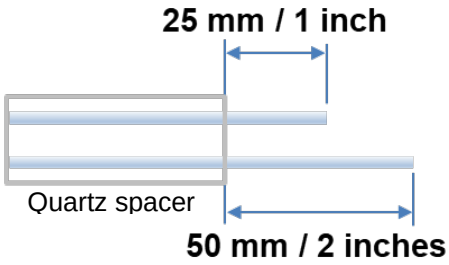
   Note: Rotate the lamp so that the wire from the far end is underneath the quartz bulb.



6. Slide the lamp spacer over the lamp wires and push it against the end of the lamp.



7. Check the lamp wire lengths and, if necessary, cut the lamp wires to the lengths shown, measured from the end of the lamp spacer.

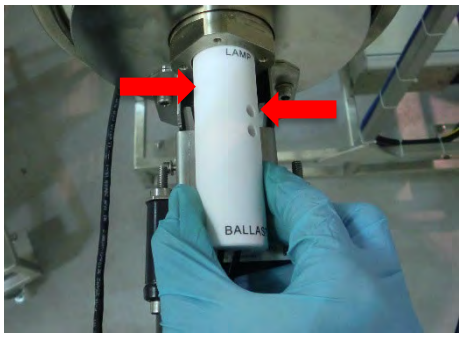


8. If necessary, strip 10 mm (0.4 inches) of outer insulation from the end of each wire. If preferred, securely crimp a ferrule to each wire for ease of connection to the terminals in the ceramic block. Trim any excess wire from the ends of the ferrules.



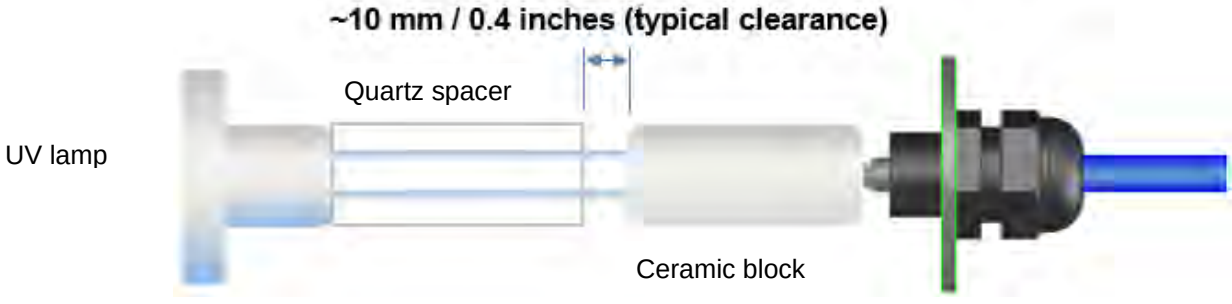
9. Connect the lamp wires to the terminals in the ceramic block at the end marked LAMP (indicated with red arrows).

   It does not matter which way around the wires are connected.



10. Push the ceramic block and spacer into the end of the thimble.

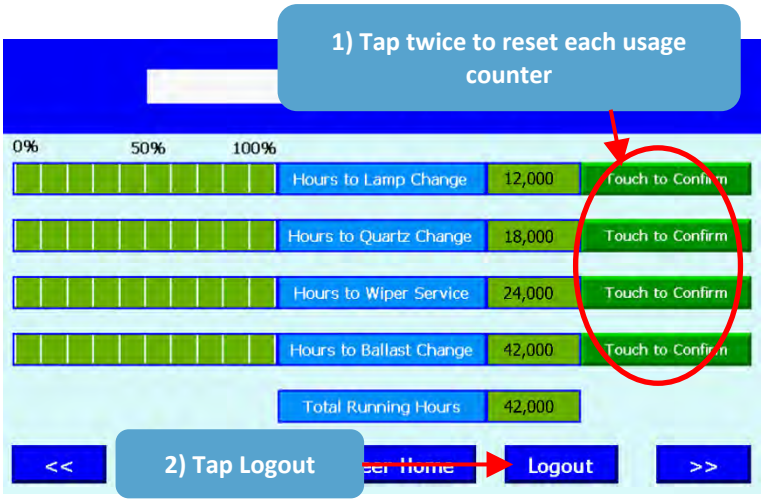
    When installed, the lamp position should be set by the combined length of the spacer and the ceramic block as shown. **Note that the quartz spacer must be pressed against the end of the lamp.**



11.	Screw the lamp connection (keyhole plate or UVShield™) onto the quartz nipple.
12.	Reset the counter for remaining lamp service hours by following the procedure in Figure 29 below.

Figure 29: Procedure for Resetting Usage Counters for Consumables

Follow the procedure in Figure 8 to enter the Engineer menu and scroll to screen 14.



You can also reset the lamp and quartz usage counters using screen 6 of the Supervisor menu.

If the blue lamp cable needs replacement, ensure that the connection inside the cabinet is correct; refer to the system drawings.

5.4. Quartz Thimble Replacement

It is good practice to check the thimble's cleanliness when replacing the lamp.

ALWAYS ISOLATE AND DRAIN THE CHAMBER BEFORE ATTEMPTING TO REMOVE THE THIMBLE, AND STAND TO ONE SIDE OF THE CHAMBER WHILE UNSCREWING THE QUARTZ NIPPLE (THIMBLE FITTING).

TOUJOURS ISOLEZ ET VIDANGEZ LA CHAMBRE AVANT DE RETIRER LA GAINE DE QUARTZ, ET TENEZ-VOUS SUR CÔTÉ DE LA CHAMBRE TOUT EN DÉVISSANT LA BAGUE D'APPUI.

5.4.1. Removing the quartz thimble

1. Ensure that the system is completely switched off, the cooling period has elapsed and the power isolator switch is locked off.
2. Remove the UV lamp (section 5.3).
3. Close and lock off the inlet and outlet valves on pipework connected to the unit, if available, to avoid draining down a large section of pipework, and follow your internal procedures to safely depressurise the isolated section of pipework.
4. Remove the drain plug or open the drain valve if one is fitted.

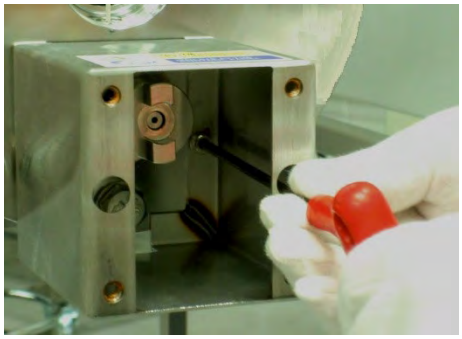
   Note: Depending on the system's location, it may be necessary to have a suitable container available to catch the water released from the drain point.



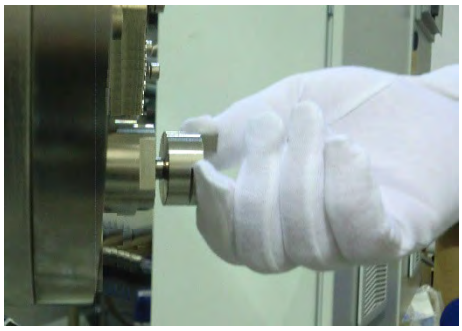
System	Chamber Volume
PQ EO 4	10 l / 2.6 U.S. gal
PQ EO 6	24 l / 6.3 U.S. gal
PQ EO 8	46 l / 12.2 U.S. gal
PQ EO 10	81 l / 21.4 U.S. gal
PQ EO 12	135 l / 35.7 U.S. gal

5. Remove the wiper motor and support box from the end plate (if fitted).



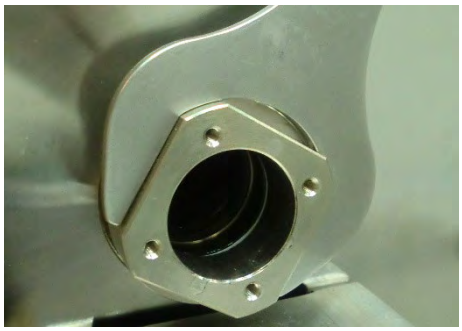


6. If a wiper is fitted, turn the wiper motor coupling anticlockwise until it stops, at which point the wiper yoke will be parked on the support bracket at the far end of the chamber.



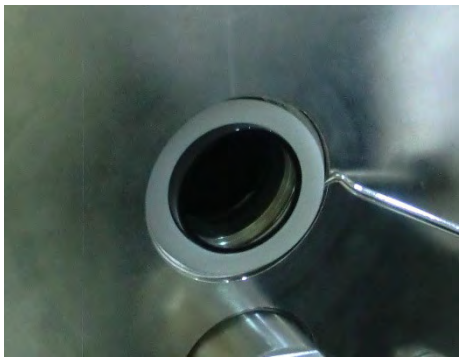
7. **Standing slightly to one side of the chamber**, remove the quartz nipple using a spanner.

   DO NOT use a pipe grip as this can damage the quartz nipple. Wear clean gloves when handling the parts. Take care not to lose the white backing rings.



8. Remove the small backing ring from the quartz nipple, and the larger backing ring from the end of the thimble.

   Do not pull the thimble out at this stage.



9. Check the thimble is free to move. If it is not then loosen the thimble from the seal by gently rocking it, and then remove the O-ring. If it is difficult to remove the O-ring from the thimble, apply a small amount of water for lubrication.

   Do not pull the thimble out at this stage.

10. The thimble is supported at both ends but not in the middle. Use a length of plastic piping to stop the thimble dropping and breaking as you remove it from its support at the far end.

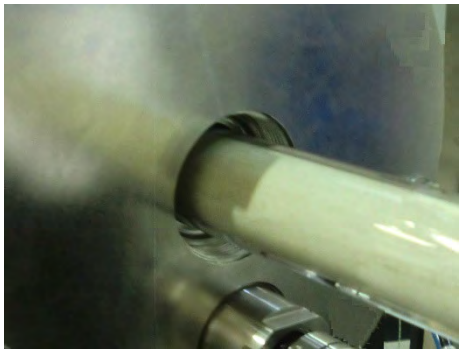
    Prepare the length of plastic pipe with an OD similar to, but less than, the ID of the quartz thimble (30–35 mm (1.2–1.4 inches) is suitable) and approximately 0.5 m (1.6 ft) longer than the chamber length. Clean the outside of the plastic pipe by wiping with IPA.

    Insert the plastic pipe into the thimble until it reaches the far end.



11. Hold the end of the thimble with one hand and gently withdraw it from the chamber, while applying a slight downward pressure on the plastic pipe with the other hand to support the thimble and prevent it from dropping and striking the chamber wall.

    Rotate the thimble as you withdraw it to reduce the resistance of the wiper ring. Be aware that the support provided by the wiper yoke will suddenly disappear as you withdraw the thimble.

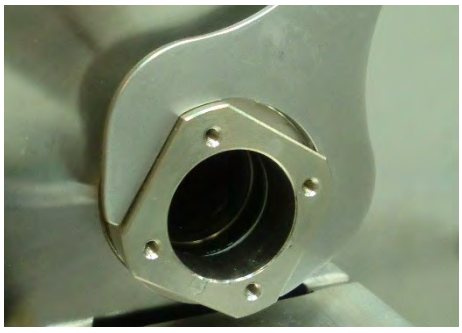


A Quartz Removal Tool (p/n 160431-0038) is available from your supplier to assist with the removal of the quartz thimble. This tool is used as follows:

1. Turn the knurled handle at the end of the tool fully anticlockwise.



2. **Standing slightly to one side of the chamber**, remove the quartz nipple using a spanner.



3. Insert the quartz removal tool into the quartz thimble as far as it will go (until the end of the quartz thimble touches the shoulder on the tool).

4. Tighten the knurled handle clockwise to grip the inner wall of the quartz thimble.	
5. Carefully pull the tool and withdraw the thimble from the chamber.	

5.4.2. Inspecting and cleaning the thimble

The quartz thimble forms the hydraulic barrier between the UV lamp and the liquid being treated. If the thimble is to be refitted then it should first be cleaned. It is essential that the thimble is kept clean, otherwise there will be a reduction in the UV light transmitted to the liquid which will reduce the effectiveness of the treatment.

- If the outside of the thimble is soiled then it may be initially cleaned using a plastic scouring pad and liquid detergent, followed by a final rinse and clean with clean water.
- If there are iron or carbonate deposits from hard water, 5% citric acid may be used to remove these deposits with the aid of a plastic scouring pad. The thimble should then be rinsed with clean water.
- If the thimble cannot be cleaned, it must be replaced.

Solarisation (UV-induced darkening) affects thimbles exposed to UV light. The discolouration will eventually start to affect the transmission of UV light through the quartz. The darkening may only be discernible in a side-by-side comparison with a new quartz component. Thimbles also become brittle due to solarisation and may break more easily if knocked on their edges or when tightening the quartz nipples.

Follow the recommended service lifetimes for thimbles and replace components even if clean to reduce the likelihood of breakages and degraded system performance.

Suivez les durées de vie de service recommandées pour les viroles en quartz et remplacez les éléments même s'ils sont propres pour réduire le risque de bris et de performances du système dégradées.

➢ Remove, clean and refit the UV sensor housing when the quartz thimble has been cleaned or replaced (section 5.5).

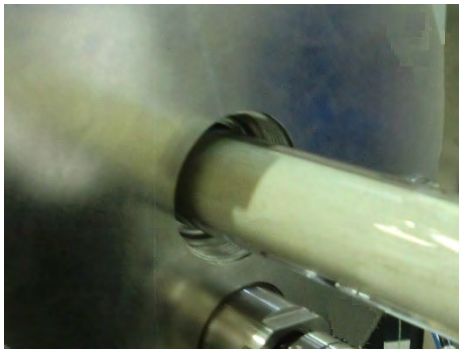
5.4.3. Fitting the thimble

	Make sure that the inside of the quartz thimble is dry before it is installed in the chamber.

1. Insert the length of plastic support piping into the thimble.

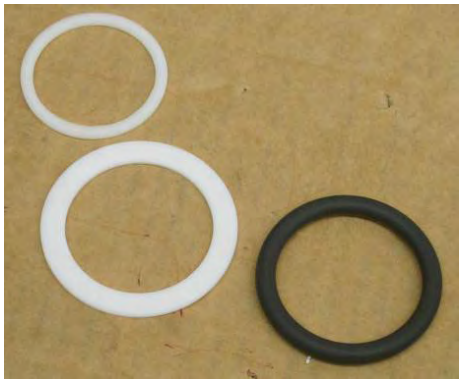
2. Wet the outside of the quartz thimble to help it slide through the wiper ring. Hold the quartz thimble with one hand while applying a supporting downward pressure on the plastic pipe with the other hand and gently feed the quartz thimble into the chamber. Carefully push the thimble through the wiper ring.

   There is a socket installed at the opposite end of the chamber to support the sealed end of the thimble. The thimble should be fitted into this socket until it is supported but not fully pushed into place. Remove the plastic pipe.



3. Fit a new O-ring onto the end of the quartz thimble. Hold the thimble with one hand while fitting the O-ring with the other hand.

   We recommend that new O-rings and backing rings should be used each time a thimble is refitted.



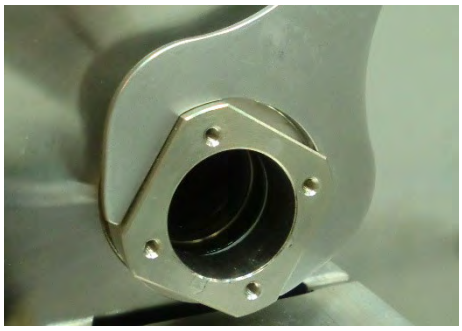
4. Fit a new backing ring against the O-ring and push it into place.

5. Fit a new backing ring into the quartz nipple. The backing ring will protect the end of the quartz thimble from damage.

   Note: This backing ring is slightly larger than the ID of the fitting so should be fitted as shown and then pushed into place. Since it is a ‘positive’ fit, it is self-retaining.



6. Fit and tighten the quartz nipple using a spanner. **DO NOT use a pipe grip as this can damage the nipple.** Tighten the nipple carefully until a definite stop is felt, or to a torque setting of 45 Nm.



7. Replace the UV sensor housing assembly if it has been removed (section 5.5.3).

8. Refit the drain plug or close the drain valve if one is fitted.

9. Open the inlet and outlet valves on the pipework supplying the unit and repressurise the chamber. Check the O-ring seal and inspect the inside of the thimble for any leaks. If a leak occurs, isolate and drain the chamber, remove the thimble and recheck the O-ring seals and quartz thimble sealing surfaces for debris or damage, replacing components if necessary.

10. Once the system is sealed, refit the UV lamp (section 3.4).

11. When fitting a new thimble, reset the counter for remaining quartz service hours by following the procedure in Figure 29 on page 75.

You can also use the quartz removal tool to assist with fitting a thimble by reversing the process shown in section 5.4.1.

5.5. UV Sensor

5.5.1. Removing the UV sensor and housing

The UV sensor can be removed for calibration and testing without having to drain the system by following steps 3–5 below.

1. Close and lock off the inlet and outlet valves on pipework connected to the unit, if available, to avoid draining down a large section of pipework, and follow your internal procedures to safely depressurise the isolated section of pipework.

2. Remove the drain plug or open the drain valve if one is fitted.

   Depending on the system's location, it may be necessary to have a suitable container available to catch the water released from the drain point.

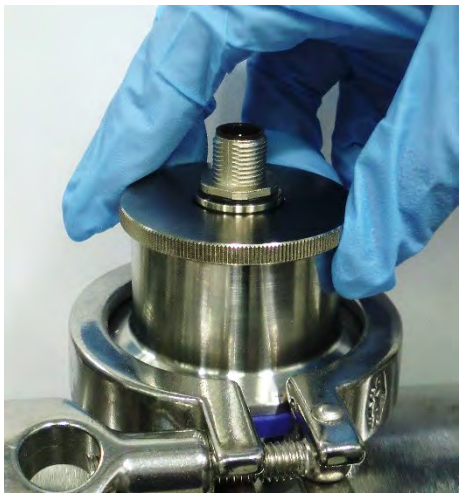


System	Chamber Volume
PQ EO 4	10 l / 2.6 U.S. gal
PQ EO 6	24 l / 6.3 U.S. gal
PQ EO 8	46 l / 12.2 U.S. gal
PQ EO 10	81 l / 21.4 U.S. gal
PQ EO 12	135 l / 35.7 U.S. gal

3. Unscrew the locking ring on the connector and disconnect the sensor lead from the sensor.

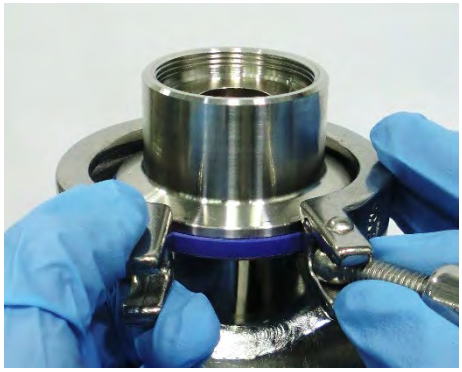
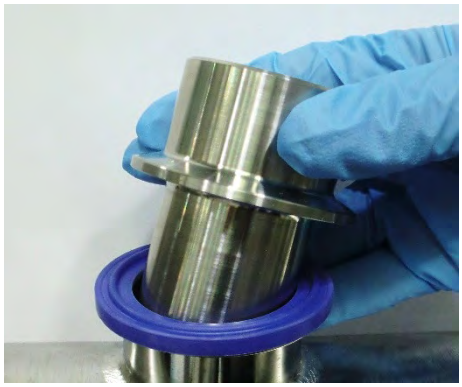


4. Unscrew the cover from the sensor housing.



5. Lift the sensor out of the housing.



6.	Undo and open the tri-clamp sanitary connector.	
7.	Withdraw the UV sensor housing from the chamber.	

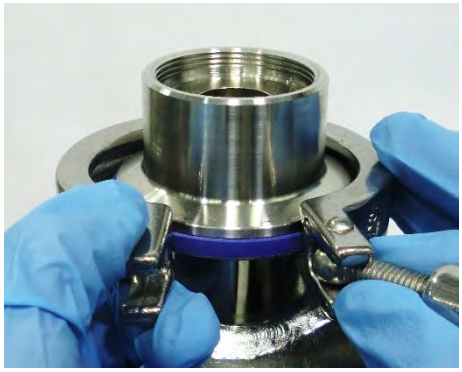
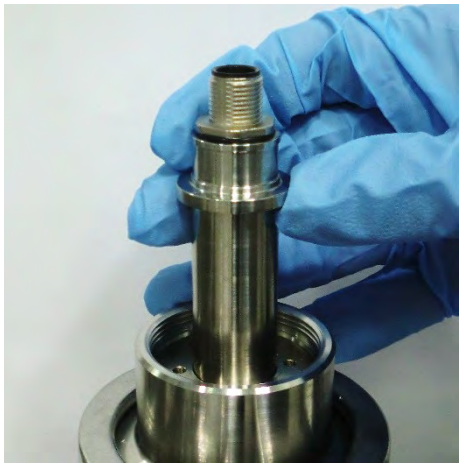
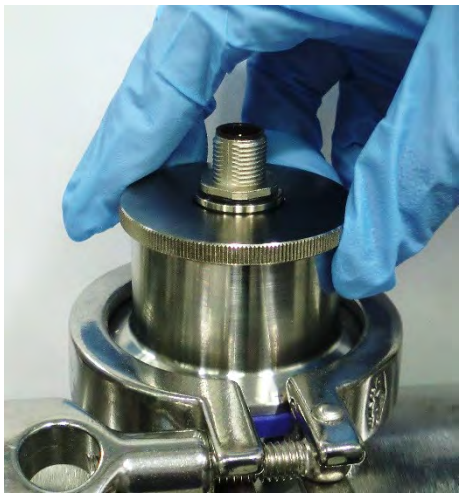
5.5.2. Cleaning the UV sensor and housing

To prevent low UV warnings, the UV sensor housing that protrudes into the chamber should be cleaned periodically with liquid detergent and, if soiled with deposits, a plastic scouring pad, in particular whenever the quartz thimble is being cleaned or replaced (section 5.4).

1.	After removal, the UV sensor and housing should be set in a clean, dry location.
2.	The UV sensor housing should be cleaned with liquid detergent and, if necessary, a plastic scouring pad. If there are deposits on the UV sensor housing which cannot be removed, it may be necessary to replace the window. A service kit is available from your supplier. Refer to sections 5.13 and 5.5.4.
3.	Ensure that the components are thoroughly dry before reinstalling them in the chamber.

5.5.3. Reinstalling the UV sensor and housing

1.	Refit the drain plug or close the drain valve if one is fitted.

2.	Place the sensor housing onto the connection on the chamber with the seal in place and refit the tri-clamp to secure it in place.	
3.	Open the inlet and outlet valves on the pipework supplying the unit and repressurise the chamber. Check the tri-clamp seal for leaks. If a leak occurs, isolate and drain the chamber, remove the sensor housing and check the seal and mating surfaces for debris or damage, replacing components where necessary.	
4.	Wherever possible, purge the sensor housing with dry nitrogen immediately before reinstalling the UV sensor.	
5.	Slide the UV sensor fully into the housing, pushing gently to open the UVGuard™ shutters inside the housing.	
6.	Screw the cover on to the sensor housing.	

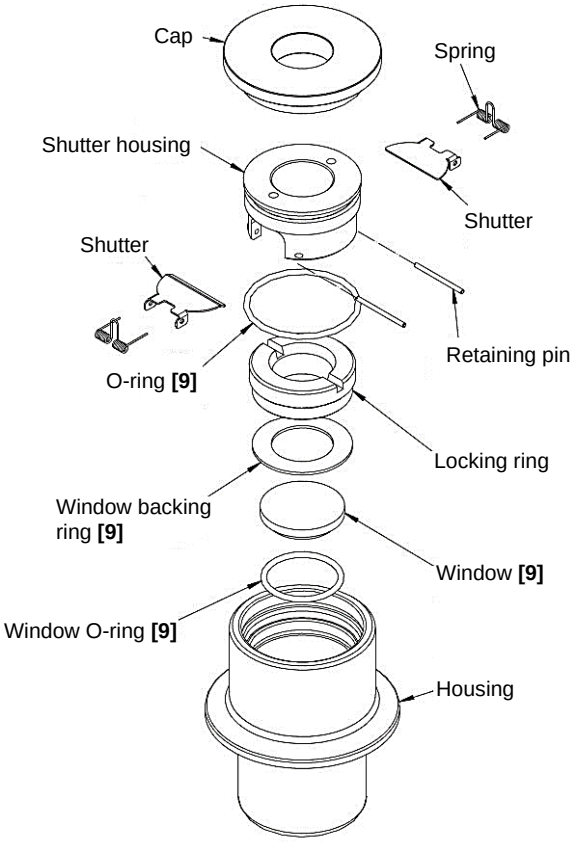
7. Reconnect the sensor cable and finger-tighten the locking ring.



5.5.4. Replacing the UV sensor housing window

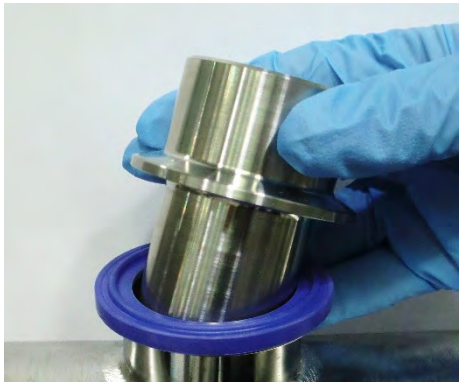
Follow the steps below to remove and replace the UV sensor housing window. The parts in the housing assembly are shown in Figure 30. Not all of these parts need to be dismantled to replace the window.

Figure 30: UV Sensor Housing Parts



Note: Figures in square brackets refer to the table of spare parts in section 5.13.

1. Remove the UV sensor housing from the chamber as described in section 5.5.1.





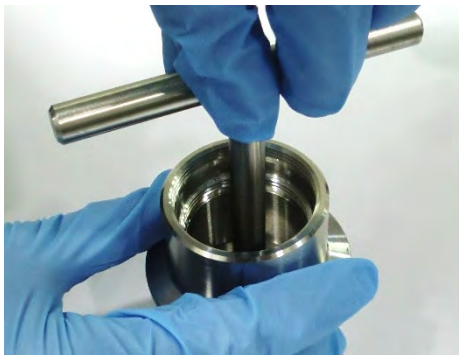
2. Unscrew the UVGuard™ shutter assembly.

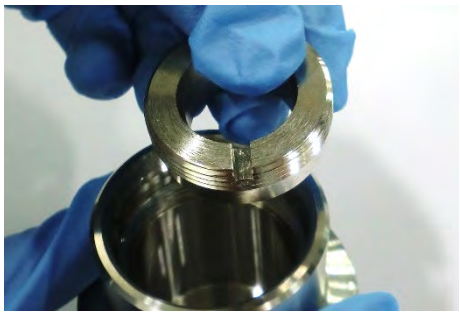
   A special tool (part number 321938) is available from your supplier.



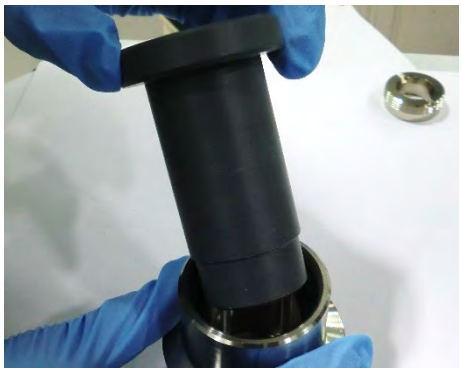


3. Unscrew and remove the slotted locking ring at the bottom of the housing.





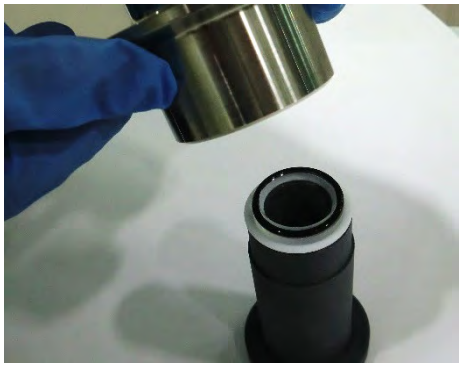
4. Gently insert the window insert tool (available from your supplier, part number 321939) so that it rests squarely on the window, and then invert the entire assembly.





5. Push down on the window carefully to release it from the housing, and then lift off the housing.





6. Remove the window, o-ring and backing ring from the insert tool.

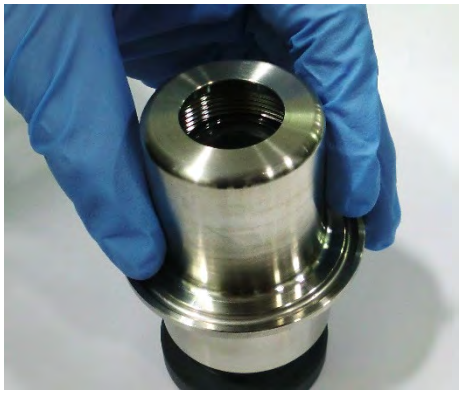
   We recommend that all these components are replaced with new parts from the sensor housing service kit (see section 5.13).

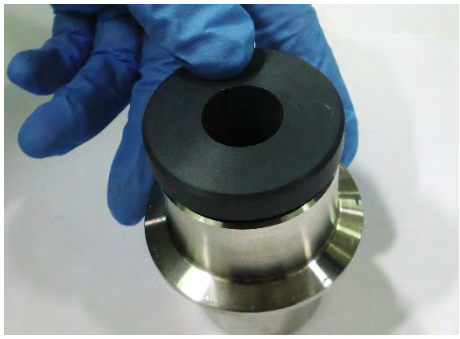
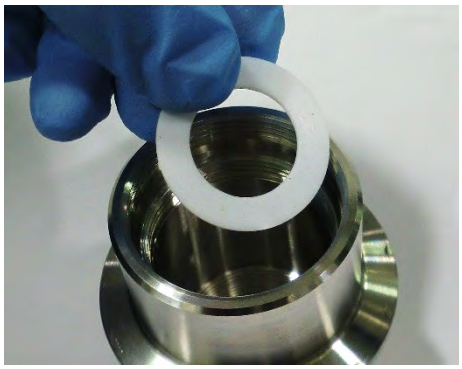
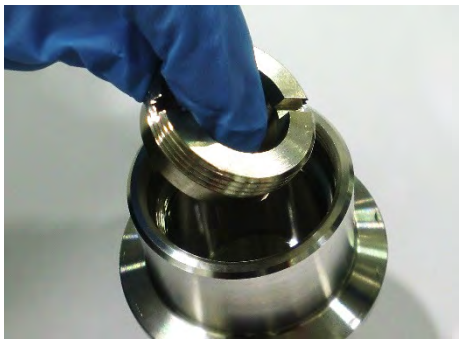
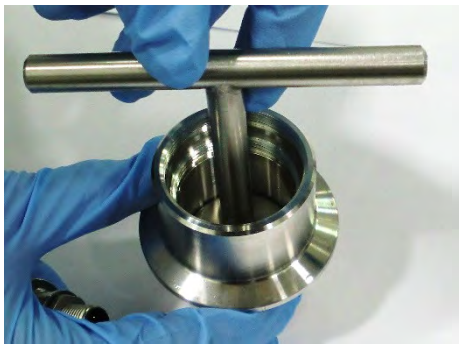


7. Push the new o-ring on to the rim of the new window and place them on to the insert tool with the O-ring at the top.



8. Carefully place the housing over the window, making sure that the window sits squarely at the end of the housing.



9.	Invert the entire assembly and lift the insert tool out of the housing.	
10.	Insert the backing ring so that it sits flat at the bottom of the housing.	
11.	Insert the locking ring and tighten with the tool until a hard stop is felt.	 

12.	Refit the UVGuard™ shutter assembly.	

13.	Reinstall the UV sensor housing in the chamber as described in section 5.5.3, and pressurize the chamber to check the housing for leaks BEFORE refitting the UV sensor.

5.5.5. UV sensor verification

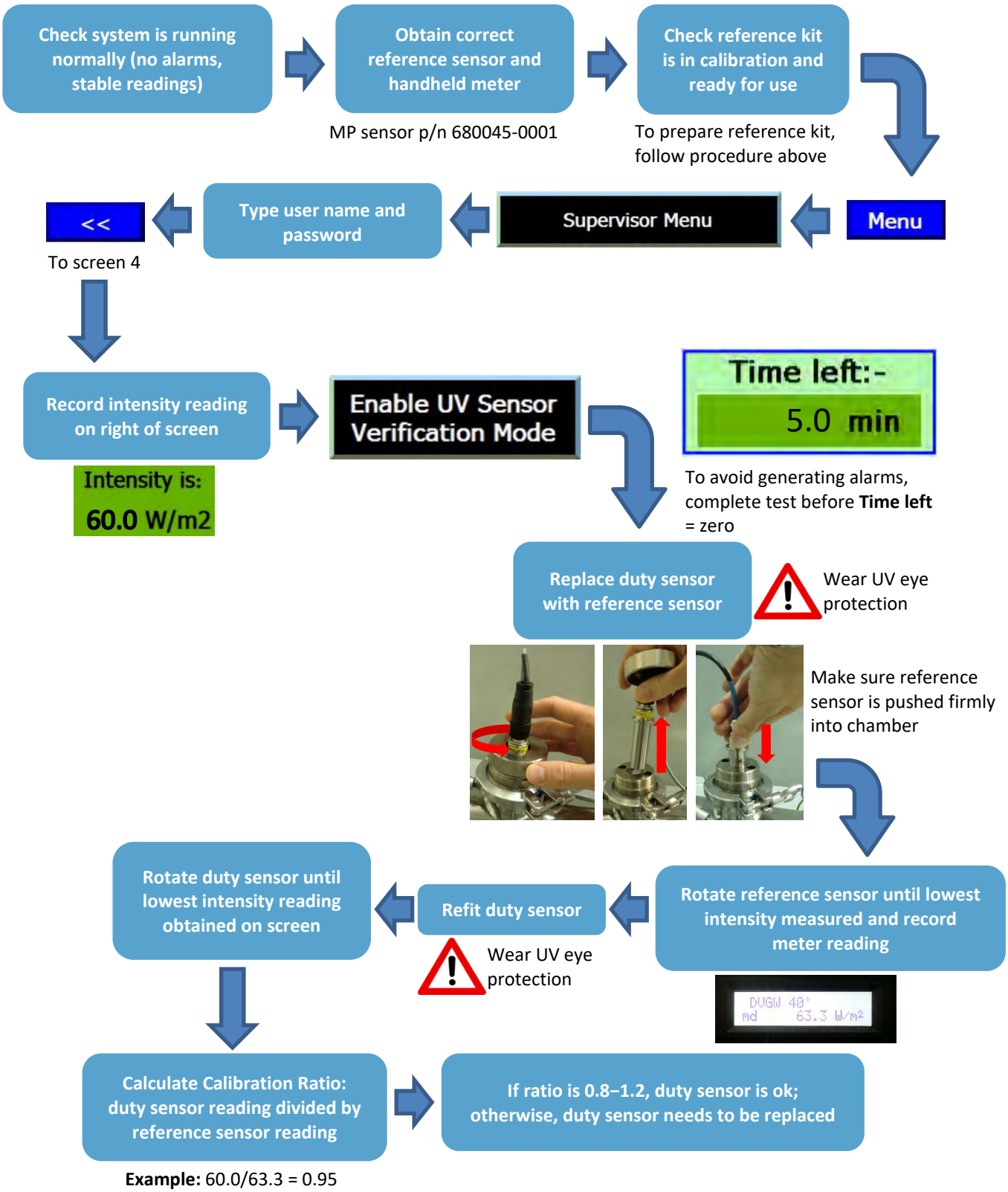
The duty UV sensor intensity measurement can be verified without the need to switch the system off. A medium-pressure reference sensor is required (p/n 680045) connected to a hand-held meter (p/n 680043). Check that the calibration of the reference sensor kit has not expired before carrying out the following procedure.

To prepare the reference sensor kit for use:

1.	Connect the reference sensor to the hand-held meter.
2.	Press ON to switch the meter on.
3.	Press ON to enter the menu options.
4.	Press ON to select **lamp type**.
5.	Press HOLD to select **medium pressure**.
6.	Press ON to return to the measurement display. 'md' should be displayed on the left of the display.
7.	Allow the meter and sensor to warm-up for 10 minutes.

Follow the procedure in Figure 31 to test the duty sensor.

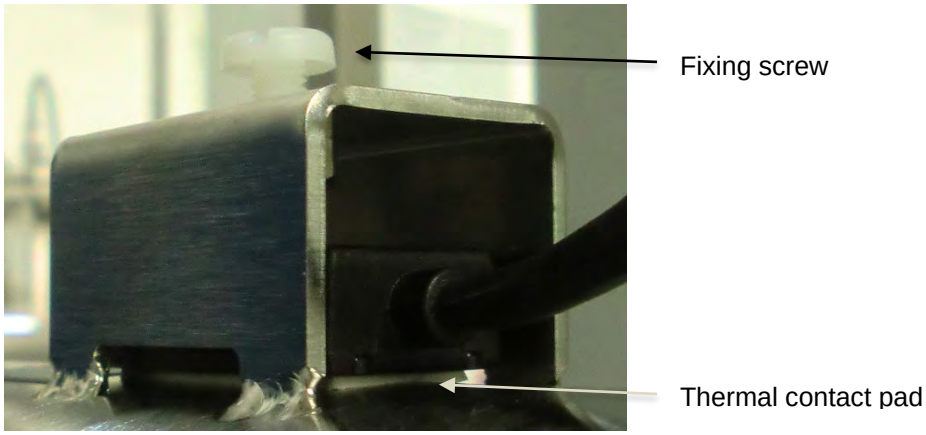
Figure 31: UV Sensor Test Procedure



5.6. Replacing the Chamber Temperature Sensor

In normal usage, the chamber temperature sensor should not require any maintenance. However, if a new sensor is fitted or if the chamber is subjected to high pressure washing, always ensure that the thermal contact pad underneath the sensor is retained or replaced to ensure that accurate temperature readings are obtained. Spare pads are available from your system supplier (section 5.13).

Figure 32: Fitting the Chamber Temperature Sensor



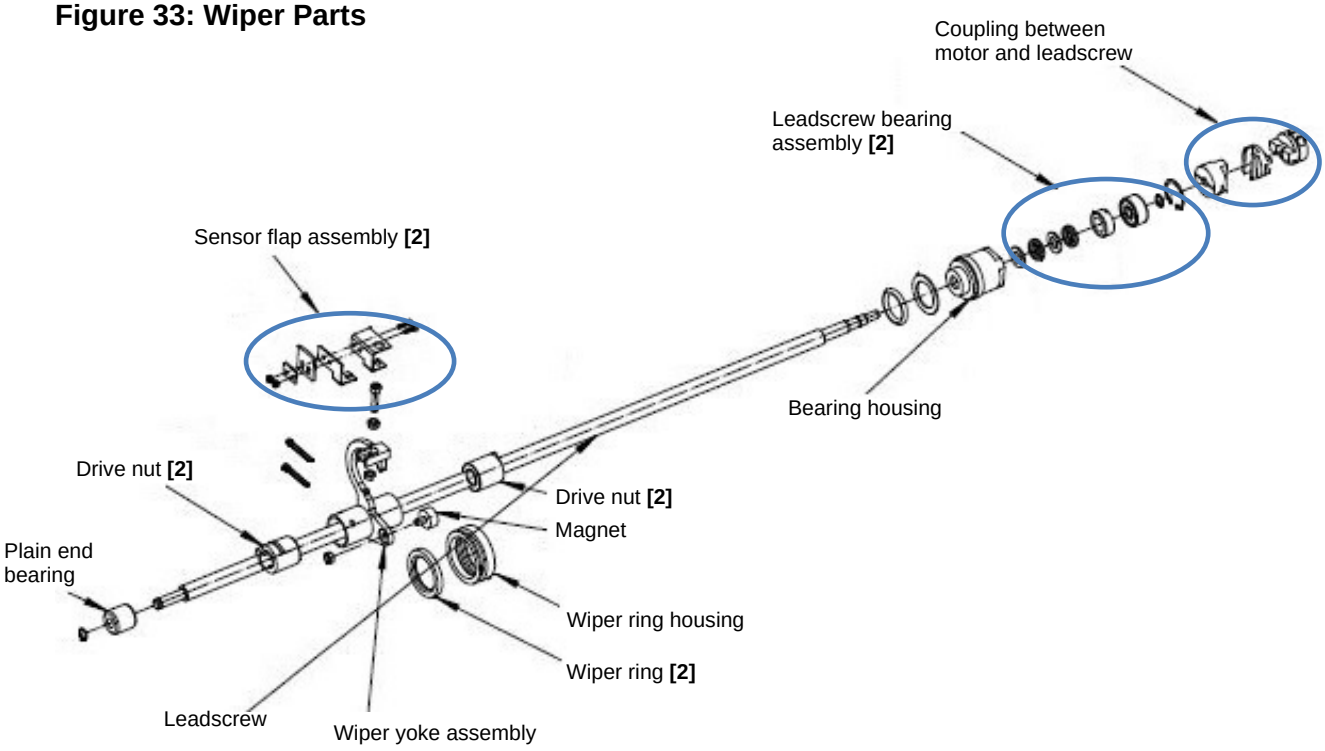
To correctly replace the chamber temperature sensor:

1.	Undo the fixing screw and remove the old temperature sensor from the chamber.
2.	Disconnect the old temperature sensor cable from the junction box as shown in the system drawings in Appendix C.
3.	Remove the old thermal contact pad from the surface of the chamber.
4.	Peel off the backing from one side of the new thermal contact pad and fix it in the recess on the metal underside of the temperature sensor.
5.	Peel off the backing from the remaining side of the thermal contact pad and use the cable to feed the sensor under the bracket.
6.	Press the sensor into position on the surface of the chamber and secure firmly using the fixing screw.
7.	Connect the new temperature sensor cable in the junction box as shown in the system drawings in Appendix C. Attach the cable to the bracket to keep it away from the chamber surface.

5.7. Auto-Wiper (option)

The auto-wiper system requires servicing every 6–12 months depending on the process liquid environment. Figure 33 shows an exploded view of the auto-wiper with the key parts identified.

Figure 33: Wiper Parts



Note: Figures in square brackets refer to the table of spare parts in section 5.13.

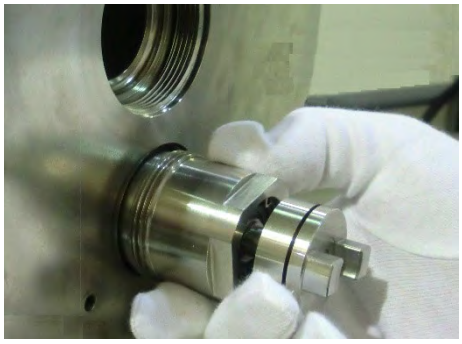
For auto-wiped systems, remove the thimble before removing the auto-wiper mechanism (refer to section 5.4.1).

Pour les systèmes à nettoyeur, retirer la gaine de quartz avant de retirer le mécanisme de nettoyage automatique (reportez-vous à la section 5.4.1).

5.7.1. Removing the auto-wiper assembly

This procedure follows the removal of the quartz thimble (section 5.4.1).

1. Unscrew the bearing from the end plate and carefully pull the leadscrew towards you until the yoke reaches the end plate.

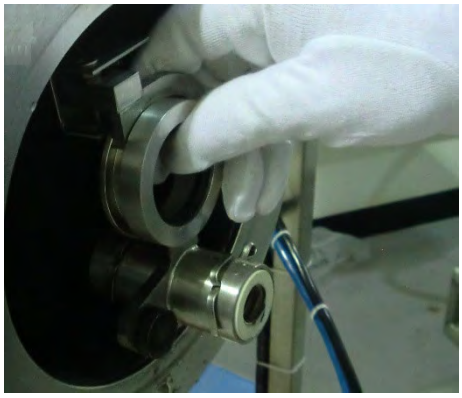


2. Unscrew the leadscrew from the wiper yoke and withdraw it from the chamber.



3. Remove the end plate from the chamber. Check the end plate O-ring seal for debris or damage and clean or replace as necessary.

4. Remove the wiper yoke from the chamber for servicing.



Do not attempt to remove the end plate without first removing the bearing housing. Otherwise, the leadscrew may be damaged due to the weight of the end plate.

N'essayez pas de déposer la plaque d'extrémité avant de déposer le logement de roulement. Dans le cas contraire, la vis sans fin peut être endommagée en raison du poids de la plaque d'extrémité.

5.7.2. Servicing the leadscrew bearing

1. Undo the grub screw and remove the self-aligning coupling from the leadscrew.



2. Remove the small external circlip from the leadscrew shaft and unscrew the bearing housing assembly.



3. Remove the O-ring and backing ring from the bearing housing assembly.

4. Remove the large internal circlip and discard.

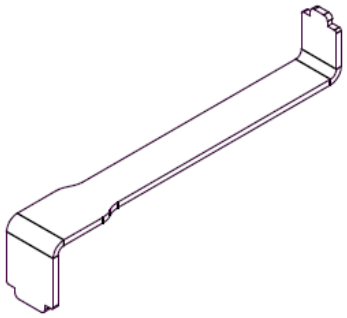


5. Use a 2 mm hex head driver to remove the appropriate tap from the wiper bearing service tool.

   Use the small adaptor for D EO H 4 and 6 systems.



   An alternative tool is available specifically for removing and replacing the retaining nut.



6. Stand the bearing housing (with the bearing face down) on a support to allow the bearing to be tapped out. Place the tap in the centre of the bearing and strike the end to push the bearing out of the housing.

   A large socket adaptor makes a suitable support (24 mm (0.9 inches) for the small bearing housing and 32 mm (1.26 inches) for the large housing). Alternatively, an anvil accessory is available from your supplier (p/n 321346).



7. Use one end of the bearing service tool to unscrew the slotted retaining nut. Use the remaining tap as a handle. Alternatively, use the retaining nut tool shown above.

   Remove the quad ring seals and spacers from the housing. These should be discarded as they may be damaged.



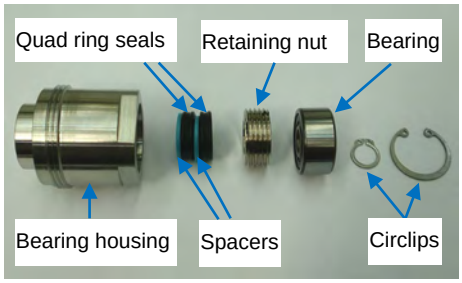


8. Clean the bearing housing. Liberally apply synthetic grease (such as Super Lube® Multi-Purpose Grease) on to the new quad ring seals.

   Note: This is the only component of the system where a lubricant other than water should be used.

   Fit new seals and spacers into the bearing housing as shown. Only use new components from the wiper service kit. Tighten the retaining nut.

   Note: The small bearing assemblies contain a third spacer which sits between the quad ring seal and the retaining nut.



9. Insert a new bearing into the bearing housing, and secure in place using a new large circlip from the wiper service kit.

   Note: Do not reuse the old circlip (circlips should only be used once).





10. Replace the bearing on the leadscrew and fit a new external circlip.



11. Fit the coupling for the motor onto the leadscrew.

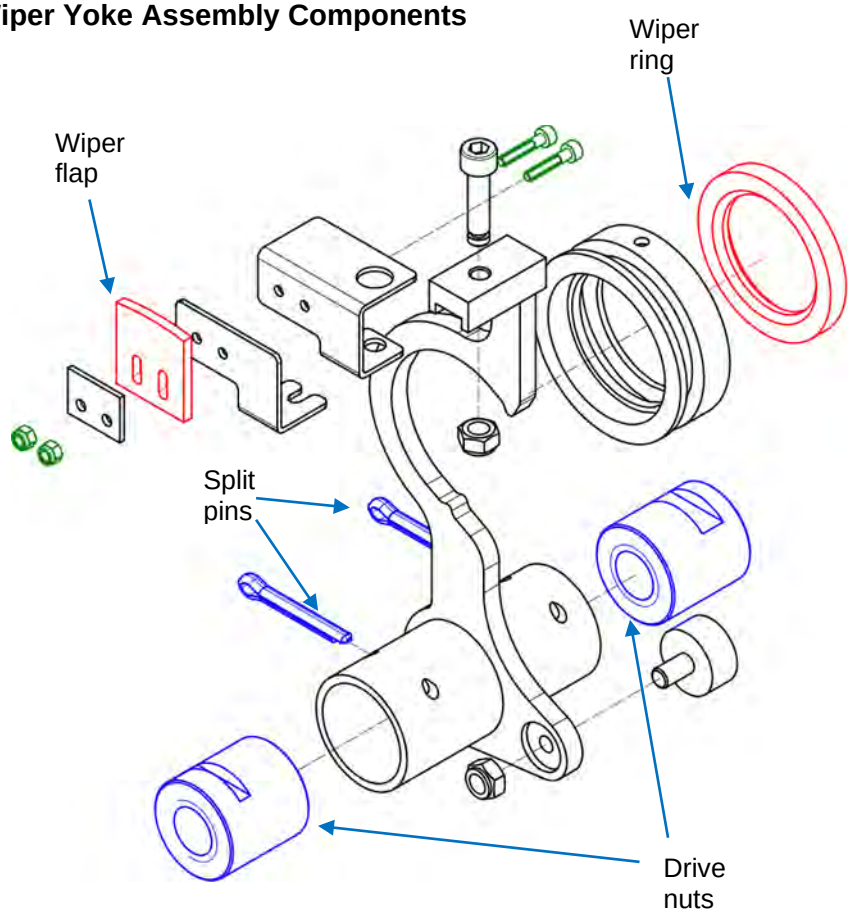


5.7.3. Servicing the wiper yoke

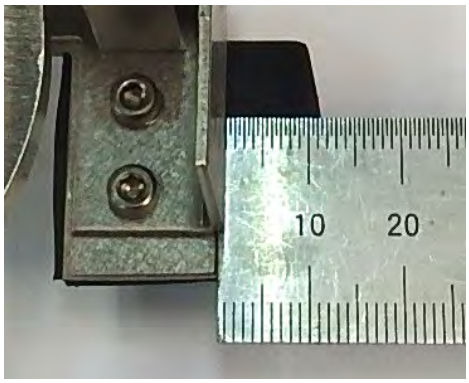
The components of the wiper yoke are shown in Figure 34. The parts involved in the servicing procedure are coloured as follows:

- The wiper ring and wiper flap (for the UV sensor window) are shown in red. These parts should be replaced every time the yoke is serviced.
- The bolts and nuts that hold the wiper flap in place are shown in green.
- The drive nuts and their split pins are shown in blue. These parts should also be replaced at every service.

Figure 34: Wiper Yoke Assembly Components



Use the following procedure for servicing the wiper yoke:

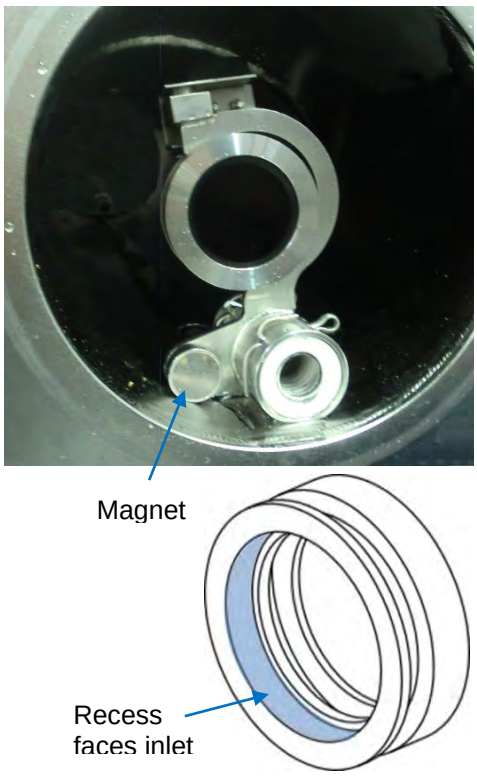
1.	Measure the distance that the wiper flap protrudes from the yoke and note the result. Unscrew the two socket bolts with a 3 mm hex head tool and remove the wiper flap.	
2.	Fit a new wiper flap in the same position as measured previously.	

3.	Use a suitable tool (such as a hex head driver) through the hole in the wiper ring housing to push out the wiper ring. Push a new wiper ring into the housing.	
4.	Use pliers to straighten and withdraw the split pins securing the two drive nuts in place, and discard.	
5.	Remove the drive nuts and discard.	
6.	Insert new drive nuts into the yoke assembly, ensuring that the grooves on the drive nuts are lined up with the holes for the split pins.	
7.	Tap new split pins into the holes to secure the drive nuts, and bend the ends around the housing.	

5.7.4. Refitting the auto-wiper assembly

1. Place the yoke assembly just inside the chamber, ensuring that it is correctly oriented.

   The magnet must be on the left side of the yoke, and the wiper ring housing recess must face towards the inlet quartz and wiper support at the other end of the chamber.



2. Fit the end plate, ensuring that it sits symmetrically on the chamber flange.

3. Insert the leadscrew into the chamber through the lower port in the end plate and screw it into the drive nuts of the yoke until it can be seen passing through the yoke.

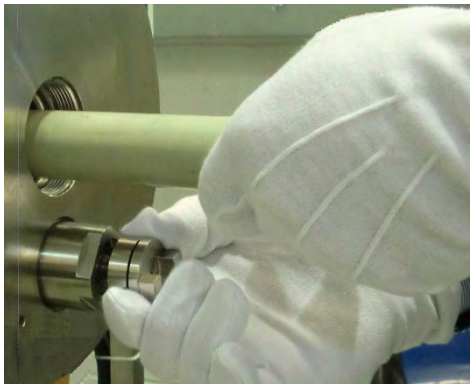


4. Push a length of plastic tube through the wiper ring to support the yoke, and then push both the tube and the leadscrew down to the end of the chamber.



5. Insert the leadscrew into the plain end bearing at the inlet, and screw the leadscrew bearing into the end plate.

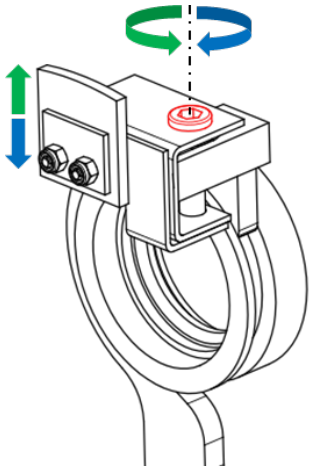
6. Use the plastic tube to align the yoke with the support bracket, and turn the motor coupling anticlockwise until the yoke is parked on the bracket.



7. Refit the quartz thimble (refer to section 5.4.3), taking care when pushing the thimble through the wiper ring. Wet the thimble with water to help lubricate the ring as it slides over the thimble.

8. Remove the UV sensor housing (refer to section 5.5.1) and turn the leadscrew manually until the yoke assembly is visible through the port.

9. Adjust the wiper flap position up or down using the hex screw on the top of the yoke.



10. Insert the sensor housing and turn the leadscrew manually to check the effectiveness of the wiper flap as it passes across the sensor housing window.

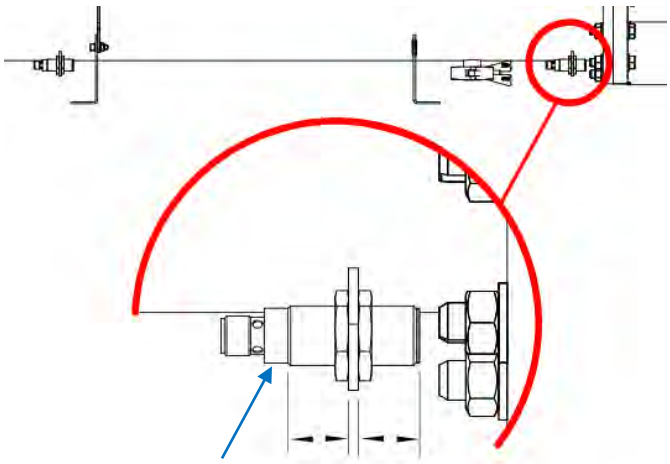
    The wiper flap should clearly deform as it passes across the window, without bending over and putting unnecessary strain on the flap at the securing bolts.

11.	Reinstall the sensor housing (refer to section 5.5.3).
12.	Repressurise the chamber and check for leaks. If leaks occur, check the thimble, wiper seals and sealing surfaces for debris or damage.
13.	Refit the motor support box and the wiper motor.
14.	Reset the counter for remaining wiper service hours using screen 14 of the Engineer menu (follow the procedure in Figure 29 on page 75).

5.7.5. Replacing wiper limit switches

In the unlikely event that you need to replace a limit switch on the chamber, it is good practice to position the switch on the bracket so that approximately equal lengths of thread can be seen on either side of the bracket (Figure 35). This sets the travel of the wiper yoke to the correct length.

Figure 35: Position of Limit Switch after Replacement



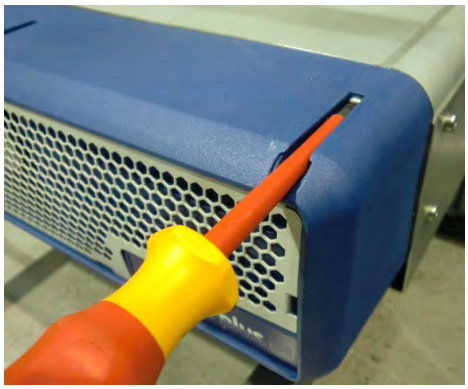
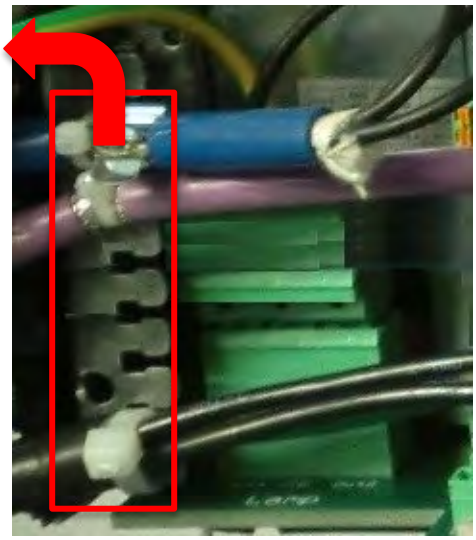
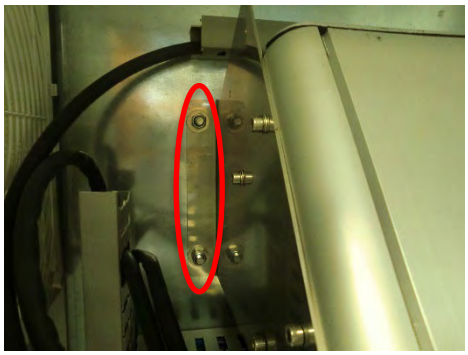
5.8. Replacing the Ballast

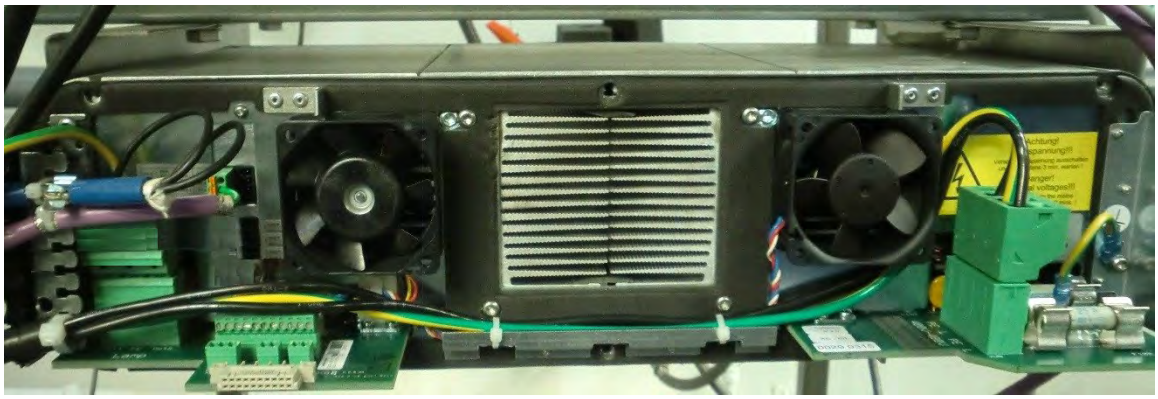


SWITCH OFF AND ISOLATE THE ELECTRICAL POWER SUPPLY BEFORE REPLACING THE BALLAST.

ÉTEINDEZ ET ISOLEZ L'ALIMENTATION ÉLECTRIQUE AVANT DE REMPLACER LE BALLAST.

1.	Ensure the system is switched off at the door isolator and is disconnected from the mains supply.

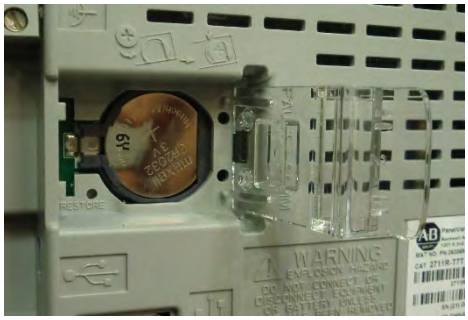
2.	After waiting a minimum of five (5) minutes, open the cabinet and use a T20 Torx screwdriver to remove the six screws on the front cover of the ballast. Pull the front cover carefully along the guide rails to remove it.	
3.	Disconnect the cables from the terminals inside the ballast.	
4.	Unscrew the support bracket holding the cables. Remove the access port gland plate and lift the cables away from the ballast.	
5.	While supporting the ballast, unscrew the M8 bolts holding each of the ballast brackets to the back wall of the cabinet.	
6.	Remove the ballast from the cabinet.	
7.	Use a hex head driver to remove the support brackets from the ballast.	
8.	Attach the support brackets to the new ballast. Ensure that the correct bracket is fitted to each side of the ballast (these are marked).	

9.	Fit the ballast to the back wall of the cabinet.
10.	Remove the front cover of the new ballast. Remove the cable support bracket from the new ballast and replace it with the bracket attached to the cables (see step 4). Connect the cables to the ballast terminals as before. The connections to the ballast are as follows (also refer to the system drawings in Appendix C): • Black – mains power from control cabinet to ballast. • Grey – communication from ballast to control cabinet. • Blue – power from ballast to UV lamp. 
11.	Refit the correct blanking plugs on the unused access ports (refer to the system drawings in Appendix C).
12.	Slide the front cover along the guide rails on each side of the ballast, making sure the cover is properly aligned to avoid damage to connectors inside the ballast. Attach the cover securely with the six screws.
13.	Switch the power on to the system and turn on the door isolator.
14.	Press and hold the SET button on the ballast for at least 5 seconds. The screen will display **Profibus**.
15.	Press SET repeatedly until **Modbus** is displayed.
16.	Press and hold SET until the address is displayed. The first digit will be flashing.
17.	Press SET repeatedly until **0** is displayed.
18.	Press and hold SET until the second digit is flashing.
19.	Press SET repeatedly until **1** is displayed.
20.	Press and hold SET until the screen returns to the main display. Check that the display reads **Modbus OK 01h**.
21.	Switch the door isolator off and then on again to apply the new Modbus address.
22.	Reset the counter of remaining ballast cycles in the UV Touch software using screen 14 of the Engineer menu (follow the procedure in Figure 29 on page 75).

5.9. Replacing the Touchscreen Battery

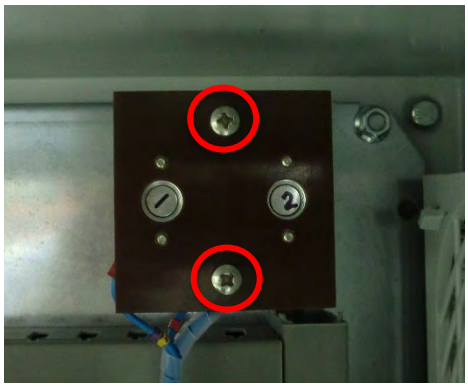
The lithium battery in the touchscreen is used to maintain the system's real-time clock in the event of a power failure.

	Wear an electrostatic discharge wristband when replacing the battery to prevent damage to the sensitive electronic components in the touchscreen. **Portez un bracelet à décharge électrostatique lorsque vous remplacez la batterie pour éviter d'endommager les composants électroniques sensibles de l'écran tactile.**

1.	Switch off the power using the main isolator on the cabinet.
2.	Squeeze the catch on the clear plastic cover on the rear of the touchscreen and pull to open. 
3.	Use a small flat-headed screwdriver to carefully remove the battery from the holder.
4.	Fit a new CR2032 battery with the positive terminal pointing outwards and close the cover.
5.	Switch on the power using the main isolator.
6.	Dispose of the used battery in accordance with the manufacturer's instructions and local regulations.

5.10. Replacing the Cabinet Temperature Sensors

The two temperature sensors located in the top right corner of the cabinet can be replaced if either of them develops a fault.

1.	Remove the two screws holding the mounting plate in position.	
2.	Disconnect the wires from the appropriate sensor terminals on the rear of the mounting plate.	
3.	Unscrew the two screws holding the faulty sensor to the plate. **Note:** Sensor 1 is the warning sensor (50ºC / 122ºF) and sensor 2 is the trip sensor (60ºC / 140ºF).	
4.	Reverse the steps above to fit the new sensor. **Note:** Ensure that the correct wires are connected to each sensor; refer to the appropriate circuit diagram in Appendix C for details.	

5.11. Maintenance of the Cooling Fans / Air Conditioner

Air flow must be maintained through the cabinet to prevent the ballast from overheating. The required frequency of maintenance of the cooling fans or air conditioner will depend on the air quality in the area where the UV system is operating.

1.	Switch off the power using the main isolator on the cabinet.

2. For fan-cooled cabinets, unclip the tops of the covers and pull gently so that they swing open. Lift out the fibre filter and replace with a new filter. Close the cover.



OR

For air-conditioned cabinets, slide the aluminium air filter out of the housing through the slot in the side of the unit (**note the orientation**).



Flush thoroughly with warm water from the exhaust side and allow to drain. Optionally, spray both sides of the filter with RP Super Filter Coat adhesive. Replace the filter inside the housing.



3. Switch on the power using the main isolator.

5.12. Replacing the Surge Protection Module

A surge protector is located to the left of the main power supply terminals in the cabinet to protect the ballast. The unit contains three modules to protect up to three supply phases; in this UV system, only two are used.

If a surge protection module fails, a warning is displayed in the software (section 6.2).

1. Switch off the power using the main isolator on the cabinet.

2.	Open the cabinet and inspect the surge protector to see which module has failed (shown by a red indicator).	
3.	Pull the failed module out of the surge protector and push a replacement module into the vacant position until it clicks into place. **Note:** You can use the right-hand module for a replacement as this is not used for surge protection in these cabinets.	
4.	Close the cabinet door and switch on the power using the main isolator.	

5.13. Spares List

ONLY USE SPARES RECOMMENDED BY THE SYSTEM MANUFACTURER TO AVOID COMPROMISING SAFETY AND PERFORMANCE, AND/OR INVALIDATING THE WARRANTY.

UTILISEZ SEULEMENT LES PIÈCES DE RECHANGE RECOMMANDÉES PAR LE FABRICANT DU SYSTÈME POUR ÉVITER DE COMPROMETTRE LA SÉCURITÉ ET LES RÉSULTATS ET/OU D'INVALIDER LA GARANTIE.

Ref.	Description	Part number
	UV lamps:	
	PQ EO 4, 6, 8, 10	130076-0335 (4 kW)
	PQ EO 12	130076-0500 (6 kW)
	Lamp spacers:	
	PQ EO 4	320292-0245
	PQ EO 6	320292-0280
	PQ EO 8	320292-0350
	PQ EO 10	320292-0430
	PQ EO 12	320292-0510
1	**Quartz thimbles (doped):**	
	PQ EO 4	321180-0857
	PQ EO 6	321180-0890
	PQ EO 8	321180-0980
	PQ EO 10	321180-1060
	PQ EO 12	321180-1301
2	**Service kits (wiped / unwiped):**	
	PQ EO 4	180046-1038 / 180045-1038
	PQ EO 6, 8, 10, 12	180046-0038 / 180045-0038
3	**End plate seals:**	
	PQ EO 4	410032-0842
	PQ EO 6	410032-0853
	PQ EO 8	410032-0860
	PQ EO 10	410032-0866
	PQ EO 12	410032-0870
	Ballasts:	
	PQ EO 4, 6, 8, 10	650065-2400 (4 kW)
	PQ EO 12	650065-2600 (6 kW)
	Wiper parts:	
4	Wiper motor	420010-0001
5	Gear box	420010-0002
	Wiper motor lead	160446-0002/03/10/20/30 (Length in metres)
6	Limit switch	620067
	Limit switch lead	710208-0002
	Accessories / replacements:	
	Lamp terminal block	160458
	Surge protection module	610091-0040
7	Quartz fitting (nipple)	320775
8	UV sensor	680050-2000
	UV sensor lead	710211-0002
9	UV sensor housing service kit	180060-3000
10	Temperature sensor and lead	170033-0001
	Thermal contact pad	321206
	UVShield™ unit (option)	160447-0001
	Filter mats for fan and exhaust filters (fan-cooled cabinets)	750024-5565-01

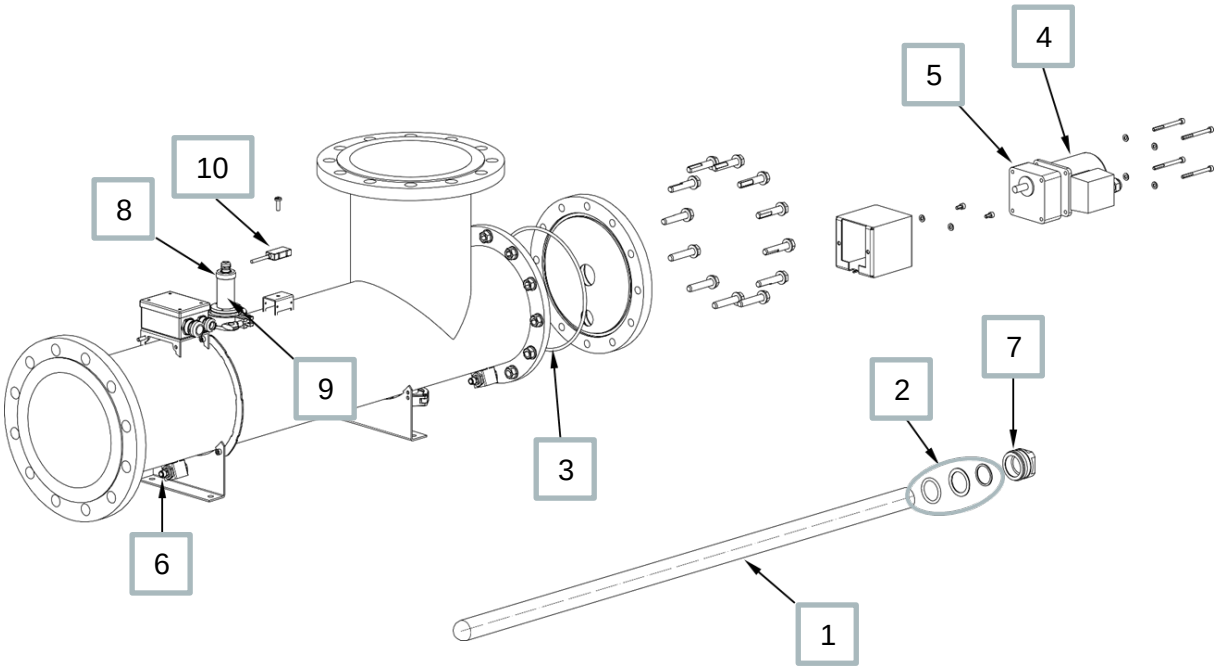
Ref.	Description	Part number
	Replacement cabinet temperature warning sensor (50ºC / 122ºF)	610074-0050-01
	Replacement cabinet temperature trip sensor (60ºC / 140ºF)	610074-0060-01
	Service tools:	
	Spanner	321354
	Quartz removal tool	160431-0038
	Sensor housing window removal tool	321938
	Sensor housing window insert tool	321939
	Wiper bearing service tool	160454-0001
	Wiper bearing retaining nut tool	322103
	Anvil (for bearing removal)	321346

Notes:

Recommended stock holding: 2 lamps, 1 thimble and 1 service kit (replace as used).

Reference numbers refer to exploded drawings in Figure 27, Figure 30, Figure 33 and below.

The service kit (ref. 2) contains all the components needed to renew the quartz thimble seals and the parts on the auto-wiper (if fitted).



5.14. Clean-in-Place

A Clean-In-Place (CIP) process can be run on the plant without bypassing the UV system, providing that the CIP function has been enabled. When the system is in CIP Cycle mode:

- The wiper is parked and disabled if a wipe cycle is in progress.
- Temperature alarms are disabled.
- The bleed valve (if fitted) will not operate (even if the chamber temperature exceeds the bleed temperature).
- The lamp is switched off.

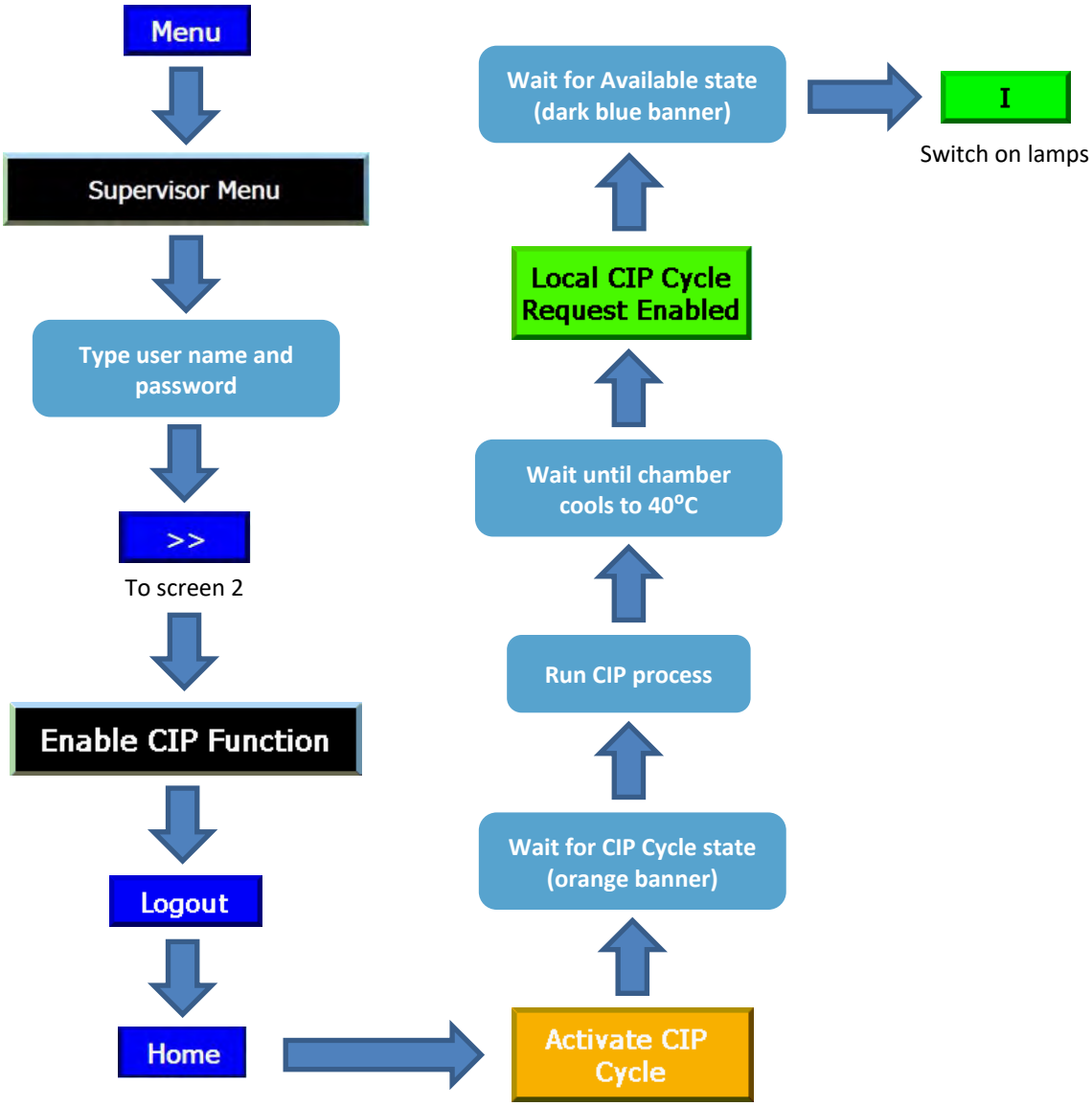
To run a CIP process in Local mode using the control panel, follow the procedure shown in Figure 36.

You can also use the digital inputs and outputs to control the system when performing CIP (in either Local or Remote mode):

1. Tap the **Enable CIP Function** button in either screen 2 of the Supervisor menu or screen 8 of the Engineer menu.
2. Send a HIGH (closed) signal to the Remote CIP Inhibit interface terminal.
   Wait until the system state changes to CIP Cycle and the OK to CIP digital output switches to HIGH (closed).
3. When the OK to CIP signal is received by your master control system, run the Clean-In-Place process in the plant.
4. Once CIP is completed, allow the UV system to cool down to below 40ºC and then switch the Remote CIP Inhibit signal to LOW (open).
   The system will return to the Available state, and the OK to CIP output will be switched off.
5. Restart the lamp either using the buttons on the Home screen (in Local mode) or the appropriate Remote signal (continuous or pulsed); refer to section 4.3.

Refer to Table 1 in section 3.5 and the drawings in Appendix C for details of the input and output connections.

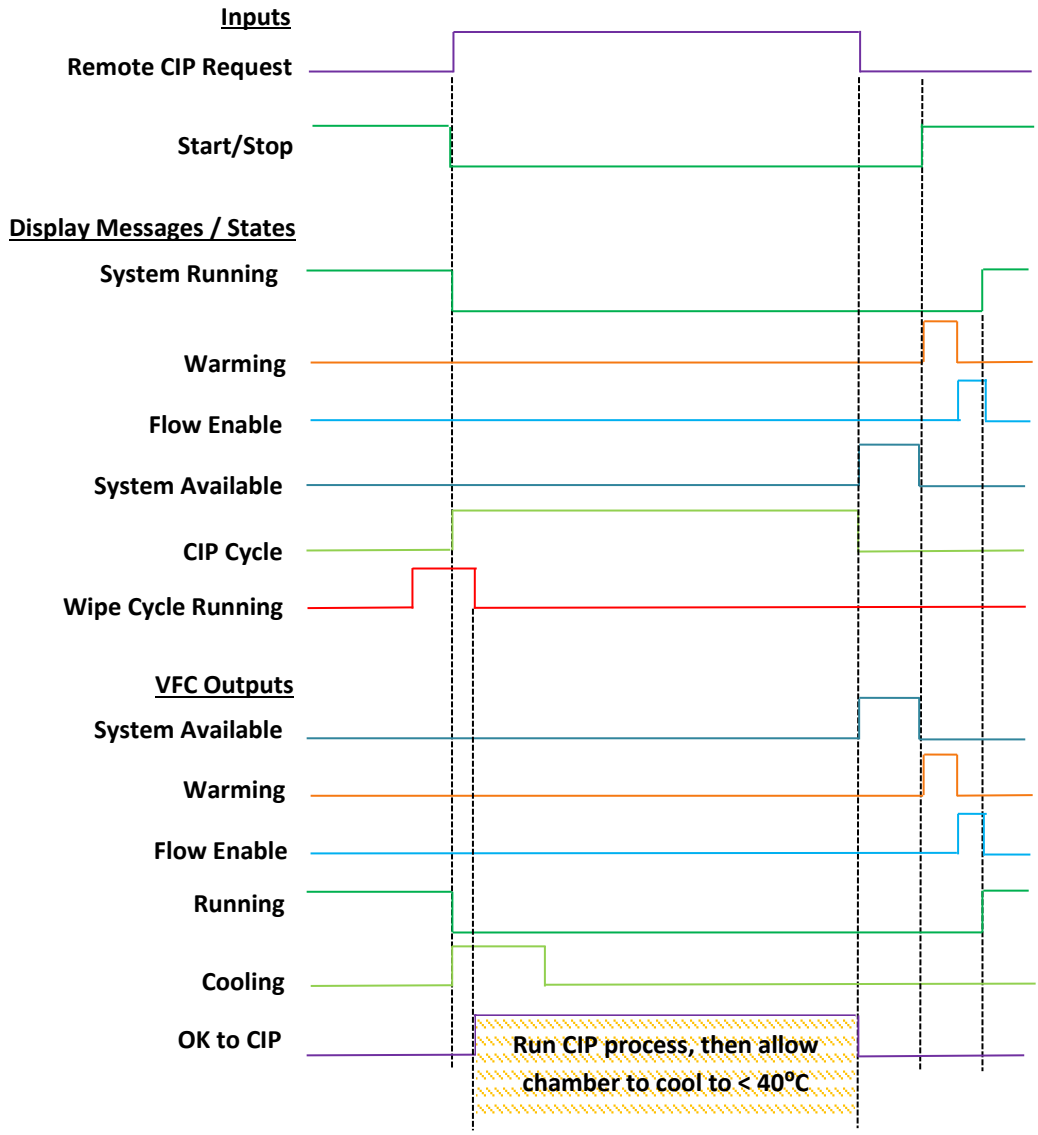
Figure 36: CIP Procedure (Local Mode)



The inputs, outputs and system states during CIP are illustrated in Figure 37.

For guidelines and recommendations for performing CIP on your UV system, refer to the guidance from your supplier (p/n 190103-XXXX-ISS).

Figure 37: Clean-in-Place Timing Diagram



6. Alarms and Fault-Finding

If you encounter problems when operating your UV system, use the flowcharts on the following pages to identify what steps to take to find the cause of the problem. For the shaded steps, refer to the corresponding section in the fault-finding table on page 118. Instructions to clean and replace components are described in the Maintenance section of this manual.

	**ENSURE ELECTRICAL POWER IS ISOLATED BEFORE REMOVAL OR REPLACEMENT OF ANY COMPONENTS. REFER TO THE SAFETY SECTION.** **ASSUREZ-VOUS QUE LE COURANT ÉLECTRIQUE EST ISOLÉ AVANT ENLÈVEMENT OU REMPLACEMENT DES TOUS ÉLÉMENTS. CONSULTEZ LA SECTION SÉCURITÉ.**
	**ELECTRICAL MEASUREMENTS MUST BE CARRIED OUT BY QUALIFIED PERSONNEL.** **MESURES ÉLECTRIQUES DOIVENT ÊTRE EFFECTUÉES PAR DU PERSONNEL QUALIFIÉ.**

6.1. Fault Flowcharts

Figure 38 gives fault-finding options for several faults that may occur when switching on the system after installation. Figure 39 gives the steps that should be followed if a warning or trip alarm is displayed when the system is switched on. Figure 40 gives a procedure to isolate and correct the cause of a low or failed UV intensity warning.

Figure 38: System Fault Flowchart

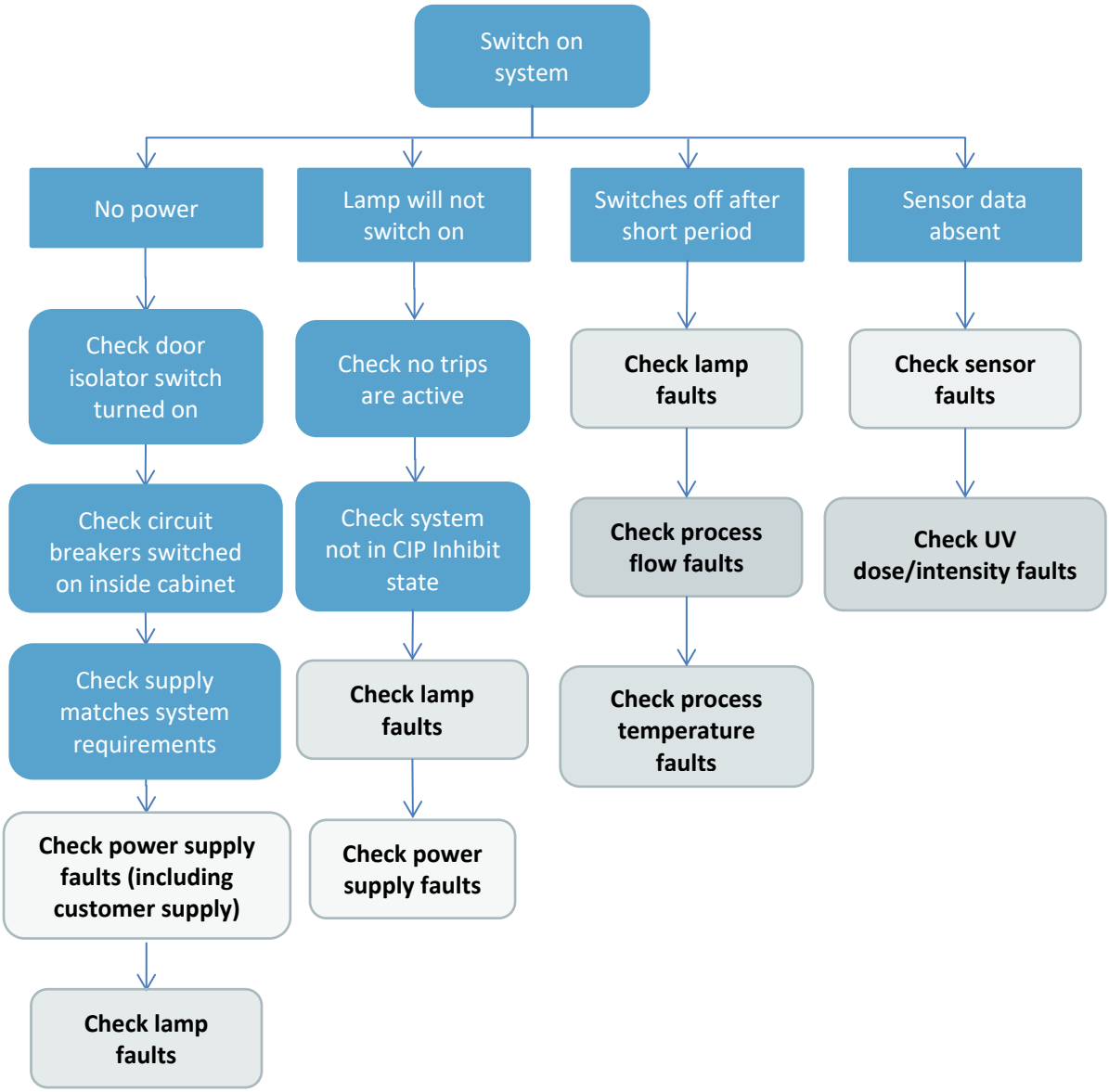


Figure 39: System Warnings and Trips Flowchart

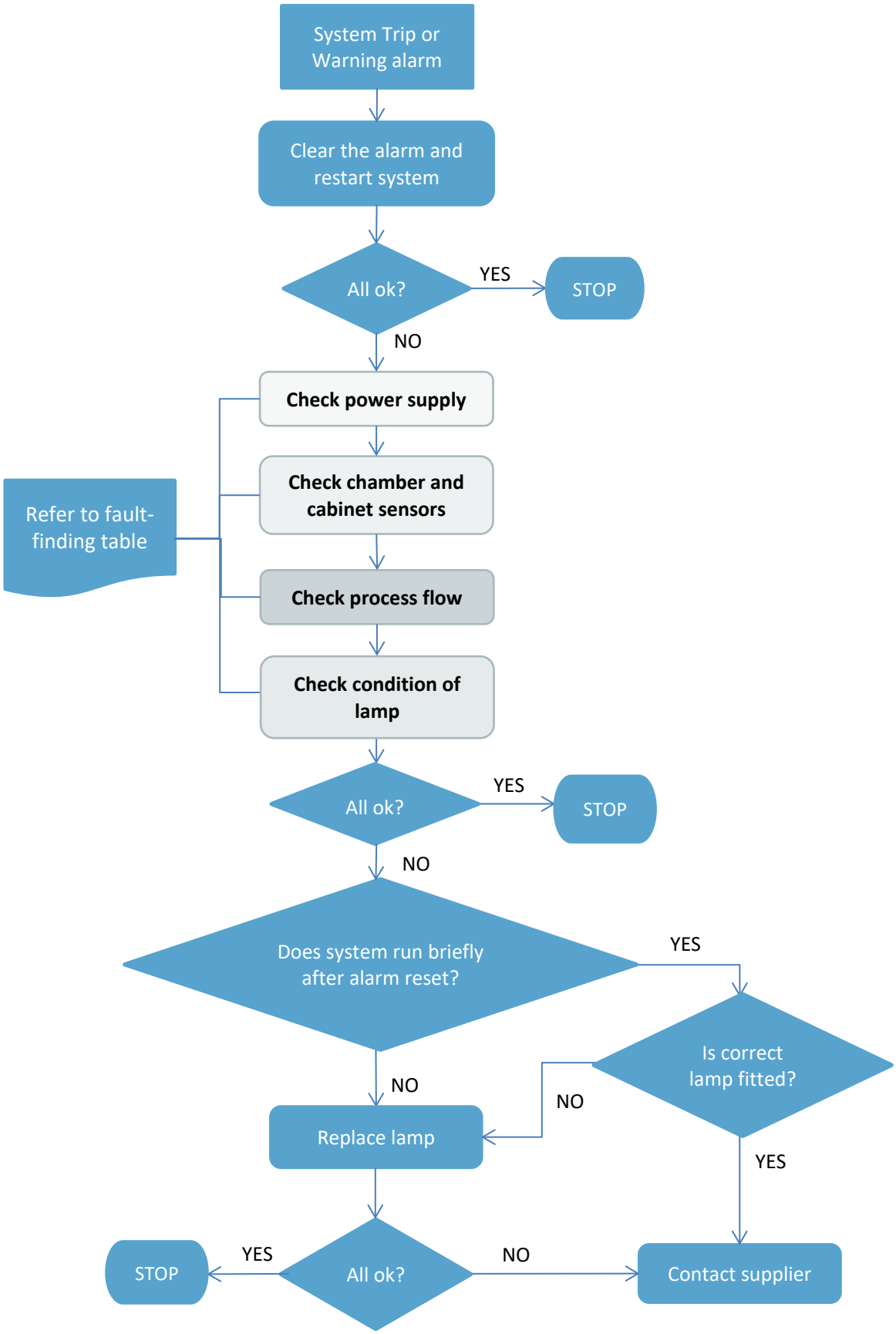
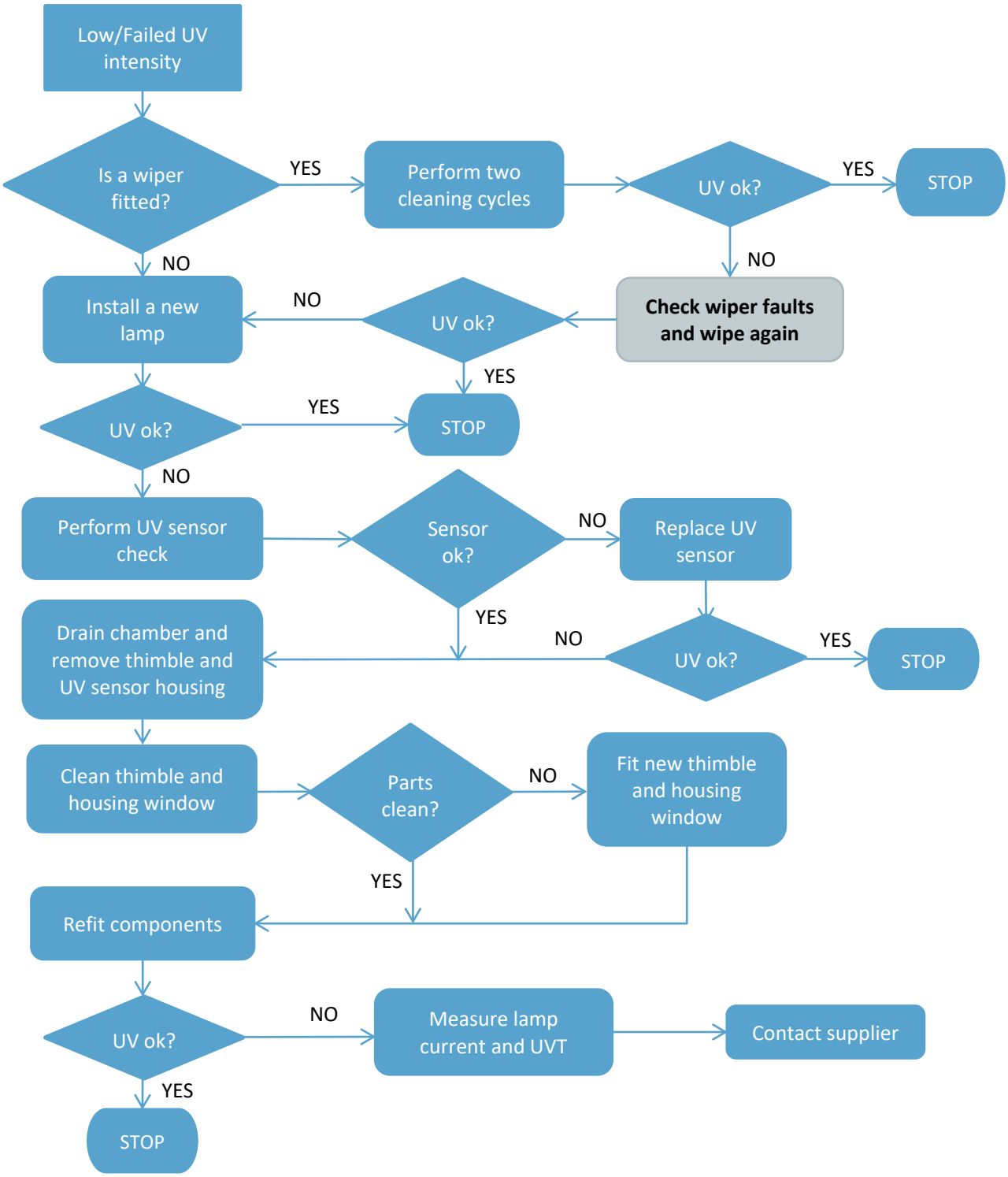


Figure 40: Low/Failed UV Intensity Fault Flowchart



6.2. System Alarms

Table 3 lists the common alarms and possible causes, together with advice on what to check to confirm the cause and suggested actions to rectify the problem.

Table 3: Fault-Finding Table

Alarm Message	Cause	Check	Action
[Software does not load.]	Control software corrupted.		Inform supplier representative of software malfunction.
[Earth leak.]	Live wire connected to earth.	Live wiring circuits in system.	Rectify wiring fault.**
Lamp cover removed [option]	UVShield™ unit at lamp connector loose or removed (lamp power cut off).	UVShield™ unit is in place and firmly connected to chamber. Connections to junction box.**	Tighten screws on UVShield™. Check terminal connections.**
[Lamp will not switch on (no alarm shown).]	System is in CIP Cycle state.	CIP Cycle displayed on control panel.	Remove CIP Request signal once CIP completed (section 5.14).
[Display screen not lit.]	Loss of 24 V supply.	Electrical mains supply is on.	Switch on electrical supply.
		If a circuit breaker is open.**	Reset circuit breaker.
		24 V power supply and connection to controller.	Replace power supply/wiring connection.
		If isolator is off.	Switch on isolator.
Warning: surge protection failure	Failure of surge protection module.	Red indicator shown on surge protector next to mains power terminals.	Replace failed module (section 5.12).
Analogue input <4mA (Temperature/ Intensity/Flow/ Transmittance)	Loss of signal from temperature/UV/flow/ transmittance sensor.	Sensor is connected correctly. Continuity of sensor lead.	Ensure lead is fully connected to the sensor. Ensure terminal connections in control are wired correctly.** Replace lead.
	System configuration.	Sensor setup in Engineer menu.*	Set up sensor input in Engineer menu (section 3.6).*
Water leak detected [option]	Chamber water leak detected (if leak detector is installed).	Condition of quartz thimble and seals with lamp removed.	Replace quartz thimble and seals (section 5.4).
Lamp change due soon	Lamp run hours approaching recommended limit.		Replace UV lamp (section 5.3).
Lamp failed to warm up OR Power failure	Lamp failure.	UV lamp (premature failure of the lamp is the most likely cause of a lamp fault). Connection to lamp and from lamp to the control.	Ensure lead is fully connected to the UV lamp. Replace UV lamp (section 5.3) and/or lead as necessary.
	Ballast failure.	Supply to ballast.	Replace ballast.
	Contactor failed to energise.	Contactor has energised.	Replace contactor.
Quartz change due soon	Quartz run hours approaching recommended limit.		Replace quartz thimble (section 5.4).

Ballast supply fault (K1 not energised)	Critical fault preventing main contactor from being energised.	Check all alarm messages displayed for faults with lamp, wiper, UV Shield (if fitted), temperature safety switch.	Only replace ballast once other causes ruled out.
*All ballast messages****	Refer to section 6.3.		
Temperature > warning level	Loss or low flow of process liquid.	Flow rate/restriction in flow.	Check fault clears with improved conditions.
	Process liquid getting hot.	Process.	
Temperature > trip level	Loss or low flow of process liquid.	Flow rate/restriction in flow.	Correct flow/process conditions.
	Process liquid too hot.	Process.	
	Failure of bleed valve (if installed).	Bleed valve connection.**	Replace failed bleed valve/connection.
Temperature safety switch open	Loss or low flow of process liquid.	Flow rate/restriction in flow.	Correct flow/process conditions.
	Process liquid too hot.	Process.	
	Loss of temperature sensor signal.	As for “Loss of signal” in “Analogue Input <4mA” alarm.	As for “Loss of signal” in “Analogue Input <4mA” alarm.
	Failure of bleed valve (if fitted).	Bleed valve connection.**	Replace failed bleed valve/connection.
Warning: Cabinet overtemp	Cabinet internal temperature greater than 50ºC / 122ºF.	Cabinet ventilation system (cooling fan or air conditioner).	Remove and clean fan and vent filters (replace if necessary). Check ventilation system operating correctly (air should be blown **into** the cabinet). Adjust limit temperature inside cabinet (if air conditioner installed).
Trip: Cabinet overtemp	Cabinet internal temperature greater than 60ºC / 140ºF.		
Warning: Low Dose (RED)	Low UV.	Cleanliness of quartz thimble, UV sensor quartz and sensor housing window.	Clean quartz components (section 5.4.2).
		Age and condition of UV lamp.	Replace UV lamp (section 5.3).
		UV sensor is operating correctly.	Perform UV sensor verification (section 5.5.5).
	Calculated dose (VRED) below alarm set point.	Alarm set point in Engineer menu.*	Reset alarm set point in Engineer menu.*
	Change in constituents in the process stream.	Process conditions, e.g., UV transmittance of process liquid.	Check fault clears with improved conditions.

Trip: Low Dose (RED)	Low UV.	Cleanliness of quartz thimble, UV sensor quartz and sensor housing window.	Clean quartz components (section 5.4.2).
		Age and condition of UV lamp.	Replace UV lamp (section 5.3).
		UV sensor is operating correctly.	Perform UV sensor verification (section 5.5.5).
	Calculated dose (VRED) below alarm set point.	Alarm set point in Engineer menu.*	Reset alarm set point in Engineer menu.*
	Failure of signal from UV sensor.	UV sensor and sensor housing window for cleanliness.	Clean UV sensor and housing window (section 5.5.2).
		As for “Analogue Input <4mA” above.	As for “Analogue Input <4mA” above.
Warning: Low Dose (Ave)	Low UV.	Cleanliness of quartz thimble, UV sensor quartz and sensor housing window.	Clean quartz components (section 5.4.2).
		Age and condition of UV lamp.	Replace UV lamp (section 5.3).
		UV sensor is operating correctly.	Perform UV sensor verification (section 5.5.5).
	Calculated average dose below alarm set point.	Alarm set point in Engineer menu.*	Reset alarm set point in Engineer menu.*
	Failure of signal from UV sensor.	UV sensor and sensor housing window for cleanliness.	Clean UV sensor and housing window (section 5.5.2).
		As for “Analogue Input <4mA” above.	As for “Analogue Input <4mA” above.
Trip: Low Dose (Ave)	Low UV.	Cleanliness of quartz thimble, UV sensor quartz and sensor housing window.	Clean quartz components (section 5.4.2).
		Age and condition of UV lamp / UV sensor.	Replace UV lamp (section 5.3) / UV sensor (section 5.5).
		UV sensor is operating correctly.	Perform UV sensor verification (section 5.5.5).
	Calculated average dose below alarm set point.	Alarm set point in Engineer menu.*	Reset alarm set point in Engineer menu.*
	Failure of signal from UV sensor.	UV sensor and sensor housing window for cleanliness.	Clean UV sensor and housing window (section 5.5.2).
		As for “Analogue input <4mA” above.	As for “Analogue Input <4mA” above.
Flow exceeds maximum design value	Process flow too high.	Flow conditions.	Reduce process flow.
	System configuration.	Loaded recipe and flow meter 4 mA and 20 mA settings in Engineer menu.*	Load correct recipe in Engineer menu (section 4.11).* Enter correct calibration values for sensor (section 3.6).
	Loss of flow signal.	Flow meter connections.**	Replace flow meter/cable if required.

Transmittance below minimum design value	Transmittance reduced due to change in process stream constituents.	Check process conditions, e.g., transmittance of process liquid.	Check fault clears with improved conditions.
	System configuration.	Loaded recipe and UVT meter 4 mA and 20 mA settings in Engineer menu.*	Load correct recipe in Engineer menu (section 4.11).* Enter correct calibration values for sensor (section 3.6).
Wiper service due soon	Wiper approaching count limit.		Service the wiper (section 5.7).
Wiper failed to reach/leave home limit switch -B1 OR Wiper failed to reach/leave away limit switch -B2	Faulty drive motor.	Electrical short / open circuit.	Check wiring to motor.**
	Wiper seized.	Wiper bearing and leadscrew for pitting or abrasions. Wiper runs freely in and out of bearing seal housing.	Replace as necessary (section 5.7).
	Faulty limit switch.	Limit switch is fitted to chamber. Wiring is terminated correctly.**	Test operation by placing magnet next to the limit switch body. Replace if necessary (section 5.7).
Communication lost to Primary/SCADA	Exchange of data between PLC and primary controller or SCADA system has failed.	Communications cables between control cabinet and primary controller or SCADA system.	Check or replace cables.
Data logging: SD card not installed	Data logging enabled but SD card not inserted into port.		Insert SD card (section 4.10).
Data logging: SD card access error	Data logging enabled but SD card malfunctioning.		Replace SD card (section 4.10).

* Password required to access the menu. Refer to the warning in section 2.1.
** Refer to drawings in Appendix C.
*** Check ballast display and refer to section 6.3 for further information.

6.3. Ballast Troubleshooting

Table 4 lists the message that may be displayed by the ballast and gives details of possible causes. This information should be relayed to your supplier's Technical Support representative.

Table 4: Ballast Display Messages

Message	Description	Actions
ELC Ready	Ballast is ready for operation.	
[Display off]	No power supply to ballast. Display defect.	Check cable between cabinet and ballast.
DC Link Undervoltage Error	Internal DC link voltage is low.	Check that circuit breakers are closed. Check that main contactor has correct voltage at inputs and outputs. Check internal ballast fuses. Check supply voltage.
Mains Voltage Error	Mains voltage drop or fluctuation detected.	Check supply to ballast.
Fan Error	Fan has stopped functioning.	
Overtemperature Error	Ambient or internal ballast temperature too high.	
Internal Error	Communication failure between internal components.	Switch system off and on using door isolator.
	Power switches not functioning in zero voltage switching mode.	
	Internal low voltage power supply fault.	Check cable between cabinet and ballast.
Over curr.	Internal overcurrent.	Check lamp and cable connections for short-circuit.
Measurement	Internal measurement error.	
Bus Error	Bus connection has failed.	
X6CA Failure	Internal communication has failed.	
Overvoltage Error	Lamp voltage too high.	Check connections to lamp.
Lamp Voltage High Error	Lamp voltage exceeds critical threshold.	Check lamp type and connections; replace lamp.
Earth Fault Error	Low insulation resistance in lamp circuit.	Check connections to lamp.
Lamp Power Low Error	Lamp has stopped.	Check connections to lamp; replace lamp.
No Modbus	Modbus connection not detected.	

6.4. System Event Sequence

Figure 41 displays the sequence of events in the system during start up and shut down under normal operating conditions, and in the case of a reduction in measured UV intensity (UVI) below the system set point (SP). It can be used to help with fault-finding and to check that inputs and outputs are working correctly.

Figure 41: Timing Diagram for Normal Operation and Low UV Events (not to scale)

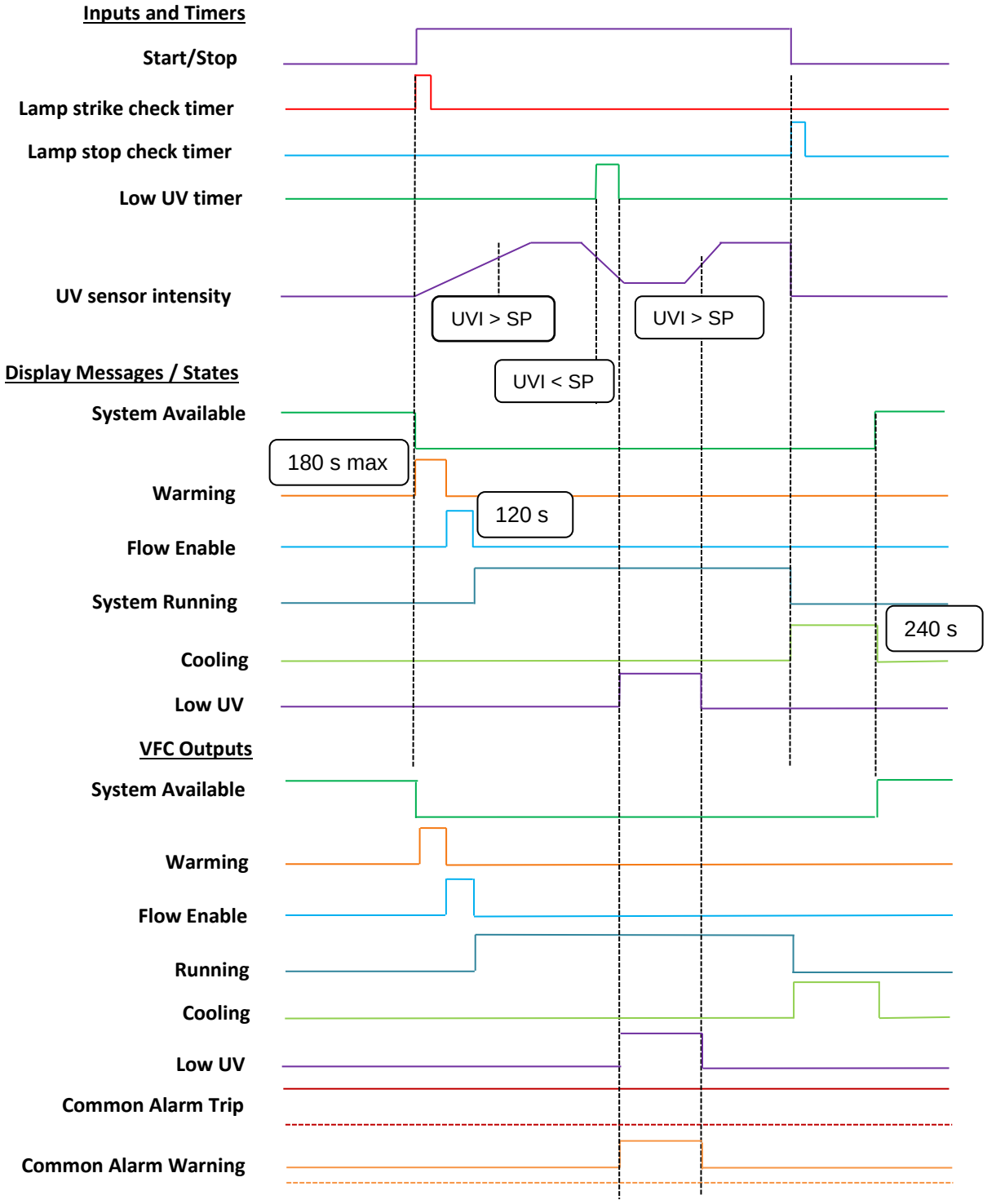
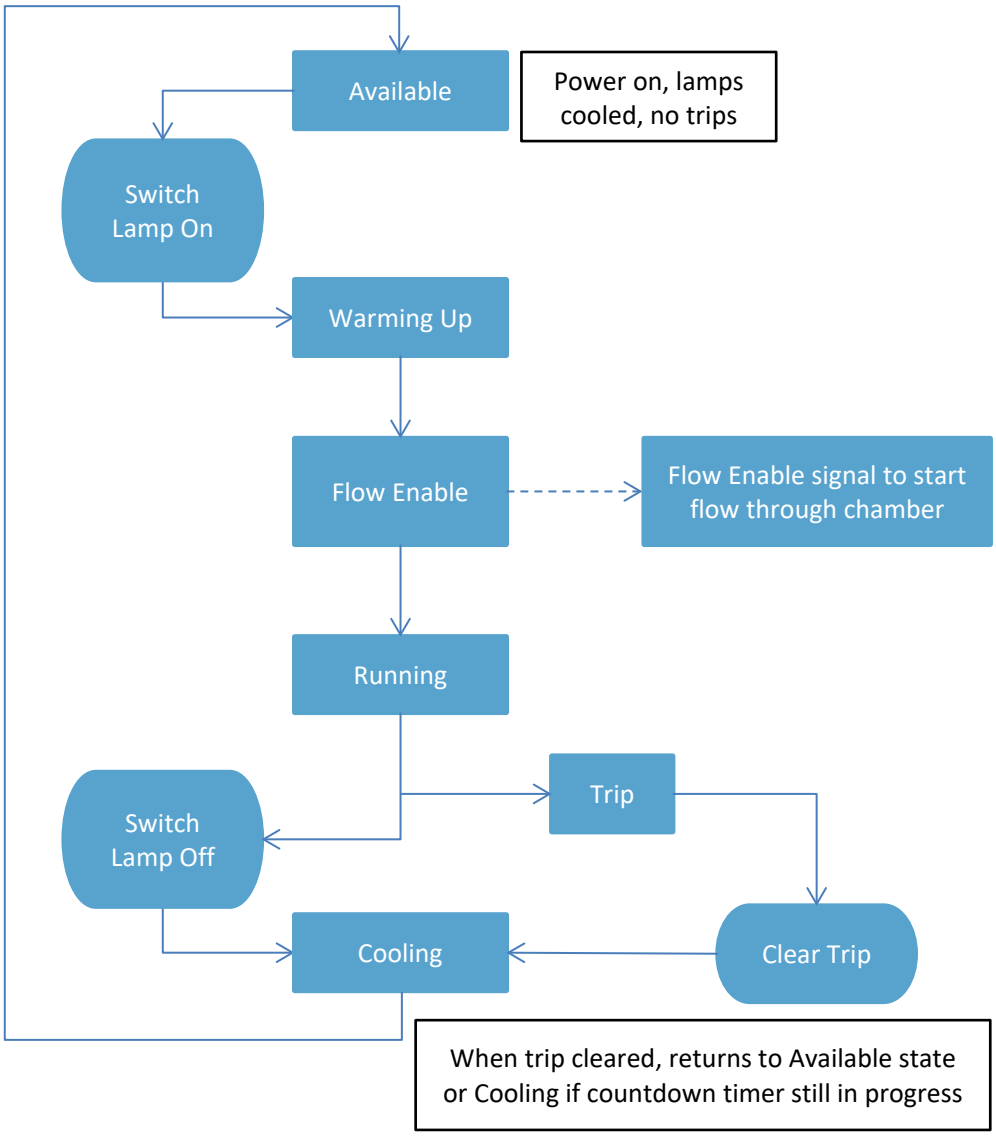


Figure 42 shows the sequence of system states during operation.

Figure 42: Operational State Sequence



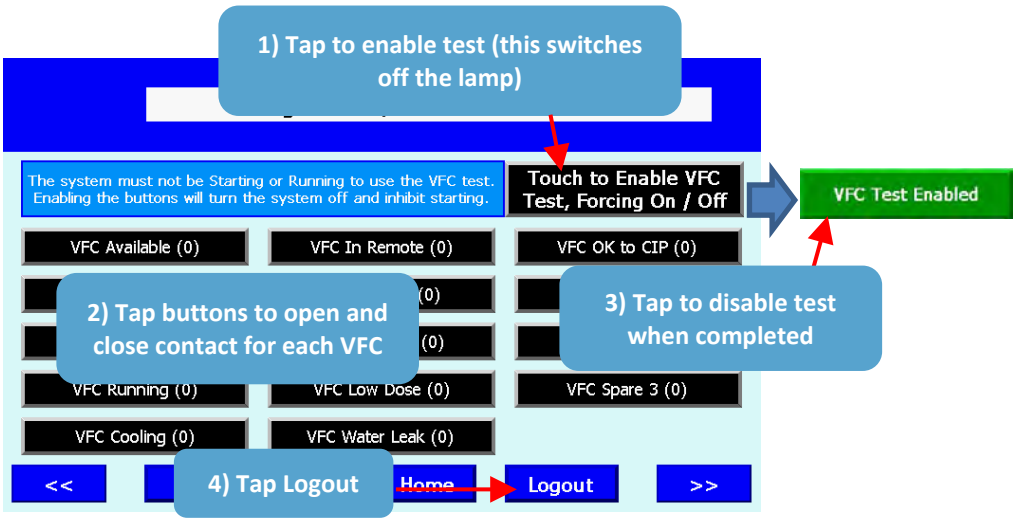
6.5. Testing Volt-Free Contacts

The UV system's volt-free contacts can be tested by opening and closing each contact individually using screen 15 of the Engineer menu. Use this to troubleshoot communication problems with the system by changing the state of each contact in turn.

Follow the procedure in Figure 43. Enabling this feature will switch off the lamp and prevent it being restarted.

Figure 43: Procedure for Testing VFCs

Follow the procedure in Figure 8 to enter the Engineer menu and scroll to screen 15.



Note: Leaving the screen will disable the VFC test and allow the lamp to be switched on.

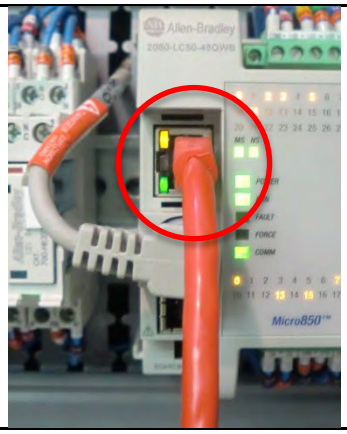
7. Ethernet Communications

A site control (SCADA) system can retrieve information from the UV system using an Ethernet connection. This information can be used to recreate the software displays at another location, or as data for logging and trending in a process control system. When in Remote Comms mode, the SCADA system can also switch the lamp on and off and perform several other operations.

7.1. Establishing Communication

The control system must be configured to communicate with the cabinet control PLC via a local area network (LAN):

1. Connect the PLC to the LAN using the RJ-45 Ethernet port available inside the cabinet.
   The IP address of the PLC is **192.168.10.88**.



7.2. Sending and Receiving Data

Data can be exchanged using Ethernet or Modbus TCP/IP communications, as shown in Table 5. When using Modbus, the holding register addresses shown in the table are used to read or write the data. Data can be read by using the 'read holding registers' function code (3); commands are sent using the 'write holding registers' function code (16). All data is formatted as DINT 32-bit signed data.

Table 5: Ethernet Tags and Modbus Addresses

Tag Name	Description	Modbus Addresses	Notes
SCADA_Data_Read[0].0	Digital output for safety shutdown	400200-1	
SCADA_Data_Read[0].1	System in Remote mode (VFC/Pulsed/Comms)		
SCADA_Data_Read[0].2	System in Available state		
SCADA_Data_Read[0].3	System in Warming state		
SCADA_Data_Read[0].4	System in Flow Enable state		
SCADA_Data_Read[0].5	System in Running state		
SCADA_Data_Read[0].6	System in Cooling state		
SCADA_Data_Read[0].7	System tripped (fail-safe)		
SCADA_Data_Read[0].8	System water leak (fail-safe)		
SCADA_Data_Read[0].9	System warning		
SCADA_Data_Read[0].10	System low UV dose warning		
SCADA_Data_Read[0].11	Bleed output		
SCADA_Data_Read[0].12	OK to CIP signal		
SCADA_Data_Read[0].13	Wiper moving forward		

SCADA_Data_Read[0].14	Wiper reversing		
SCADA_Data_Read[1]	UV transmittance	400202-3	Values are multiplied by 100
SCADA_Data_Read[2]	Flow	400204-5	
SCADA_Data_Read[3]	UV intensity	400206-7	
SCADA_Data_Read[4]	Temperature	400208-9	
SCADA_Data_Read[5]	Average dose	400210-11	
SCADA_Data_Read[6]	Validated RED	400212-13	
SCADA_Data_Read[7]	RED	400214-15	
SCADA_Data_Read[8]	Log reduction achieved	400216-17	
SCADA_Data_Read[9]	System Current State: 1 = Available 2 = Warming 3 = Flow Enable 4 = Running 5 = Cooling 6 = Tripped 7 = Cooling Down (Power-up) 8 = CIP	400218-19	
SCADA_Data_Read[10]	Start/stop mode: 1 = Local 2 = Timed 3 = Remote VFC 4 = Remote Pulsed 5 = Remote Comms	400220-21	
SCADA_Data_Read[11]	Warming up time remaining	400222-23	Values are multiplied by 100
SCADA_Data_Read[12]	Flow Enable time remaining	400224-25	
SCADA_Data_Read[13]	Cooling down time remaining	400226-27	
SCADA_Data_Read[14]	Ballast target power for lamp	400228-29	
SCADA_Data_Read[15]	Actual ballast power	400230-31	
SCADA_Data_Read[16]	Actual ballast voltage	400232-33	
SCADA_Data_Read[17]	Actual ballast current	400234-35	
SCADA_Data_Read[18]	Hours remaining until lamp change due	400236-37	
SCADA_Data_Read[19]	Lamp switch on cycles	400238-39	
SCADA_Data_Read[20]	Hours remaining until quartz sleeve change due	400240-41	
SCADA_Data_Read[21]	Hours remaining until wiper service due	400242-43	
SCADA_Data_Read[22]	Wipe cycles	400244-45	
SCADA_Data_Read[23]	Hours remaining until ballast change due	400246-47	
SCADA_Data_Read[24]	Ballast switch on cycles	400248-49	
SCADA_Data_Read[25]	Communication counter reading from PLC to Primary/SCADA	400250-51	
SCADA_Data_RW[0].0	Start/stop the system	400400-1	
SCADA_Data_RW[0].1	Reset alarms		
SCADA_Data_RW[0].2	Set reduced dose		
SCADA_Data_RW[0].3	Send CIP request		
SCADA_Data_RW[1]	Set flow value	400402-3	Values must be multiplied by 100
SCADA_Data_RW[2]	Set UV transmittance value	400404-5	
SCADA_Data_RW[3]	Set minimum RED setpoint	400406-7	
SCADA_Data_RW[4]	Set minimum reduced RED setpoint	400408-9	
SCADA_Data_RW[5]	Set minimum AVG setpoint	400410-11	

SCADA_Data_RW[6]	Set minimum reduced AVG setpoint	400412-13
SCADA_Data_RW[7]	Set temperature warning value	400414-15
SCADA_Data_RW[8]	Set bleed temperature value	400416-17
SCADA_Data_RW[9]	Set temperature trip value	400418-19
SCADA_Data_RW[10]	Communication counter reading from Primary/SCADA to PLC	400420-21

8. Effect of UV Light on Process Liquid

8.1. Ultraviolet Light

Ultraviolet light is electromagnetic radiation of wavelength 10 to 400 nm. It lies beyond the violet end of the visible spectrum and precedes the X-ray band.

The UV region is further divided into four sub-bands:

- UV-A 315–400 nm
- UV-B 280–315 nm
- UV-C 200–280 nm
- Vacuum UV 10–200 nm

The UV-C light produced by the system will disrupt an organism's DNA (or RNA in the case of a virus), thereby preventing replication and rendering the organism harmless.

8.2. Intensity

The intensity of the UV light reaching each organism depends on the UV output of the lamp, the age of the lamp and thimble, and the transmittance of UV light through the process liquid. The intensity decreases over time as the lamp ages. Transmittance is reduced as the turbidity of the liquid increases, and also as the concentration of any contaminant which can absorb UV radiation increases.

The lamp is sized for the required duty, based on its expected end-of-life performance. However, if the properties of the process stream change or the quartz thimble in the UV chamber gets coated with contamination, the performance of the system may prematurely decline. The automatic wiping mechanism for the quartz thimble will help maintain system performance under these fouling conditions.

The UV sensor supplied with the system provides continuous data to the control panel for monitoring system performance. This will indicate a fault that might mean the UV lamp requires changing, the quartz thimble needs cleaning, or the properties of the process liquid have changed.

8.3. Dose

The dose (or fluence) is the product of the intensity of the light incident on the sample and the exposure time. It is commonly expressed in units of mJ/cm^2. The dose is related to the inactivation effect; the value required to achieve a reduction in the number of organisms varies with the species concerned.

The exposure of the process liquid to the UV radiation is a function of the time it takes for the process liquid to pass through the treatment chamber, and the design of the chamber. The innovative design of the chamber was optimized using computational fluid dynamics (CFD). The UV system has been selected based on the process liquid flow rate, the concentration and type of organism that may be present and the nature of the process liquid, to ensure that the appropriate dose of UV light is achieved.

9. Lamp / Quartz Breakage During Operation

In the unlikely event that a quartz sleeve fails during operation, an earth leak alarm or (if the option is included in your system) a water leak alarm will be displayed on the control panel. If the lamp breaks as a result, then a lamp failure or low UV alarm will also be triggered. Although sudden quartz and lamp breakages are rare, we recommend that a water leak alarm in particular is treated as a potential source of contamination of your process.

If you do not have an automatic divert valve[1] in your process, the lamp and quartz materials could be released into the process stream. In this case, please take the following steps as soon as a failure is identified:

1. Shut down the process.
2. Quarantine any product that has been made with water treated by the UV since the failure occurred.
3. Isolate and drain the UV chamber.[2]
4. Open the chamber and inspect the failed lamp and quartz sleeve/thimble.
5. Ascertain the extent of the damage and material loss. Photographic records are always useful for future review.

Take the following steps to place the system back into service:

1. Drain the process pipework, tanks, chambers etc.[2]
2. Flush out all areas of the process downstream from the UV chamber at high velocity.[2]
3. Examine the pipework, especially any areas where debris could be trapped (valves, fittings, low spots, filters, etc.) to ensure that all quartz pieces have been removed.
4. If applicable, run a CIP process on the affected pipework. Follow any other local regulations for cleaning the pipework in your process.
5. Once you have satisfied yourself that the system is clean, perform a rinse and test the liquid to check that it complies with your water quality requirements.[3]
6. Investigate the cause of the failure and take measures to prevent a reoccurrence.
7. Check that service replacement schedules for quartz sleeves and lamps have been followed. Old quartz can become brittle and could be the root cause of a failure.
8. Contact your supplier for further advice if necessary.
9. Order the spare parts required to repair the UV chamber from your supplier. Remember to replace all electrical parts that have been exposed to the process liquid to prevent future issues.
10. Once the parts are available, contact your supplier to arrange for a service engineer to visit and recommission the system.

Notes:

1. With high risk processes where the product cannot be quarantined or is intended for immediate consumption, consider installing with an automatic dump valve downstream that is activated by the low UV or water leak alarm signals. There needs to be sufficient pipe length between the UV chamber and dump valve to ensure that the valve diverts the flow before contaminated water reaches the valve. This length will depend on the process flow velocity and the time taken for the valve to respond and operate.
2. Water drained or flushed from the system may contain traces of quartz and/or mercury. Waste should be handled according to your local regulations.
3. Particular attention should be paid to quartz particles and mercury. A medium-pressure lamp contains ~150 mg of mercury and an amalgam lamp contains ~15 mg.

10. Decommissioning

1.	Ensure the system is thoroughly clean. We recommend that a CIP cycle is run where possible (section 5.14).
2.	Perform an assessment of the safest method of supporting and lowering the items while they are detached from the supporting fixtures. Use suitable and properly maintained lifting equipment operated by trained personnel.
3.	Dispose of the components according to local regulations. Pay particular attention to the UV lamp which contains a small quantity of mercury. On request, your supplier representative can supply details of suitable disposal companies.

10.1. Temporarily Suspending Operation

If you intend to suspend operation of the UV treatment system *and switch off the flow of liquid through the chamber* for more than a few days, it is important to follow the steps below to reduce the chance of corrosion occurring inside the chamber and avoid faults when the system is restarted.

	**It is not recommended to leave the UV chamber standing full of stationary water, as this can cause corrosion and fouling to begin more rapidly than when water is flowing. The risk depends on many factors which are specific to each application. Please assess the risk as part of your system's maintenance procedures and contact your system supplier if you have any questions.**

Before the system is shut down:

1.	Isolate and completely drain the UV chamber.
2.	Flush the chamber out thoroughly with clean water. This is especially important if you are treating contaminated water (such as in wastewater systems) or water containing additives (such as in swimming pool systems).
3.	Allow the chamber to dry.

Before refilling the chamber and restarting the system:

1.	Inspect the inside of the chamber to check for any fouling of surfaces by limescale or other deposits. Pay particular attention to the quartz sleeves/thimbles, the UV sensor housing and the auto-wiper mechanism (if fitted).
2.	If there are any signs of fouling on the surfaces, we recommend that the chamber is given a complete service to clean or, if necessary, replace components BEFORE the system is switched on. Failure to do so may cause the system to malfunction when it is restarted.
3.	If the ETA Plus ballasts in the cabinet have been unused for more than 2 years, elevated leakage currents may arise on the internal electrolytic capacitors which can damage the ballast. As a preventative measure, apply 115–230 VAC to the ballast main supply for one hour to reform the capacitors BEFORE reconnecting the normal power supply and operating the ballast.

Appendix A: System Record Table

Use this table to record lamp changes, modifications, updates, etc.

Date	Action

Appendix B: Declarations of Conformity

FM29365

EU DECLARATION OF CONFORMITY

We:

Hanovia Ltd (t/a Nuvonic), 780 Buckingham Avenue, SLOUGH, Berkshire SL1 4LA, United Kingdom

(EU Address: Berson Milieutechniek B.V., Tarasconweg 2, Unit A1.29 / Postvak 25, 5627 GB EINDHOVEN, Netherlands)

declare under our sole responsibility that the products listed below:

Ultraviolet Fluid Treatment Systems PureLine PQ EO 4/6/8/10/12

to which this declaration relates are in conformity with the provisions of the EU Electromagnetic Compatibility (EMC) Directive 2014/30/EU and comply with the following standards:

EN 61326-1:2013	Electrical equipment for measurement, control and laboratory use – EMC requirements
EN 61000-3-2:2019	Limits for harmonic current emissions (equipment input current < 16A per phase)
EN 61000-3-3:2013 + A1:2019	Limitation of voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current < 16A

The products are also in conformity with the provisions of the EU Low Voltage Directive 2014/35/EU and comply with the following standard:

EN 61010-1:2010 + A1:2019	Safety requirements for electrical equipment for measurement, control and laboratory use

The products are also in conformity with the provisions of the EU Pressure Equipment Directive 2014/68/EU and comply with the following standards:

EN 13445-1:2014 + A1:2014	Unfired pressure vessels - Part 1: General
EN 13445-2:2014 + A3:2018	Unfired pressure vessels - Part 2: Materials
EN 13445-3:2014 + A8:2019	Unfired pressure vessels - Part 3: Design
EN 13445-4:2014	Unfired pressure vessels - Part 4: Fabrication
EN 13445-5:2014 + A1:2018	Unfired pressure vessels - Part 5: Inspection and testing

Issued: 1st April 2023, Slough, UK

Signed for and on behalf of Hanovia Ltd (t/a Nuvonic)

Andy Sidwell – Technical Director

formerly Aquionics, Berson, Hanovia and Orca GmbH

FM29365

UK DECLARATION OF CONFORMITY

This declaration is issued under the sole responsibility of the manufacturer:

Hanovia Limited (t/a Nuvonic)
780 Buckingham Avenue, SLOUGH, Berkshire, SL1 4LA, UK

and hereby declares that:

Ultraviolet Fluid Treatment Systems PureLine PQ EO 4/6/8/10/12

have been demonstrated to fulfil the objectives referred to in the listed sections of the following UK Statutory Instruments:

SI 2016 No. 1101*	The Electrical Equipment (Safety) Regulations 2016
SI 2016 No. 1091*	The Electromagnetic Compatibility Regulations 2016
SI 2016 No. 1105*	The Pressure Equipment (Safety) Regulations 2016

*As amended by EU Exit Regulations

and hereby declares that the equipment named above is in conformity with the relevant United Kingdom designated standards:

EN 61010-1:2010 + A1:2019	Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements
EN 61326-1:2013	Electrical Equipment for Measurement, Control and Laboratory Use - EMC Requirements - Part 1: General Requirements
EN 61000-3-2:2019	Limits for harmonic current emissions (equipment input current < 16A per phase)
EN 61000-3-3:2013 + A1:2019	Limitation of voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current < 16A
EN 13445-1:2014 + A1:2014	Unfired pressure vessels - Part 1: General
EN 13445-2:2014 + A3:2018	Unfired pressure vessels - Part 2: Materials
EN 13445-3:2014 + A8:2019	Unfired pressure vessels - Part 3: Design
EN 13445-4:2014	Unfired pressure vessels - Part 4: Fabrication
EN 13445-5:2014 + A1:2018	Unfired pressure vessels - Part 5: Inspection and testing

The UKCA marking is applied accordingly.

Signed:

Name: Andy Sidwell
Position: Technical Director
Location: Slough, UK
Date: 1st April 2023

PURITY WITH SCIENCE

NUVONIC
formerly Aquionics, Berson, Hanovia and Orca GmbH

EU Declaration of Conformity

We, Hanovia Ltd (t/a Nuvonic), declare under our sole responsibility that the product listed below:

Medium Pressure UV Lamps

to which this declaration relates is in conformity with the provisions of the EU Electromagnetic Compatibility (EMC) Directive 2014/30/EU and complies with the following standards:

EN 55015:2013 + A1:2015	Limits and methods of measurement of radio disturbance characteristics of electrical lighting and similar equipment
EN 61547:2009	Equipment for general lighting purposes – EMC immunity requirements
EN 61000-3-2:2019 + A1:2021	Limits for harmonic current emissions (equipment input current < 16A per phase)
EN 61000-3-3:2013 + A1:2019	Limitation of voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current < 16A

The product is also in conformity with the provisions of the EU Low Voltage Directive 2014/35/EU and complies with the following standard:

EN 61010:2010 + A1:2019	Safety requirements for electrical equipment for measurement, control and laboratory use – Part 1: General requirements

The product is also in conformity with the EU Restriction of the Use of Certain Hazardous Substances (RoHS) Directive 2011/65/EU and complies with the following standard:

EN 63000:2018	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances

Issued: 1st April 2023, 780 Buckingham Avenue, SLOUGH, Berkshire SL1 4LA, United Kingdom

(EU Address: Berson Milieutechniek B.V., Tarasconweg 2, Unit A1.29 / Postvak 25, 5627 GB EINDHOVEN, Netherlands)

Signed for and on behalf of Hanovia Ltd (t/a Nuvonic)

CE

Andy Sidwell – Technical Director

910530-0001-01

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PURITY WITH SCIENCE

NUVONIC
formerly Aquionics, Berson, Hanovia and Orca GmbH

United Kingdom Declaration of Conformity

This declaration is issued under the sole responsibility of the manufacturer:

Hanovia Ltd (t/a Nuvonic), 780 Buckingham Avenue, SLOUGH, Berkshire SL1 4LA, United Kingdom

and hereby declares that the product:

Medium Pressure UV Lamps

has been demonstrated to fulfil the objectives referred to in the listed sections of the following UK Statutory Instruments:

SI 2016 No. 1101*	The Electrical Equipment (Safety) Regulations 2016
SI 2016 No. 1091*	The Electromagnetic Compatibility Regulations 2016
SI 2012 No. 3032*	The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012

* As amended by EU Exit Regulations

and hereby declares that the equipment named above is in conformity with the relevant United Kingdom designated standards for electrical and electronic equipment:

EN 61010:2010 + A1:2019	Safety requirements for electrical equipment for measurement, control and laboratory use – Part 1: General requirements
EN 55015:2013 + A1:2015	Limits and methods of measurement of radio disturbance characteristics of electrical lighting and similar equipment
EN 61547:2009	Equipment for general lighting purposes – EMC immunity requirements
EN 61000-3-2:2019 + A1:2021	Limits for harmonic current emissions (equipment input current < 16A per phase)
EN 61000-3-3:2013 + A1:2019	Limitation of voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current < 16A
EN 63000:2018	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances

The UKCA marking is applied accordingly.

Issued: 1st April 2023, Slough, UK

Signed for and on behalf of Hanovia Ltd (t/a Nuvonic)

Andy Sidwell – Technical Director

UK CA

910530-0001-01

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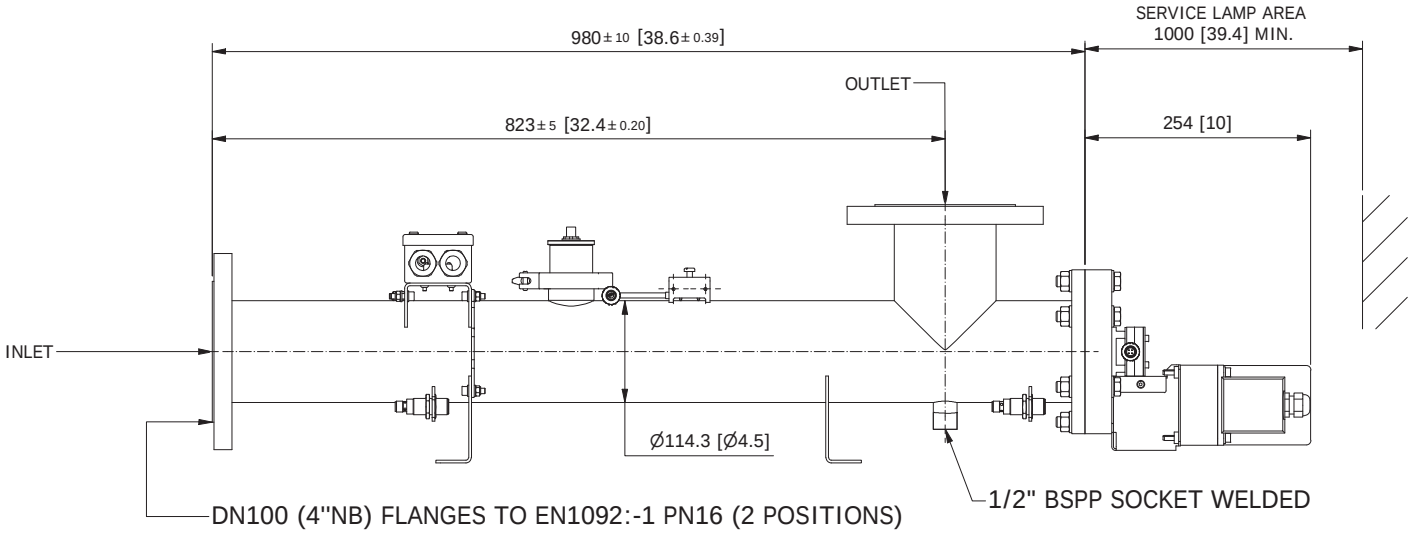
USA
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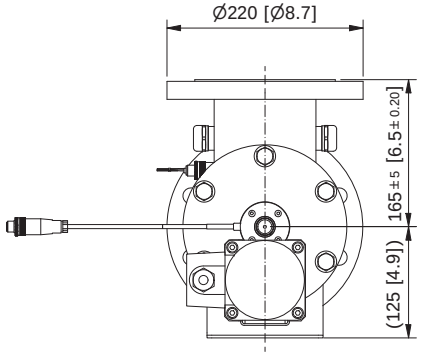
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Tel: +86 (0) 21 61617 9599
Email: apac@nuvonicuv.com

Appendix C: Drawings

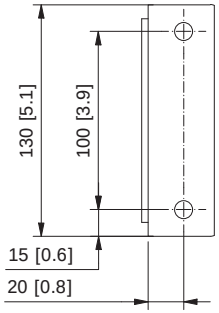
Chamber and Cabinet Drawings (all models)

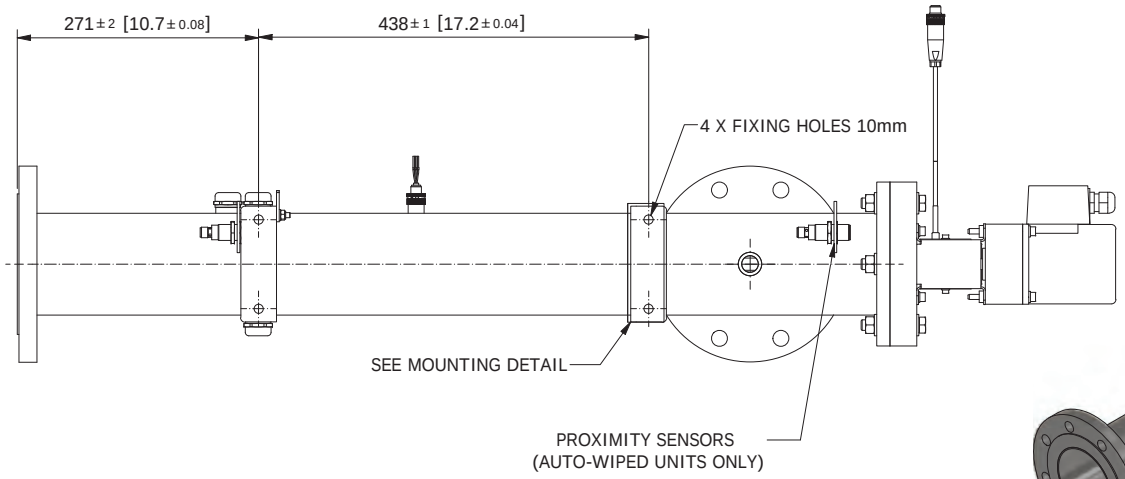
THIRD ANGLE PROJECTION

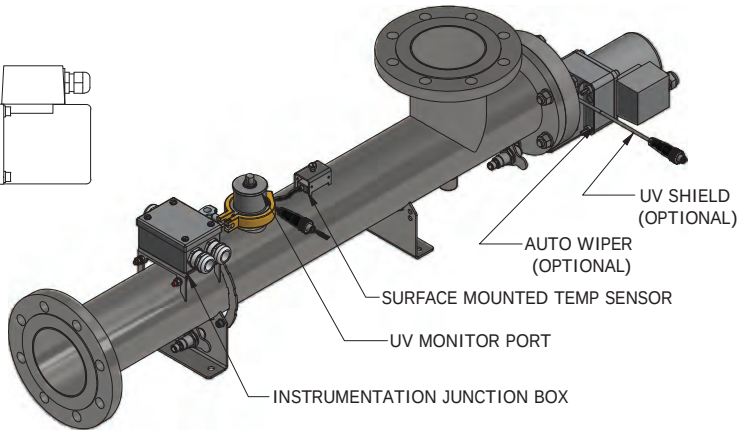




FEET MOUNTING DETAIL







MAX OPERATING PRESSURE	6 BAR (87 PSI)
OPTIONAL PRESSURE	10 BAR (145 PSI)
CHAMBER WEIGHT DRY	30 kg (66 lbs)
CHAMBER WEIGHT IN OPERATION	43 kg (94 lbs)

NUVONIC

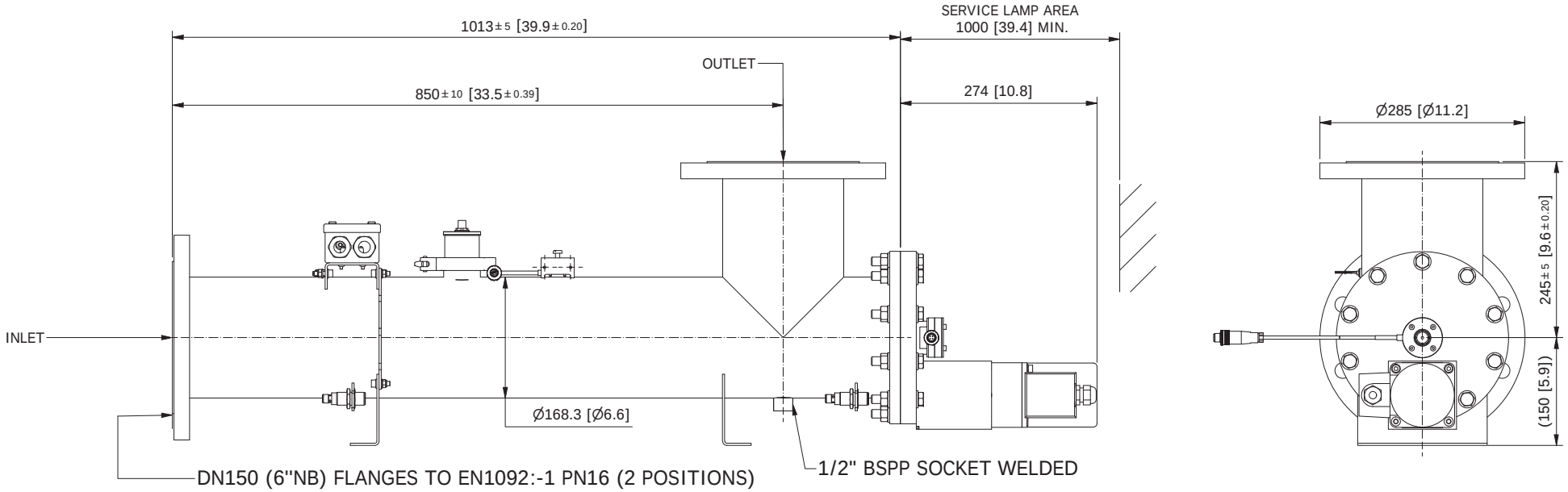
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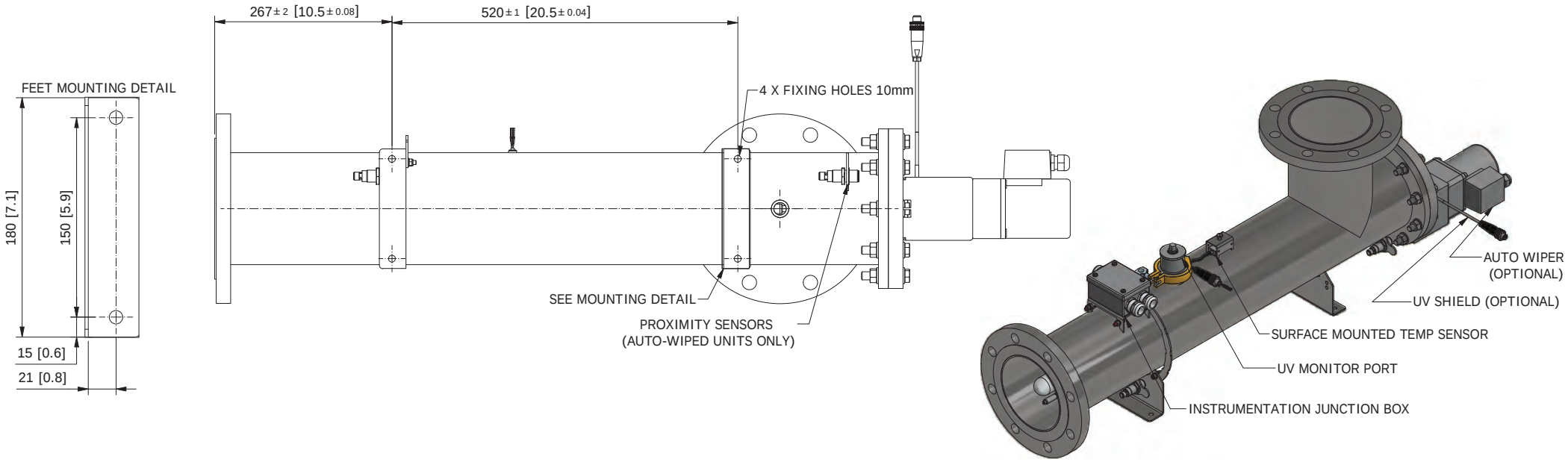
DRN	MM	DATE	05/03/2024
CKD	R.H	APPRD	R.H

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DRAWING No:-
820352-2G03A-01

THIRD ANGLE PROJECTION





MAX OPERATING PRESSURE	6 BAR (87 PSI)
OPTIONAL PRESSURE	10 BAR (145 PSI)
CHAMBER WEIGHT DRY	44 kg (97 lbs)
CHAMBER WEIGHT IN OPERATION	68 kg (149 lbs)

nuvonic

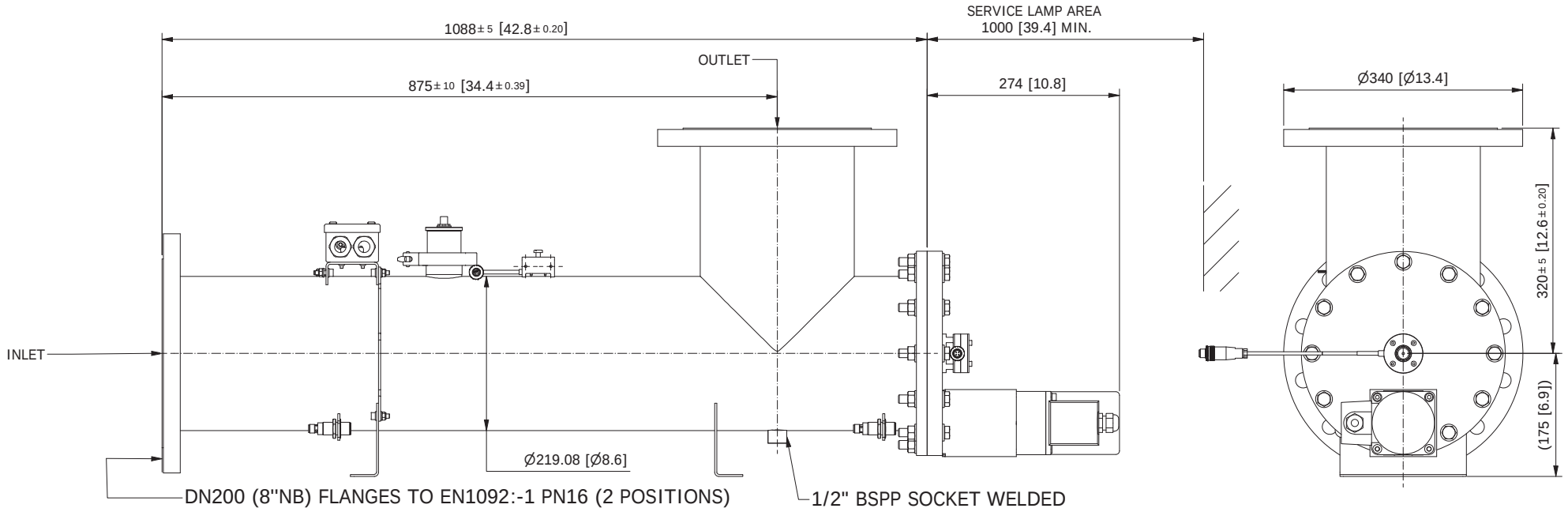
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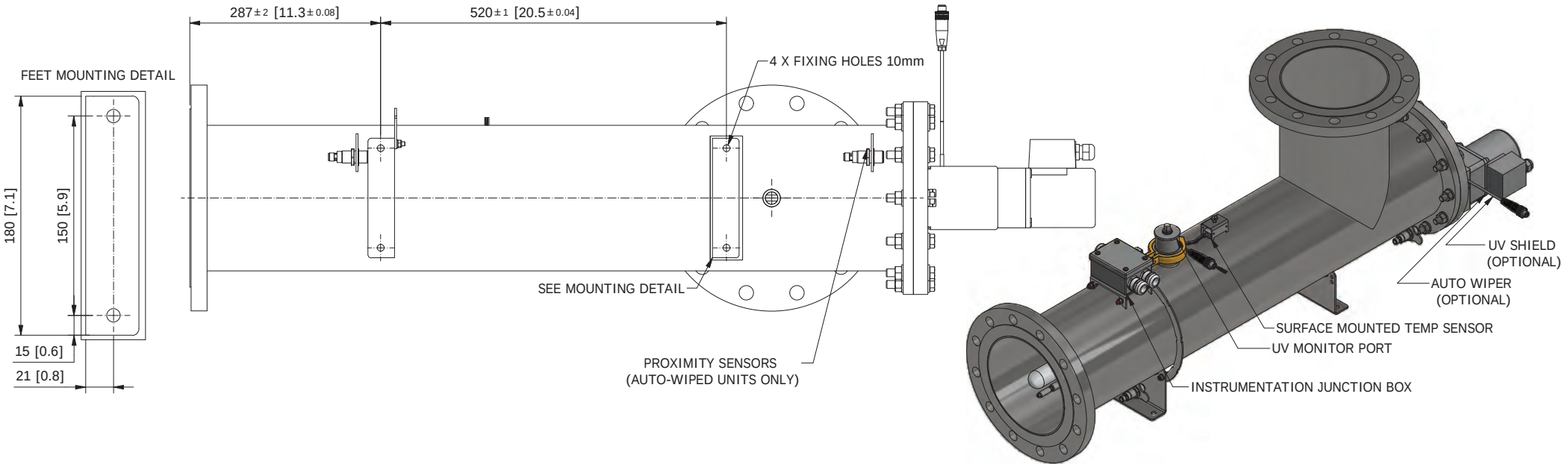
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CKD	R.H	APPRD	R.H

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DRAWING No:-
820352-2K03A-01

THIRD ANGLE PROJECTION





MAX OPERATING PRESSURE	6 BAR (87 PSI)
OPTIONAL PRESSURE	10 BAR (145 PSI)
CHAMBER WEIGHT DRY	65 kg (143 lbs)
CHAMBER WEIGHT IN OPERATION	111 kg (244 lbs)

NUVONIC

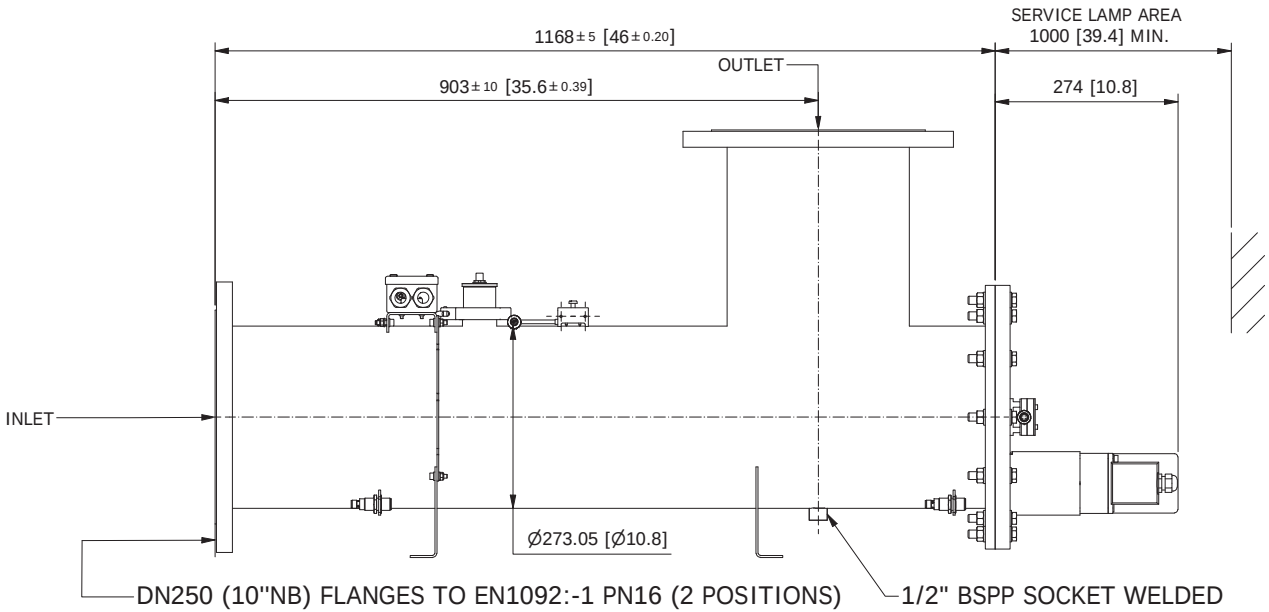
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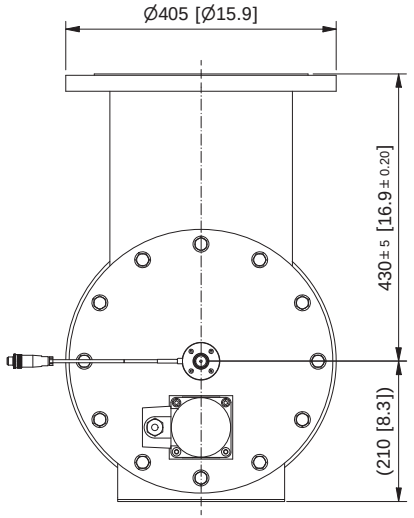
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CKD	R.H	APPRD	R.H

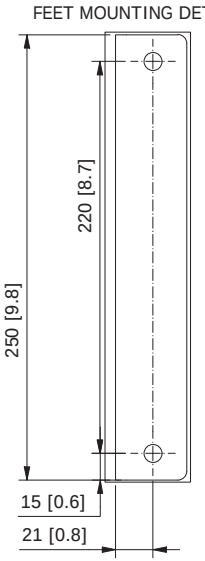
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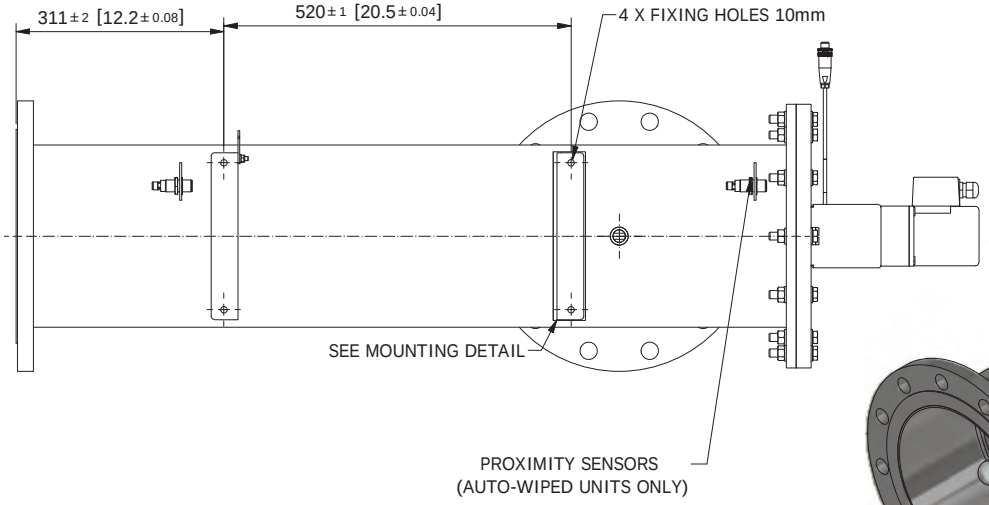
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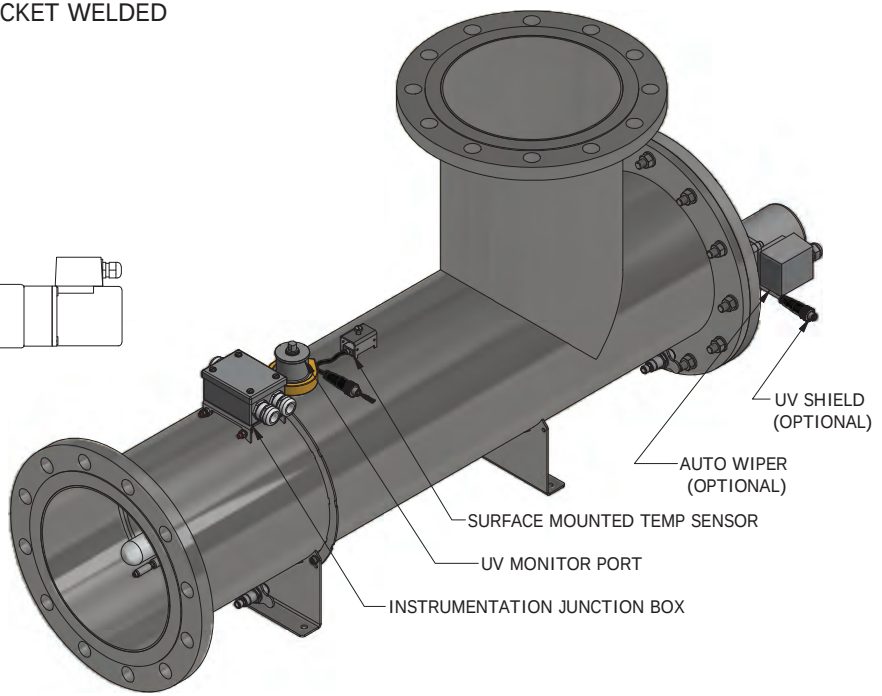
THIRD ANGLE PROJECTION











MAX OPERATING PRESSURE	6 BAR (87 PSI)
OPTIONAL PRESSURE	10 BAR (145 PSI)
CHAMBER WEIGHT DRY	96 kg (212 lbs)
CHAMBER WEIGHT IN OPERATION	176 kg (388 lbs)

NUVONIC

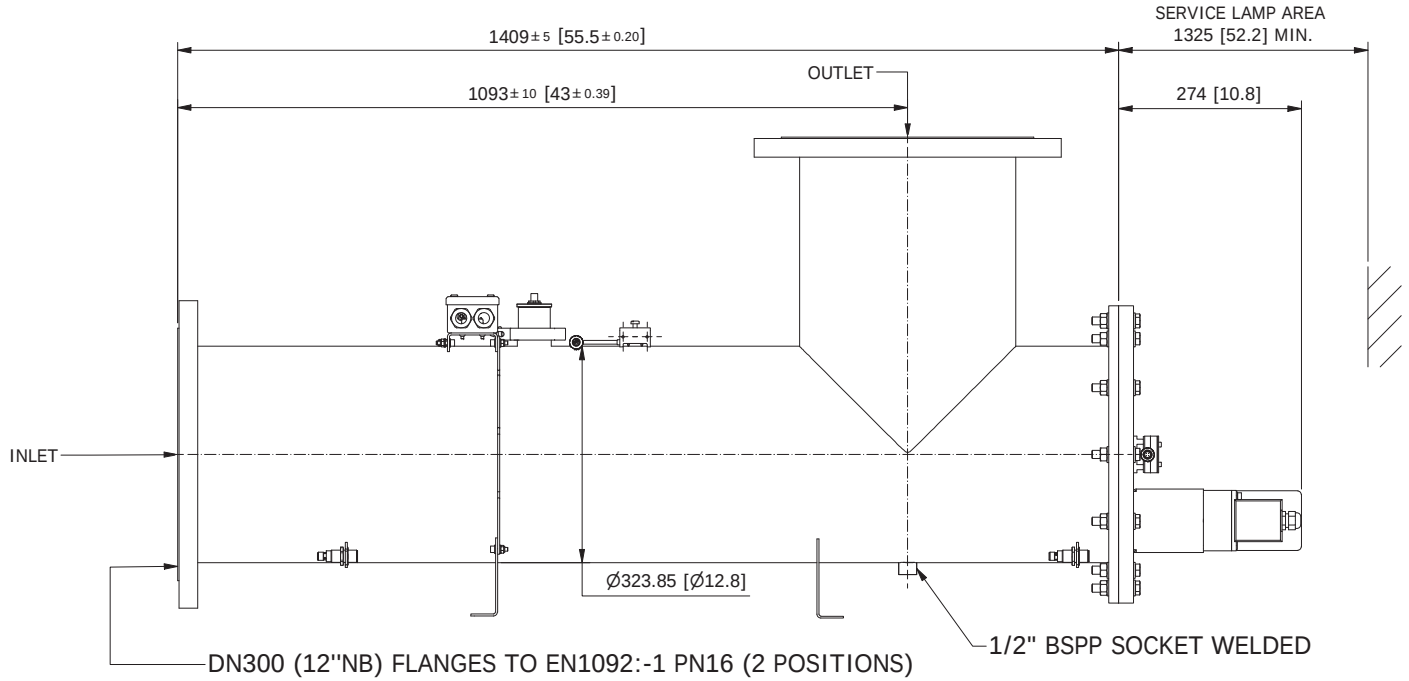
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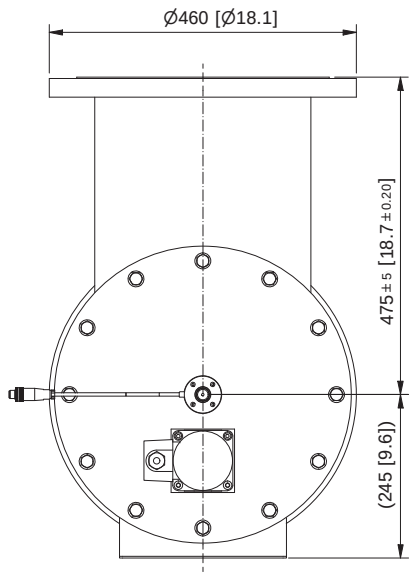
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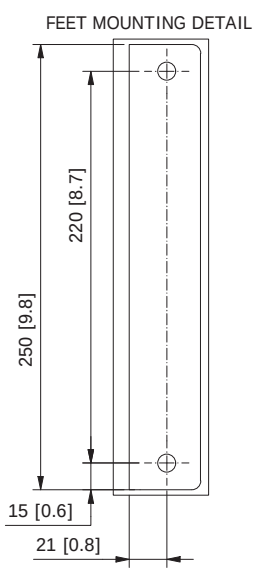
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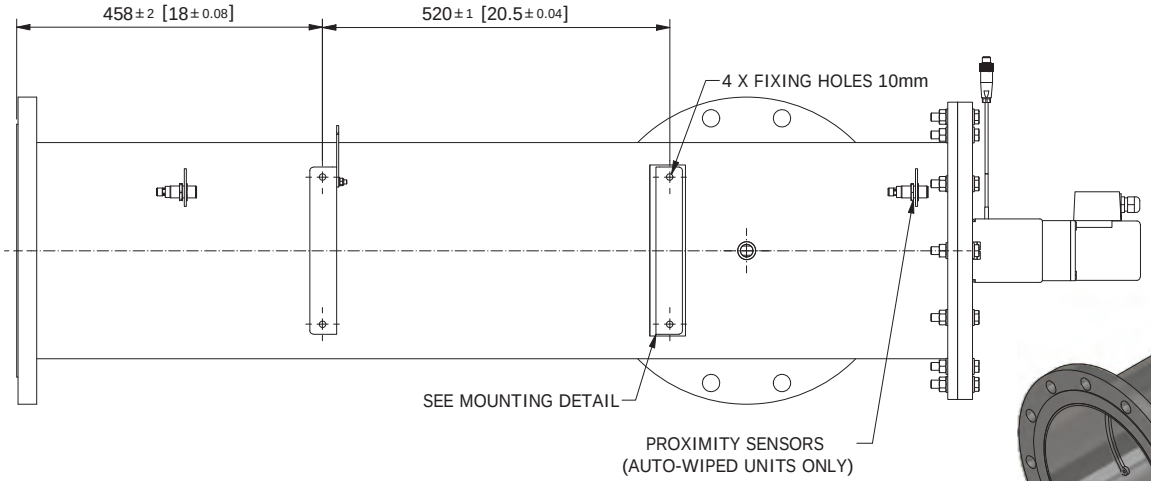
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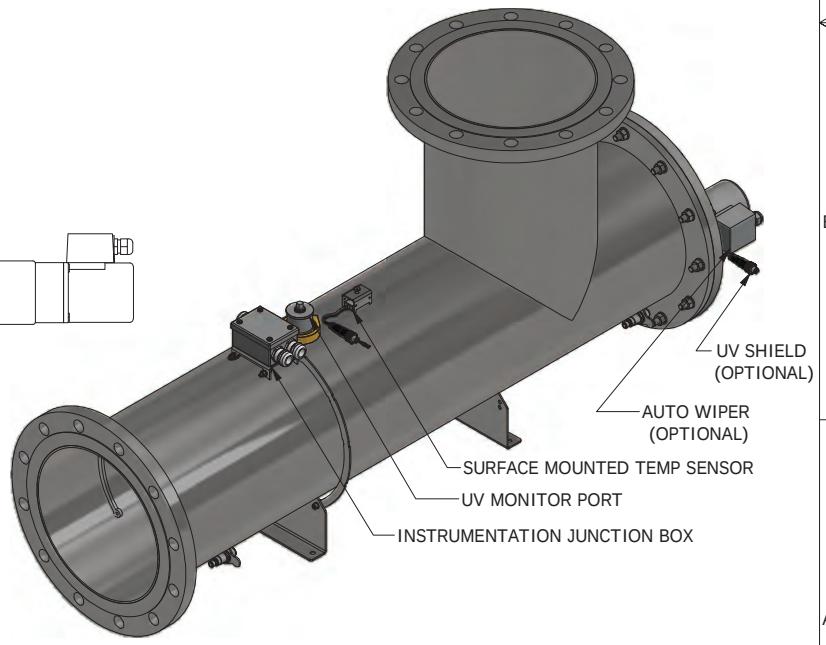
THIRD ANGLE PROJECTION











MAX OPERATING PRESSURE	6 BAR (87 PSI)
OPTIONAL PRESSURE	10 BAR (145 PSI)
CHAMBER WEIGHT DRY	145 kg (319 lbs)
CHAMBER WEIGHT IN OPERATION	279 kg (616 lbs)

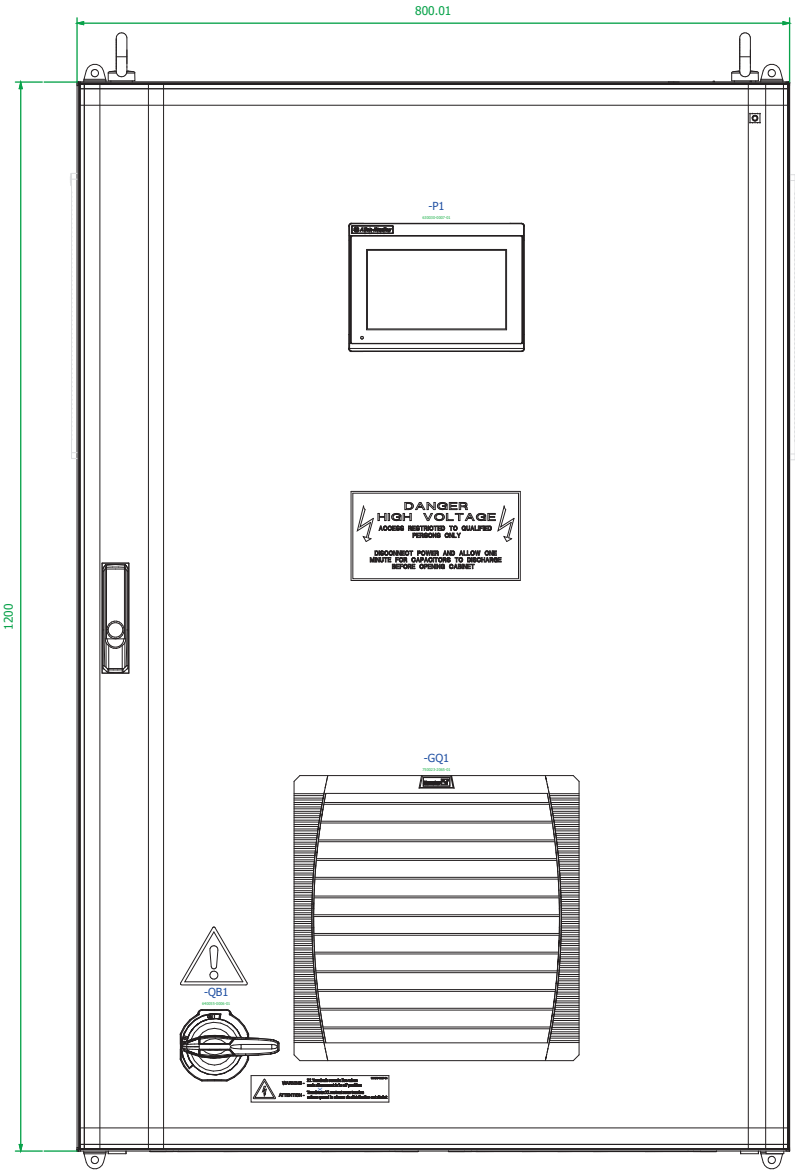
nuvonic

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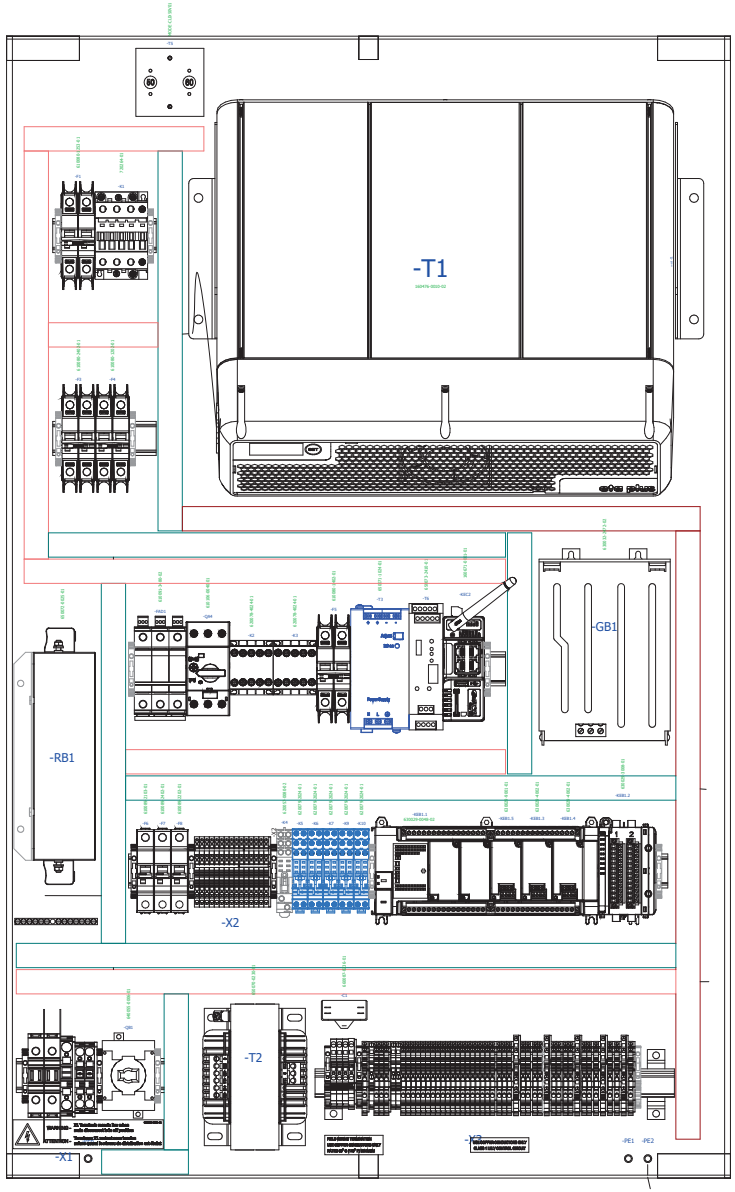
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DRAWING No:-
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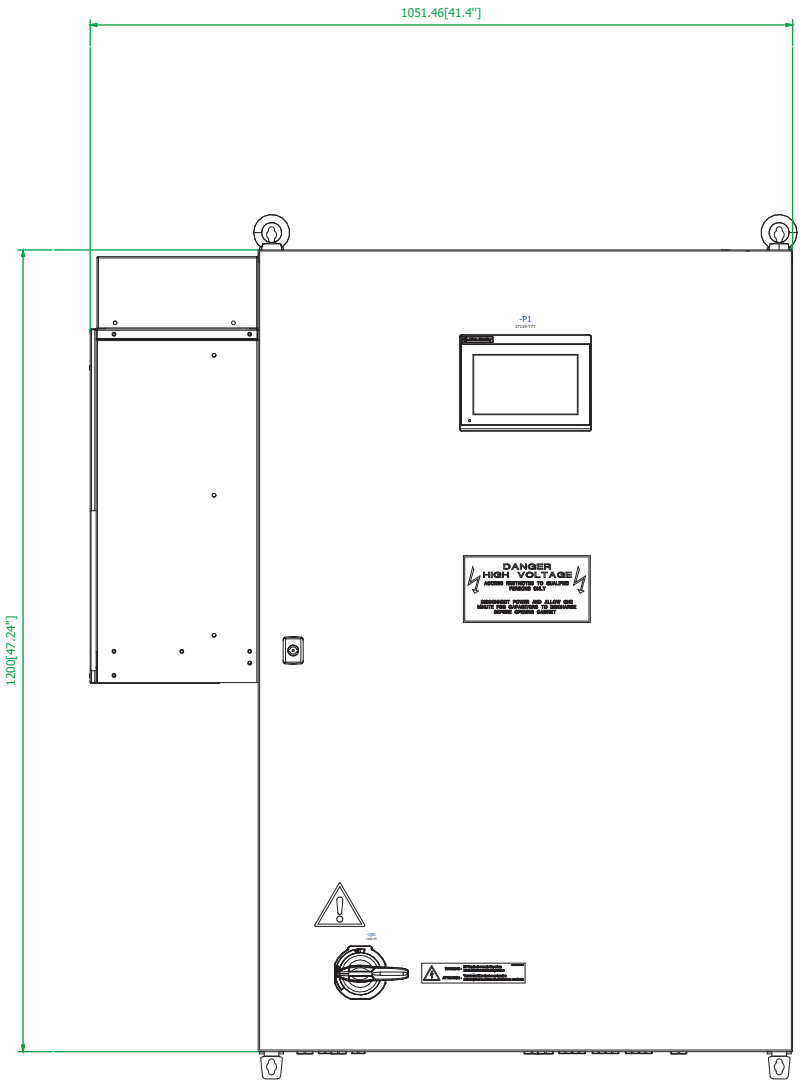


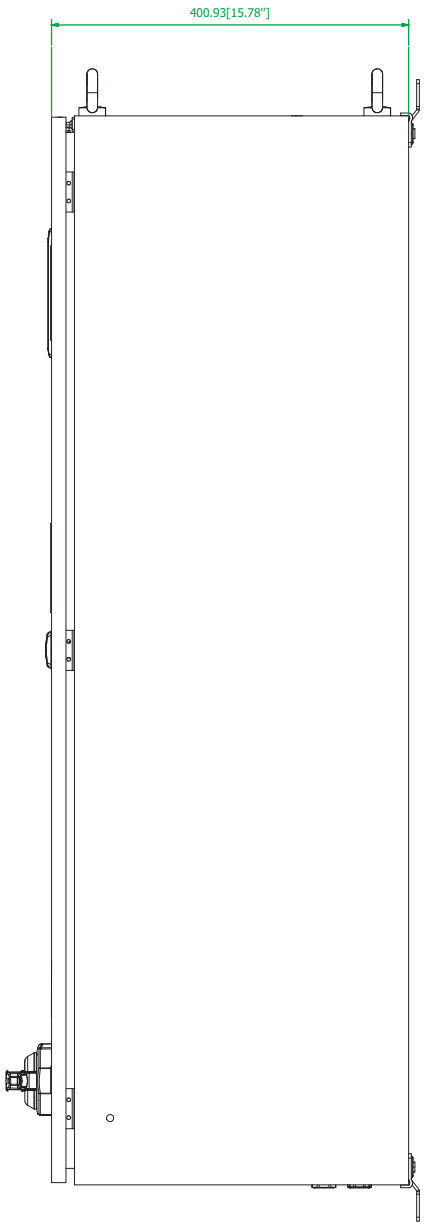
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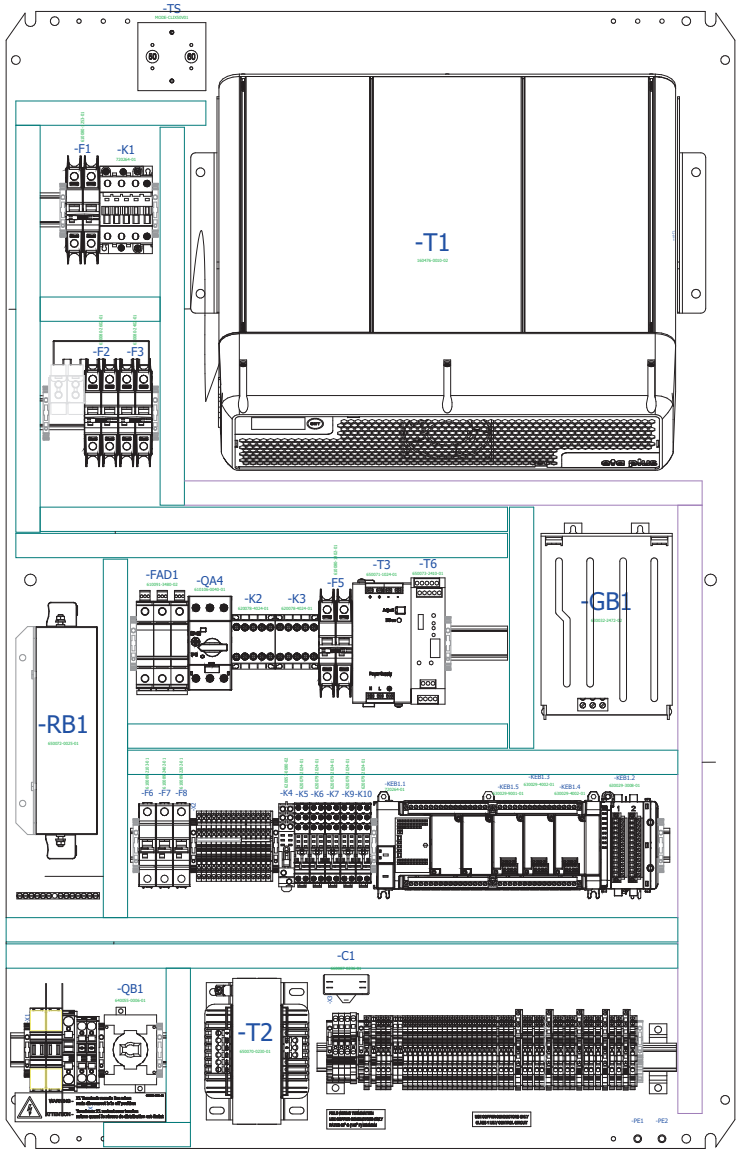
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&MTL5/57

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Revision name	ECN number	Revision date	Revision creator	Drawn by Daniel.Jones	Approved by FHQ		Date 25/07/2024		Page 58 of 58







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Revision name	ECN number	Revision date	Revision creator	Drawn by DANIEL.JONES	Approved by FHQ		Date 20/11/2024		Page 57 of 57

Circuit Diagrams